Threat Modeling, Attack Path Analysis, and Detection Techniques for IIoT Systems

A Thesis Submitted

In the Partial Fulfilment of the Requirement for the Degree of

Doctor of Philosophy



Submitted By

Kumar Saurabh

(PWC2017001)

Under the supervision of

Prof. O. P. Vyas

&

Late Prof. Shirshu Varma

to the

Department of Information Technology

Indian Institute of Information Technology, Allahabad

CERTIFICATE



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY ALLAHABAD

(A center of Excellence in Information Technology Established by Govt. of India)

CERTIFICATE FROM SUPERVISOR

It is certified that the work contained in the thesis titled **"Threat Modeling, Attack Path Analysis, and Detection Techniques for IIoT Systems,"** by **Kumar Saurabh, PWC2017001** has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

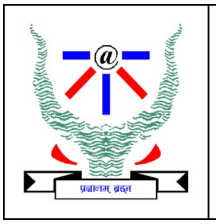
Date:

Place: IIIT Allahabad, Prayagraj

(Prof. Dr. O.P. Vyas)

(Late Prof. Shirshu Verma)

Counter Signed By Dean (Academics)



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY

ALLAHABAD

(A center of Excellence in Information Technology Established by Govt. of India)

CANDIDATE DECLARATION

I, **Kumar Saurabh**, Roll No. **PWC2017001**certify that this thesis work entitled **"Threat Modeling, Attack Path Analysis, and Detection Techniques for IIoT Systems"** is submitted by me in Complete fulfillment of the requirement of the Degree of Philosophy in Department of Information Technology, **Indian Institute of Information Technology, Allahabad.**

I understand that plagiarism includes:

1. Reproducing someone else's work (fully or partially) or ideas and claiming it as one's own.

2. Reproducing someone else's work (Verbatim copying or paraphrasing) without crediting.

3. Committing literary theft (copying some unique literary construct).

I have given due credit to the original authors/ sources through proper citation for all the words, ideas, diagrams, graphics, computer programs, experiments, results, websites, that are not my original contribution. I have used quotation marks to identify verbatim sentences and given credit to the original authors/sources. I affirm that no portion of my work is plagiarized. In the event of a complaint of plagiarism, I shall be fully responsible. I understand that my Supervisor may not be in a position to verify that this work is not plagiarized.

Name: Kumar Saurabh **Date:**

Enrolment No: PWC2017001

Department of Information Technology

IIIT-Allahabad (U.P.)

Abstract

The growing integration and mutual interdependence of Industrial Control and Production Systems (ICPS) have significantly expanded their attack surface, making them increasingly vulnerable to a sophisticated threat landscape characterized by highly orchestrated, multi-vector cyberattacks. The National Institute of Standards and Technology (NIST) Cyber Security Framework (CSF) provides a structured and comprehensive approach to manage industrial cybersecurity risks through its core functions. While the Threat modeling plays a central and foundational role in anticipating and mitigating such risks, current approaches often fall short in providing comprehensive lifecycle coverage and fail to support effective attack path analysis, especially when integrating dynamic and heterogeneous smart manufacturing technologies.

Despite the existence of numerous methodological solutions for threat modeling in Cyber-Physical Systems (CPS), current approaches are ad hoc and inefficient in providing clear insights to researchers and organizations involved in Industrial Internet of Things (IIoT) technologies. There is a conspicuous absence of a standardized, systematic approach for the identification, assessment, and prioritization of threats, in the context of IIoT, while most existing methodologies remain narrowly focused on specific industries, limiting their relevance across the diverse and rapidly evolving IIoT ecosystem. Whereas traditional Intrusion Detection Systems (IDSs) were effective in stable environments, the Machine Learning (ML) and Deep Learning (DL) based IDS, and Intrusion Prevention System (IPS) provides better results for dynamic and complex environments. Despite the inherent limitations of existing methodologies, such as high false positives, lack of real-time detection, and the need for extensive data preprocessing and feature engineering, the emergence of advanced ML and DL techniques has become exponentially critical, particularly in high-risk environments such as IIoT, where threat vectors are continuously evolving.

This study presents a systematic threat modeling and attack path analysis for Industrial IoT, combining "Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege (STRIDE)" and "Common Vulnerability Scoring System (CVSS)" based vulnerability scoring. Unlike existing research, which lacks a standardized, industry-agnostic process, this approach offers a structured method for identifying, assessing, and prioritizing threats across diverse IIoT environments. Additionally, the study showcases the practical benefits of this research through two case studies in the industrial manufacturing line and the power industry. In the proposed work, the limitations of traditional and ML based IDS mentioned above are also addressed. This research presents a suite of advanced intrusion detection frameworks for IIoT systems, addressing known and complex cyber threats through ML, DL and hybrid approaches.

An improved "Ensemble Learning Approach For Network Intrusion Detection System in Industrial IoT (ELA-NIDS)" demonstrates the efficacy of ensemble tree-based ML models, achieving up to 96.01% accuracy in classifying to attain a notable level of precision in the classification of diverse attack categories on IIoT systems. The "LSTM Based Deep Learning Model for Intrusion Detection Systems for IoT Networks (LBDMIDS)" model, leveraging LSTM variants, achieved up to 99.99% accuracy on "BoT-IoT" and "UNSW-NB15" datasets, showcasing strong temporal learning capability. The "Network Flow Based Semi Supervised Generative Adversarial Networks Model for IoT Botnets Detection (GANIBOT)", a semi-supervised SGAN based approach, demonstrated its strength in reducing labeling efforts and improving accuracy (from 45% to 59%) using "Least Absolute Shrinkage and Selection Operator (LASSO)" for feature selection. A computationally efficient Network Flow based Deep Learning Model for Distributed Denial of Service (DDoS) Attack Detection in IoT Domains (NFDLM)" maintains high DDoS detection accuracy while ensuring computationally efficient implementation via feature selection and basic ANN/LSTM models. The "Open-

Set learning based Hybrid Neural Network System for Enhancing Zero-Day Attack Detection (ZDA)" model introduces a open-set learning-based hybrid CNN-LSTM architecture, that allows the model to simultaneously analyze spatial features and time-series dynamics, making it more robust to variations and novel attack patterns, helps in effectively detection of zero-day attacks and outperforms baseline models. Lastly, the "Threat Intelligence Integration for Advanced Persistent Threats and Zero-Day Exploits in IIoT (HMS-IDS)" approach adopts a three-tier hybrid structure combining signature-based, anomaly-based, and CL-k-means clustering to achieve high detection rates across known and unknown threats.

Collectively, these models contribute a comprehensive and scalable detection strategy aligned with selected NIST CSF 2.0 functions, primarily **Identify**, **Detect**, and **Respond**, thereby enhancing IIoT cybersecurity. These are further compounded by the computational inefficiency of conventional IPS in resource-constrained IIoT environments. Current methods either cause data loss or trade accuracy for efficiency, limiting their effectiveness in real-time. The proposed hybrid model ensures high detection accuracy with minimal computational overhead. Additionally, this research envisions an open-source assessment tool to assist resource constrained IIoT systems like Small & Medium Enterprises (SMEs), in evaluating and enhancing their cybersecurity posture against emerging threats.

Keywords: Threat Modeling, Attack Path Analysis, Intrusion Detection Systems, Industrial Internet of Things, Intrusion Prevention Systems, Zero Day Attacks.

Acknowledgement

To my advisor, **Prof. O.P. Vyas**, thank you for standing by me through the long and difficult years of my PhD. Your guidance, patience, and faith in me often gave me strength when I was close to giving up. You taught me to look beyond the obvious, to see the larger purpose of research, and yet never overlook the smallest insight that could change its direction. Working under your supervision has been one of the most meaningful experiences of my life. I also express my heartfelt gratitude to **Late Prof. Shirshu Varma**, my initial advisor, whose kindness and guidance shaped the beginning of this journey. Your lesson of living life with an equal heart for everyone will remain with me forever.

I am equally thankful to my doctoral committee members, **Prof. Dr. Rahamatullah Khondokar**, **Prof. Dr. S. Venkatesan**, **Prof. Dr. Pramod Kumar (Dean-Research & Innovation)**, and **Prof. Dr. Manish Kumar (HoD-IT)**, for their thoughtful feedback, support, and valuable advice, which strengthened my research and helped me find direction when the path felt uncertain.

Special thanks to **Dr. Ranjana Vyas** for your motherly affection, kindness, and care. The warmth and second-home atmosphere you created made many difficult days easier to bear.

To the senior researchers, batchmates (especially **Manish Kumar Maurya**, and all my IIITA M.Tech. friends, especially WCE 2017-19 batchmates), thank you for filling this journey with knowledge, laughter, courage, and companionship. I am especially grateful to **Dr. Uphar Singh** for walking with me through this PhD journey, for your technical insights, honest discussions, and invaluable support during the phases when this work felt impossible.

I sincerely appreciate the support of **Prof. M.S. Sutaone** (Director, IIITA), **Prof. Dr. Pavan Chakraborty** (Dean Academics), the Research & Development

Section, all administrative officers and supporting staff at IIITA, editors, and reviewers. I am especially thankful to my friend **Ashutosh Kumar** (IoT Security Research Lab, IIITA), my last-mile DSBI labmate **Pranav Sahu**, and the fellow project students, including **Ayush Singh, Deepak Gajjala, Krishna Kaipa, P. Aditya, Saksham Sood, Vaidik Sharma, V. Vamshi, Abhishek Sharma, Tanuj, Shubham Singh**, and **Neha**, for their collaboration, friendship, and shared moments that made this journey lighter. I am also thankful to the Ministry of Education (MoE), India, and IIIT Allahabad for the doctoral fellowship and research ecosystem that made this work possible.

I am profoundly grateful to **Technische Hochschule Mittelhessen (THM), Germany**, and to **C3iHub, IIT Kanpur**, for providing the financial and mental stability that helped me carry this thesis to completion. On a special note, I thank **Prof. Sandeep K. Shukla, IIT Kanpur**, for placing his faith in me and including me in the **NCIIPC's Cyber Security Capability Maturity Model (CSCMM)** project. His mentorship gave me confidence when I doubted myself, pushed me to think critically, and helped me believe that I could aim higher.

My deepest gratitude belongs to my **parents** and **brother**, whose love, sacrifices, and silent prayers carried me through out. They may never read every page of this dissertation, but they will always be proud of. During the COVID-19 pandemic, during the pain of spinal injuries, and during the countless nights of uncertainty, their belief in me became my strength. I am especially thankful to **my loving wife, Preeti**, who stood beside me with endless patience, care, and love, even when this journey took more from us than we expected. To my adorable **daughter, Prisha**, your smile gave me reasons to continue on days when I felt completely broken. This thesis is not mine alone; it belongs to everyone who stood by me when the journey felt impossible. Finally, I thank you, the reader, for taking the time to read my work. .

"Everything happens for a reason. There is a reason you're going through a bad phase and there is a reason that you have a stroke of good luck. Accept the present, without fretting about the past or worrying about the future. It will all be good".

—**Enlightened by my father, rooted in the Bhagavad Gita** (Chapter 2, Verse 47).

Contents

List of figures **xxi**

List of Tables **xxv**

List of Abbreviations **xxx**

1 Introduction **2**

1.1 Motivation 5

1.2 Overview of an Industrial IoT based System 6

1.3 Cyberattacks Scenarios in an IIoT System 7

1.3.1 Phase 1: Compromise Within the Office Network 7

1.3.2 Phase 2: Breach Into the Operational Technology Environment 8

1.3.3 Phase 3: Exploiting External and Cloud Services 8

1.3.4 Phase 4: Operational Disruption Through Targeted Scenarios 9

1.4 Research Objectives 10

1.5 Research Contributions 10

1.6 Thesis Outline 13

2 Literature Review **15**

2.1 Adoption of Industrial Internet of Things: Overview 15

2.1.1 Infusion of IT & OT Convergence 16

2.1.2 The Hyperindustrialisation 18

2.2 CyberSecurity Lifecycle: Before & Beyond IEC 61508 19

2.2.1 ISA/IEC 62443 21

2.3 Existing Cybersecurity Frameworks for IIoT Systems 22

2.3.1 NIS & NIS2 Directive Characteristics 22

2.3.2 KRITIS Coverage 25

2.3.3 NIST Cybersecurity Framework Overview 27

2.3.4 Comparative Study of Cybersecurity Frameworks 28

2.4 Positioning Threat Modeling, Detection, & Response in the NIST Cybersecurity Lifecycle 29

2.4.1 Aligning Threat Modeling with NIST CSF 30

2.4.2 Aligning Attack Detection & Response with NIST CSF . 31

2.5 Prevalent Cybersecurity Threats in Industrial IoT 32

2.5.1 Supply chain attacks 32
2.5.2 Data Breaches 32
2.5.3 Cyber Espionage 33
2.5.4 Malware Infections 33
2.5.5 Third-Party Vulnerabilities 33
2.5.6 Phishing Scams 34
2.5.7 Identity Fraud 34
2.5.8 Cryptojacking 35
2.5.9 Active or Passive Eavesdropping 35
2.5.10 Man in the Middle (MITM) 35
2.5.11 BotNet Attack 36
2.5.12 DoS and DDoS Attacks 36
2.5.13 Advanced Persistent Threats (APTs) 38
2.5.14 Zero Day Attacks 39
2.6 Datasets and Algorithms: A Technical Landscape 40
2.6.1 Overview of Experimental Corpora 40
2.6.2 UNSW-NB15 41
2.6.3 Bot-Iot Dataset 41
2.6.4 CIC-ToN-IoT Dataset 42

2.6.5 KDD-99 cup Dataset 44
2.6.6 NSL-KDD Dataset 44
2.6.7 CIC IDS 2017 45
2.6.8 NF-ToN-IoT-v2 46
2.6.9 Machine Learning & Ensemble Learning Models: An Overview 47
2.6.10 Logistic Regression (LR) 47
2.6.11 Naive Bayes (NB) 48
2.6.12 K-Nearest Neighbors (KNN) 49
2.6.13 Support Vector Machine (SVM) 50
2.6.14 One-Class SVM (OCSVM) 51
2.6.15 Decision Tree (DT) 52
2.6.16 Random Forest 52
2.6.17 Extra Trees 53
2.6.18 XGBoost 53
2.6.19 CatBoost 54
2.6.20 Performance Metrics 54
2.7 Literature: Gaps, Problems, and Key Contributions 56
2.7.1 Threat Modeling and Attack Path Analysis 56

2.7.2 Detection of known Cyber Threats Leveraging Ensemble ML Techniques 61

2.7.3 Machine learning Approach for Botnet Attack Detection . 63

2.7.4 Deep Learning based IDS for IIoT systems 67

2.7.5 DDoS Attack Detection Leveraging ML & DL Approaches 70

2.7.6 Zero Day Attack Detection in IIoT 72

2.7.7 Intrusion Response Mechanisms in IIoT 76

3 Threat Modeling and Attack Path Analysis 80

3.1 Threat Modeling: Characteristics & Issues 80

3.1.1 Types of Threat Modeling Approaches 81

3.1.2 Threat Modeling Methodologies 82

3.2 Proposed Methodology 85

3.2.1 Architectural Setup & Asset Selection 86

3.2.2 Data Flow Diagrams 86

3.2.3 Asset Identification 87

3.2.4 Threat Modeling 88

3.2.5 Using Attack Vectors to create Attack Paths 89

3.2.6 Vulnerability Scoring 90

3.2.7 Base Metrics 91

3.2.8 Temporal Metrics 92
3.2.9 Environmental Metrics 93
3.3 Case Study: Threat Modeling of an IIoT System 94
3.3.1 Case Study I: Electric Power Systems as an Internet of Production (IoP) 94
3.3.2 Case Study II: Manufacturing Industry (IoM) 98
3.4 Identification of Key Assets with Predominant Threats 102
3.5 Chapter Summary 104

4 Detection Mechanisms for known Cyber Threats Leveraging ML Techniques 107

4.1 Significance of NIDS in Second Line of Defence 107
4.1.1 Intrusion Detection System 108
4.1.2 Signature-Based intrusion detection system 108
4.1.3 Anomaly based intrusion detection system 108
4.2 ELA-NIDS: An Ensemble Learning base 109
4.2.1 Proposed Methodology 110
4.2.2 Dataset 110
4.2.3 Data Preprocessing 112
4.2.4 Data transformation 112

4.2.5 Data transformation using clustering 112
4.2.6 Label Encoding 113
4.2.7 Data Normalization 114
4.2.8 Data Augmentation 114
4.2.9 Feature Engineering 115
4.2.10 Model Selection Criteria: 116
4.2.11 Results and Performance Analysis 117
4.3 GANIBOT: A Semi-Supervised Generative Adversarial Network Approach for DDoS Attack Detection 118
4.3.1 Formulated approach 120
4.3.2 Dataset: N-BaIoT 120
4.3.3 Feature Selection 121
4.3.4 The SGAN Model 123
4.3.5 Results 125
4.4 LBDMIDS: LSTM based Deep Learning Approach for Network Intrusion Detection System 129
4.4.1 Dataset & Outlined method 131
4.4.2 Proposed Architecture 131
4.4.3 Data Preprocessing 132
4.4.4 Feature Extraction 132

4.4.5 Z-scale Normalization 132
4.4.6 Training and Validation Process 134
4.4.7 Stacked LSTM 135
4.4.8 Bi-Directional LSTM 136
4.4.9 Model Structure and Parameters 136
4.4.10 Results and Performance Analysis 137
4.4.11 Comparison with other techniques: 140
4.5 NFDLM: Computationally Efficient DDoS Attack Detection . . . 141
4.5.1 Proposed Methodology and Experimental Setup 141
4.5.2 Data Preprocessing 143
4.5.3 Data Cleaning 144
4.5.4 Scaling and normalizing data 144
4.5.5 Handling Imbalances in Data Labels 145
4.5.6 Feature Selection and Model Building 145
4.5.7 Correlation 146
4.5.8 Mutual information 146
4.5.9 Model Selection & Training 147
4.5.10 Results & Discussion 149
4.6 Chapter Summary 152

5 Semantic Attribute based Open-Set IDS for IIoT 154

5.1 Methodology and Implementation Details 155

5.1.1 Dataset & Preprocessing 156

5.1.2 Hybrid CNN-LSTM Model 158

5.2 Experimental Setup, Results and Analysis 159

5.2.1 Experimental Setup 159

5.2.2 Results and Analysis 159

6 A Hybrid Approach for Zero-Day & Advanced Persistent Threats Detection in IIoT 164

6.1 Proposed Methodology 166

6.1.1 Data Pre-Processing 166

6.1.2 Data Sampling 168

6.1.3 Data Normalisation 169

6.1.4 Feature Engineering 171

6.1.5 Proposed Hybrid Intrusion Detection System Design . . . 172

6.1.6 Devising The the Signature-based IDS (To Identify Known Attacks) 173

6.1.7 An Anomaly-based IDS (To Cluster Unknown Attacks & Benign Data) 174

6.2 Results and Observations 179

6.2.1 Performance Evaluation of all individual ML models on CIC-ToN-IoT dataset 180

6.2.2 Performance Evaluation of Ensemble Learning models on CIC-ToN-IoT dataset 180

6.2.3 Performance Evaluation of K-means Clustering on CIC-ToN-IoT dataset 183

6.2.4 Performance evaluation of signature based on different datasets 186

6.2.5 Performance evaluation of Anomaly-Based IDS on different datasets 187

6.2.6 Testing Time comparison for Signature and Anomaly-based IDS on different datasets 187

6.2.7 Runtime Complexity 188

6.3 Chapter Summary 190

7 An ML-based Second Line of Defense for Intrusion Prevention 191

7.1 Methodology 192

7.1.1 Datapoint Duration and Threshold Calculation 193

7.1.2 Dataset & Data Preprocessing 194

7.1.3 Data Preprocessing and Normalization 195

7.1.4 Feature Selection 196

7.1.5 Observation 197
7.1.6 Selection of Models 198
7.2 Performance Analysis of the Proposed Intrusion Prevention System 199
7.2.1 Phase 1: Formation of models and calculation of threshold value 199
7.2.2 Phase 2: Prevention - Dropping of packets 200
7.3 Chapter Summary 201

8 Discussion and Future Direction 202

8.1 Insights and Observations 202
8.1.1 Unified Case Study for Attack Path Analysis and Detection Response 203
8.1.2 Cross-Chapter Comparative Summary of Proposed IDS/IPS Models 210
8.2 Contributions of this Thesis 211
8.3 Limitations of this Study 213
8.4 Future Research Directions 214
8.4.1 Open-Source IIoT Cybersecurity Assessment Tool 215
8.4.2 Dynamic Threat Modeling for Evolving IIoT Environments 216
8.4.3 Improved Generative Models for Multiclass IDS 216

8.4.4 Edge-Hardware Latency Evaluation of HMS-IDS 217

8.4.5 Adversarial Robustness and Concept Drift Mitigation . . . 217

8.5 Code and Reproducibility 218

8.6 Journal (Included in Thesis) 219

8.7 Conferences (Included in Thesis) 219

List of Figures

1.1 The CIA triad for cybersecurity (Image Ref [50]) 3

1.2 NIST V-2.0 CSF (Image Ref [13]) 4

1.3 The Architecture of a Manufacturing Plant[63] 6

1.4 Operational Disruption Through Targeted Scenarios[63] 9

1.5 Thesis-Workflow 11

2.1 “From ICT to Industry 4.0: A Conceptual Continuum” 17

2.2 IEC 62443 Cybersecurity Lifecycle [109] 22

2.3 NIS2 Directive, an extension of the former NIS Directive that aims to establish a common level of cybersecurity across the European Union (Image Source [154]) 24

2.4 NIS2 Pillars (Image Ref [51]) 25

2.5 Critical infrastructures (KRITIS) by BSI (Image Ref [25]) 26

2.6 NIST V-2.0 CSF (Image Ref [13]) 27

2.7 Malware Attacks on IIoT systems 34

2.8 Botnet Attack on IoT devices 37

2.9 DDoS attack scenario diagram 38

2.10 Attack Surface in an Industrial IoT 39

2.11 CIC-ToN-IoT-Dataset Distribution 43

2.12 Botnet Attack on IoT devices 63

3.1 Proposed Methodology 85

3.2 CVSS Metrics 91

3.3 DFD for SCADA in Electric power system 95

3.4 Purdue model for Electric power systems 95

3.5 Attack paths for SCADA in Electric power system 97

3.6 DFD for HMI in a Manufacturing Industry 99

3.7 The Architecture of A Manufacturing Plant 100

3.8 Attack paths for HMI in a Manufacturing Industry 101

3.9 Purdue model for a Manufacturing Industry 102

4.1 Workflow Diagram of the Proposed Methodology 120

4.2 SGAN Block diagram 123

4.3 SGAN evaluation metrics for multi-class classification on small (Fig. a) and Large testing (Fig. b) data without feature selection . 126

4.4 SGAN evaluation metrics for multi-class classification on small and Large testing datasets with feature selection 127

4.5 Proposed LBDMIDS architecture 131

4.6 Training vs. Validation accuracy (STACKED LSTM) 139

4.7 Training vs. Validation loss (STACKED LSTM) 139

4.8 Training vs. Validation accuracy (BI-LSTM) 139

4.9 Training vs. Validation loss (BI-LSTM) 139

4.10 NIDS accuracy comparison of Bi-LSTM and Stacked LSTM on both datasets 140

4.11 Proposed NFDLM workflow diagram 142

4.12 Bar graph showing distribution of attack category 143

4.13 Heat diagram for Correlation Coefficient feature selector 151

4.14 Graph showing ranking of features based on Mutual Information . 151

5.1 Illustration of the Training and Testing Dataset. 156

5.2 Methodology Flowchart. 157

5.3 Number of rows for each class. 157

5.4 Performance of CNN-LSTM model in detecting DDoS attacks as Zero-Day Attacks 161

5.5 Confusion Matrix for DDoS attack as Zero-Day Attack class. . . . 162

6.1 Proposed Methodology Diagram of The HMS-IDS 167
6.2 Confusion Matrix for Signature-Based IDS 181
6.3 Signature-Based IDS's Training Accuracy 181
6.4 Signature-Based IDS's Testing Accuracies 182
6.5 Signature-Based IDS's Training and Testing Time-Model-Wise . . 182
6.6 Confusion Matrix for Anomaly-Based IDS 185

7.1 IDS To IPS 192
7.2 Binary classification (UNSW-NB15) 195
7.3 Multiclass classification (UNSW-NB15) 195
7.4 Correlation matrix Binary Data 196
7.5 Feature Selection Binary Data 197
7.6 Correlation matrix Multiclass data 197
7.7 Feature Selection Multiclass Data 198
7.8 Phase 1: IPS 200
7.9 Phase 2 : IPS 200

8.1 Data Flow Diagram for Unified IIoT Case Study 204
8.2 Attack Paths for Unified IIoT Case Study 205
8.3 Proposed Workflow for Unified IIoT Cybersecurity 208

List of Tables

2.1 Industry Evolution vs. IIoT Development 18

2.3 Cybersecurity Lifecycle: Pre vs. Post IEC 61508 Era 20

2.5 Comparison of Cybersecurity Frameworks for IIoT systems 29

2.6 Mapping Threat Modeling to NIST Cybersecurity Framework (CSF 2.0) Stages 30

2.7 Intrusion Detection Across NIST CSF for IIoT 31

2.8 Response Mechanisms Across NIST CSF for IIoT 31

2.9 Distribution of different types of attacks in the dataset. 43

2.10 Class Distribution in Training and Test Sets 45

2.11 NSL-KDD Dataset Distribution 45

2.12 Daily Label of CICIDS 2017 Dataset 46

2.13 The attacks with the distributions found in the NF-ToN-IoT v2 dataset : 47

2.14 Summary of Literature Survey on Threat Assessment in Industrial IoT Systems 59

2.15 Different intrusion detection approaches 71

2.16 Different intrusion prevention and detection approaches 78

3.1 Elements of STRIDE 89

3.4 Threat summary for Electric power systems 96

3.5 Threat summary for Manufacturing Industry 100

3.6 Threat count per asset 102

3.7 Top 5 threats 103

3.8 Threat count per asset 103

3.9 Top 5 threats 104

3.2 Tactics and Techniques Representing The MITRE ATT&CK® Matrix for Industrial Control Systems (ICSs) [159] 105

3.3 Tactics and Techniques Representing The MITRE ATT&CK® Matrix for Industrial Control Systems (ICSs) (Continued) [159] . . . 106

4.1 Comparison of Datasets for Attack Detection in Industrial IoT . . 111

4.2 Comparison of the Proposed NIDS Performance Metrics with Recent Works 116

4.3 Model performance on evaluation metrics 118

4.4 XG-Boost Multiclassification performance 119

4.5 Dataset details 121

4.6 Features of N-BaIoT dataset 122

4.7 Results for Multi-class classification without feature selection . . 125

4.8 Results for Multi-class classification with feature selection 127

4.9 Effectiveness Comparison for Binary Classification on N-BaIot Dataset 129

4.10 Network Flow Feature Descriptions 133

4.11 Aggregated Flow Feature Descriptions 134

4.12 Model hyper-parameters for Stacked LSTM 135

4.13 Model hyper-parameters for Bi-LSTM 136

4.14 Stacked LSTM 137

4.15 Bi-directional LSTM 138

4.16 Stacked LSTM & Bi-directional LSTM 139

4.17 Comparison of Results on BoT-IoT and UNSW_NB-15 Datasets . 140

4.18 Models comparison 150

4.19 Effectiveness comparison 150

5.1 Detection Rate Comparison of Different Models 160

5.2 Metrics for Different Attack Classes based on CNN-LSTM model 160

6.1 Ensemble-Based Model Accuracy 183

6.2 Ensemble Model's Training Multi-Classification Report 183

6.3 Ensemble Model's Testing Multi-Classification Report 184

6.4 Anomaly-Based Model's Detection Accuracy 185

6.5 Results of Signature-Based IDS on different datasets 186

6.6 Anomaly-Based IDS Metrics for Different Datasets 187

6.7 Time comparison for different datasets 188

7.1 Dataset Features 194

7.2 Binary classification 198

7.3 Multiclass classification 199

7.4 Binary classifiers 201

7.5 Multi-class classifiers 201

8.1 Threat identification in the IIoT case study 205

8.2 Attack path mapping with IDS detection and response 207

8.3 Cross-chapter comparative summary of proposed IDS/IPS models 211

List of Abbreviations

AI	Artificial Intelligence
APT	Advanced Persistent Threat
ANN	Artificial Nueral Network
BO	Bayesian Optimization
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
CPS	Cyber-Physical Systems
DL	Deep Learning
GAN	Generative Adversarial Network
GRU	Gated Recurrent Unit
IIoT	Industrial Internet of Things
IT	Information Technology
ICPS	Industrial Control and Production System
IoCs	Indicators of Compromise
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
KNN	k-Nearest Neighbors

LSTM	Long Short-Term Memory
LASSO	Least Absolute Shrinkage and Selection Operator
MAE	Mean Absolute Error
ML	Machine Learning
MLP	Multi-Layer Perceptron
OT	Operational Technology
PLC	Programmable Logic Controller
RNN	Recurrent Neural Network
SGAN	Semi Supervised Generative Adversarial Networks
SVM	Support Vector Machine
ZDA	Zero Day Attack

Chapter 1

Introduction

The growing integration and mutual interdependence of industrial controls and production systems(ICPS) has revolutionised industrial operations, enabling real-time monitoring, automation, and data-driven decision-making. Cyberthreats to ICPS in smart factories are growing in the age of interconnected systems, which is essential for secure deployment. The CIA triad (Fig. 1.1) serves as a foundational model for cybersecurity implementation within IIoT ecosystems. Confidentiality safeguards sensitive data (such as sensor readings, control commands, and credentials) from unauthorized access; integrity ensures that data and commands remain unaltered by malicious or accidental interference; and availability guarantees continuous, real-time access to IIoT systems and information. Threat identification, detection, and protection function as key operational processes to uphold the core objectives defined by the CIA triad in securing IIoT environments.

Threat modeling helps to systematically identify and assess potential threats in a system. While it plays a central and foundational role in anticipating and mitigating such risks, current approaches often fall short in providing comprehensive lifecycle coverage and fail to support effective attack path analysis, especially when integrating dynamic and heterogeneous smart manufacturing technologies. Mod-

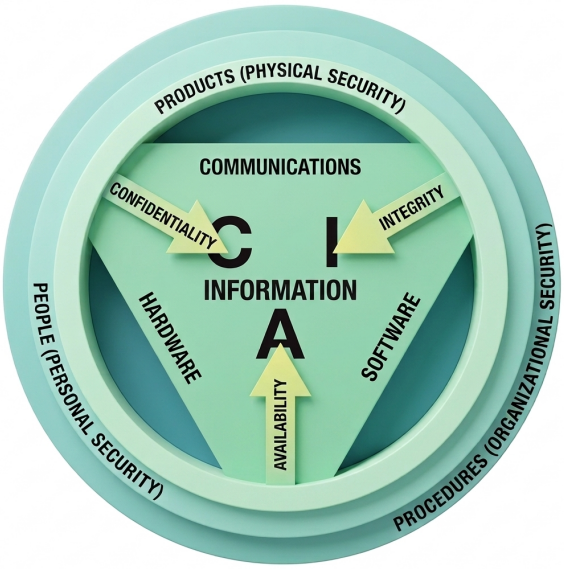


Figure 1.1: The CIA triad for cybersecurity (Image Ref [50])

ern IIoT cybersecurity demands a risk-driven, path-aware, and contextual threat modeling approach which combines traditional models with attack path analysis and impact-based prioritization to protect critical systems efficiently.

Traditional cybersecurity measures, such as signature-based IDS, are often inadequate in addressing the dynamic and evolving nature of cyber threats in IIoT environments. Signature-based systems rely on known Indicators of Compromise (IOCs), making them ineffective against zero-day attacks that exploit previously unknown vulnerabilities. In contrast, ML-based IDS offer a more robust solution by analysing network behaviour and identifying anomalies that deviate from normal patterns. These systems leverage statistical algorithms to detect complex attack patterns, enabling proactive threat detection and response. However, the integration of ML in IIoT cybersecurity is still in its nascent stages, with challenges such as high false positive rates, dataset biases, intensive preprocessing demands, limited adaptability to evolving threats, temporal dependencies, and substantial resource requirements.

This thesis proposes an integrated approach for cyberthreat modeling, attack

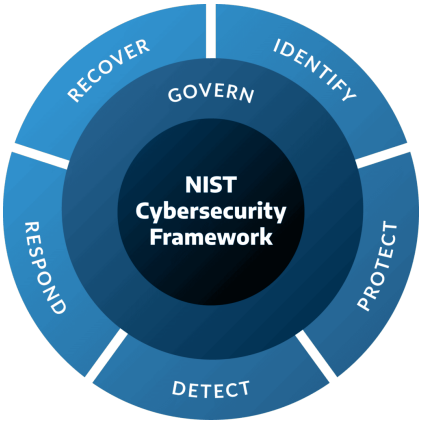


Figure 1.2: NIST V-2.0 CSF (Image Ref [13])

path analysis, defence, and protection mechanisms in IIoT systems, leveraging the NIST Cybersecurity Framework (CSF) [Fig. 1.2] as a foundational guide. The research focuses on developing a quantitative threat modeling framework that systematically identifies, assesses, and prioritises cyber threats in IIoT environments. It primarily addresses Identify, Detect, and Respond functions, while Protect, Recover, and Govern are outside the main scope and are identified as future extensions. This research also introduces an attack path analysis methodology to model potential attack vectors and their impact on IIoT systems. By leveraging the MITRE ATT&CK framework, the proposed approach maps adversarial tactics and techniques, providing a comprehensive understanding of how attackers navigate through IIoT networks. This enables the development of targeted defence mechanisms that mitigate vulnerabilities at critical points in the system. Through case studies in critical sectors such as manufacturing and power, this thesis demonstrates the practical applicability of the proposed framework.

Furthermore, by incorporating ML & DL algorithms, the proposed system enhances threat detection & protection capabilities, particularly for “Zero-Day Attacks (ZDAs)” and “Advanced Persistent Threats (APTs)”, where traditional methods often fail to address. While this research primarily focuses on threat modelling, attack path analysis, threat detection and response to various cyberattacks

like known attacks, zero-day attacks and advanced persistent threats contributing to IIoT cybersecurity, it also explores the unique challenges faced by resource-constrained IIoT systems to implement advanced cybersecurity measures and how those are particularly vulnerable to cyber threats.

1.1 Motivation

Industries form the backbone of the economy, yet they remain acutely vulnerable to cyber threats due to their broad attack surfaces and frequently constrained cybersecurity capabilities. As smart factories become more connected, they also become more vulnerable to cyberattacks. Traditional security systems can't keep up with new and unknown threats, which can disrupt operations or cause real-world damage. The increasing interdependence of ICPS has amplified cyber risks, where traditional signature-based IDS are ineffective against evolving threats such as zero-day attacks and APTs.

This thesis is motivated by the need for an integrated security solution that combines quantitative threat modeling, attack path analysis, and ML/DL-based detection & protection techniques to enable real-time, high-accuracy detection and mitigation of cyber threats in IIoT environments, particularly within resource-constrained systems. By using ML and proven cybersecurity threat modeling and attack path analysis frameworks for known, unknown, and advanced persistent cyber threats, this research aims to bridge the cybersecurity gap for IIoT systems by probing the NIST CSF and designing security solutions that empower them to defend against evolving cyber threats.

1.2 Overview of an Industrial IoT based System

A typical Manufacturing plant based on Industry 4.0 has been depicted in Fig.[1.3] For the sake of simplicity, the figure contains mainly two parts, i.e., the manufacturing plant and another administrative & control. The manufacturing plant consists of one or more assembly lines with their corresponding Human Machine Interface (HMI) operator, Engineering workstations, etc.

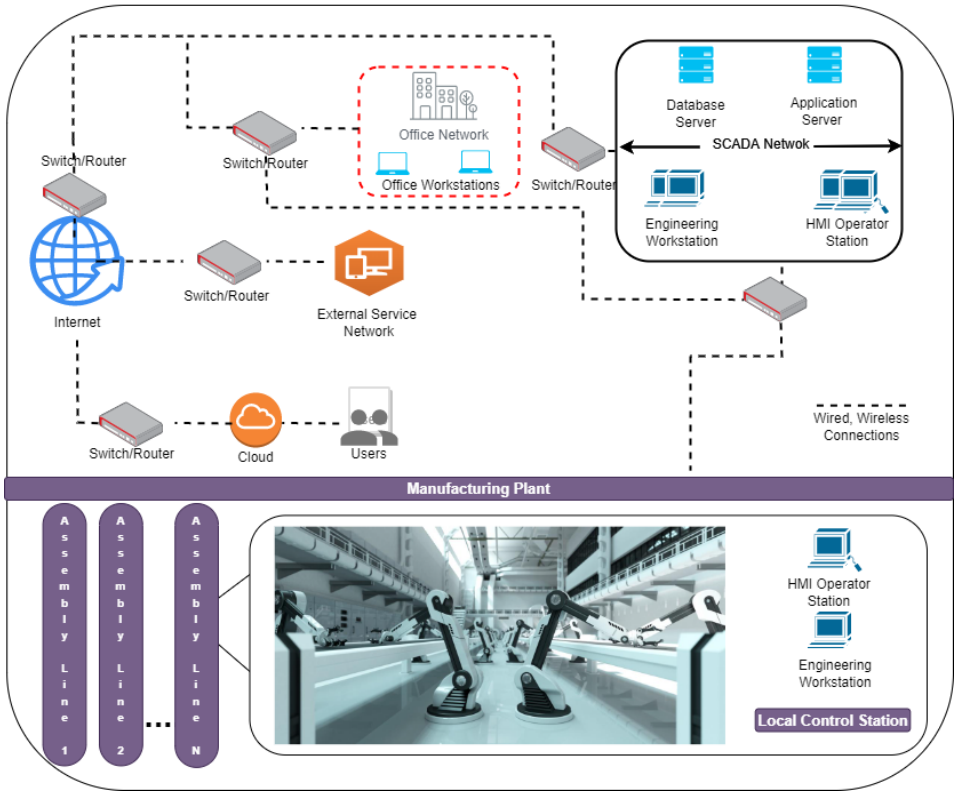


Figure 1.3: The Architecture of a Manufacturing Plant[63]

However, this convergence has also introduced significant cybersecurity challenges, as IIoT systems are increasingly targeted by sophisticated cyber threats such as Advanced Persistent Threats (APTs), zero-day exploits, DDoS and Ransomware. These threats exploit vulnerabilities in interconnected devices, commu-

nication protocols, and control systems, posing risks to critical infrastructure in manufacturing, energy, healthcare, etc.

1.3 Cyberattacks Scenarios in an IIoT System

To understand the behaviour and methodology of attackers is crucial for designing effective cybersecurity defences. The act of attackers [63] refers to the phases, techniques, and objectives pursued by malicious entities targeting IIoT infrastructures.

The following narrative presents a multi-phase attack scenario in an Industrial Internet of Things (IIoT) system, based on the conceptual diagram from Fig.1.4. It explores the tactics, progression, and implications of a coordinated cyber-attack from an attacker-centric perspective.

1.3.1 Phase 1: Compromise Within the Office Network

Attackers frequently gain their initial access by compromising a vulnerable node in the office network. This can occur through phishing emails, malicious attachments, or insider threats. Once inside, the attacker may:

- Move laterally within the office network, compromising other devices and gathering credentials.
- Target the production network to disrupt operations or extract sensitive data.
- Attempt to breach the central SCADA platform, including application servers and operator stations.

- Use compromised office nodes to communicate with Internet-based command-and-control (C2) infrastructure.

1.3.2 Phase 2: Breach Into the Operational Technology Environment

Upon extending control into the OT space, attackers begin manipulating the industrial environment directly:

- Compromised production machines attack peer devices, altering processes or spreading malware.
- They may use these machines to establish outbound communication with Internet nodes, potentially for data exfiltration or botnet activity.

1.3.3 Phase 3: Exploiting External and Cloud Services

With industrial systems relying on external service networks and cloud platforms, attackers exploit these trust zones:

- Attackers take control of a node in the external service network and begin internal disruption within that segment.
- They then target the user network, focusing on user devices with elevated privileges or administrative access.
- From this external position, attackers can re-enter the office network, bypassing traditional defenses.

1.3.4 Phase 4: Operational Disruption Through Targeted Scenarios

The attack culminates in specific, high-impact scenarios:

- **Scenario 1:** A new outsourced machine is compromised before deployment, enabling attackers to control manufacturing processes remotely.
- **Scenario 2:** A device introduced from within the organization is already compromised, giving attackers insider access to disrupt workflows.
- **Scenario 3:** Attackers spoof the status of existing devices, making machinery appear offline while it continues operating dangerously, disabling safety protocols such as emergency doors.

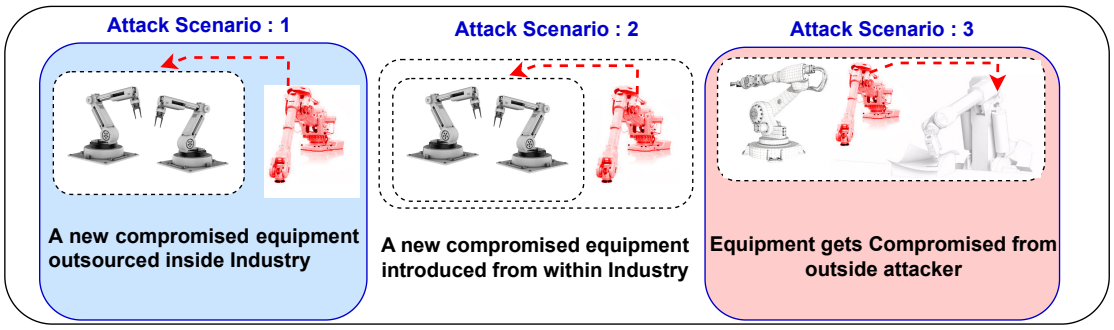


Figure 1.4: Operational Disruption Through Targeted Scenarios[63]

This narrative illustrates the sophisticated, multi-vector nature of cyber-attacks targeting industrial IoT environments. By leveraging both internal vulnerabilities and external trust channels, attackers can navigate across IT and OT layers, ultimately achieving both digital and physical disruption. Defending against such threats requires an integrated, holistic security approach encompassing network segmentation, behavioral monitoring, and real-time threat detection.

1.4 Research Objectives

The objective of this research is :

1. To know, How can we formalize and integrate cyber-threat modeling, critical-asset prioritization, and attack path analysis mechanism into a single, extensible sector agnostic architecture for ICPS?
2. To understand, Which data-driven ML and DL or novel hybrids models can learn evolving adversary behaviors across distributed IIoT topologies, while maintaining robustness against adversarial evasion and near to real time attack detection?
3. To check, How can the system balance real-time decision making at the edge with periodic, centralized model retraining to optimize detection accuracy, resource utilization, and resilience against zero-day exploits?
4. To enhance the effectiveness of intrusion response systems in detecting and mitigating cyberattacks.

Addressing these objectives will bridge the gap between siloed security tools and the dynamic defense needs of modern IIoT infrastructures, ultimately delivering a reliable, scalable, and intelligence-driven cybersecurity solution that safeguards critical industrial processes from advanced persistent threats.

1.5 Research Contributions

This research builds upon and extends existing works on machine learning-based intrusion detection & response, attack path analysis, and threat modeling in IIoT

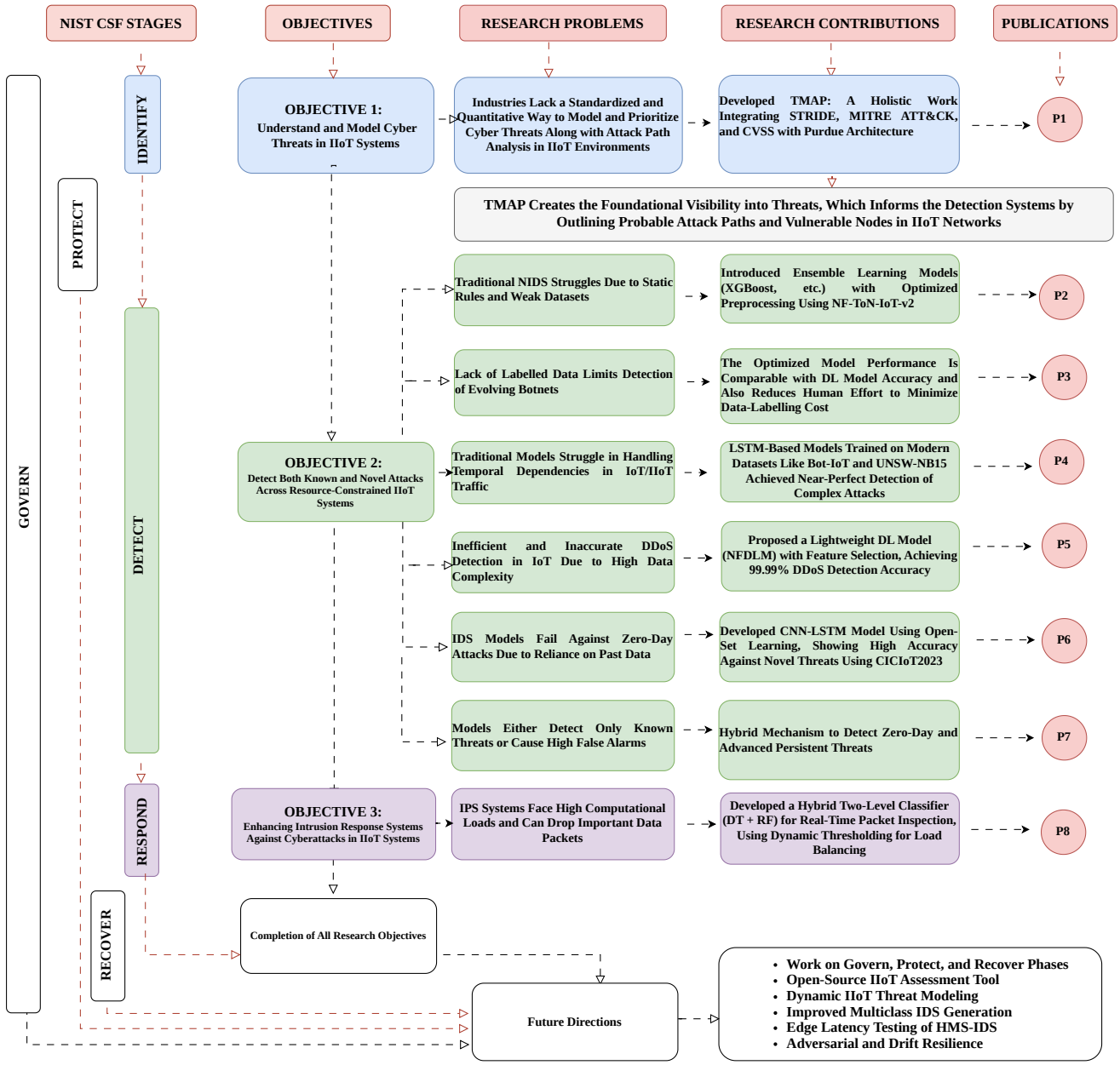


Figure 1.5: Thesis-Workflow

cybersecurity by addressing key limitations in the previous studies. This research fills critical gaps in existing literature and offers a cost-effective, real-world security solution for IIoT ecosystems, by enhancing zero-day attack detection, reducing false positives, and improving practical usability.

Research Contribution 1: TMAP: TMAP is a Standardised and Versatile Threat Modeling Approach for IIoT. Threat Modeling & Attack Path Analysis develops a structured approach for identifying vulnerabilities in IIoT systems. It integrates STRIDE, MITRE ATT&CK, and CVSS for threat prioritisation and provides industry-specific case studies in the manufacturing and power sectors.

Research Contribution 2: ELA-NIDS: Ensemble Learning for IIoT Security addresses limitations of traditional NIDS by leveraging tree-based ensemble models. It utilises the NF-Ton-Iot-v2 dataset to enhance industrial threat detection.

Research Contribution 3: Open Set Learning based approach for Zero-Day Attack Detection: Tackles zero-day detection challenges using hybrid CNN-LSTM models. A novel zero-shot learning framework is implemented to classify unknown attacks effectively.

Research Contribution 4: HMS-IDS: Hybrid Multi-Stage IDS for IIoT: combines supervised and unsupervised learning techniques to detect both known and unknown threats. It achieves high detection accuracy, reaching 99.49% for known attacks and 98.93% for zero-day threats.

Research Contribution 5: GANIBOT: Generative Adversarial Networks for Botnet Detection implements semi-supervised learning to identify stealthy botnet attacks. GANs are used to generate synthetic data, improving classification performance.

Research Contribution 6: LBDMIDS: LSTM-Based Deep Learning for IoT

Intrusion Detection leverages time-series analysis for real-time anomaly detection. It demonstrates high accuracy (99.99%) on modern IoT datasets such as Bot-IoT and UNSW-NB15.

Research Contribution 7: NFDLM: A computationally efficient DDoS Detection for IIoT proposes a feature selection mechanism and computationally efficient ANN/LSTM models. This approach effectively balances computational efficiency with high detection accuracy (99.99%).

Research Contribution 8: MLAPS: Machine Learning-Based Intrusion Prevention System introduces a two-stage classifier combining Decision Tree and Random Forest for intrusion prevention. This system reduces computational costs while maintaining a high detection accuracy of 98.4%.

1.6 Thesis Outline

This thesis is outlined in eight chapters, each addressing different aspects of the research work on IIoT systems cybersecurity. The thesis is outlined as follows.

Chapter 1 introduces the growing cybersecurity challenges in Industrial IoT (IIoT) systems due to expanded attack surfaces due to the growing integration and mutual interdependence of industrial controls and production systems(ICPS). It outlines real-world attack scenarios, defines the research problem, and sets specific objectives aligned with the NIST Cybersecurity Framework. The motivation and research objectives of the study have been presented, along with an overview of the thesis structure and its core contributions.

Chapter 2 contains a comprehensive review of existing IIoT cybersecurity frameworks, standards (NIST, NIS2, KRITIS), and the cybersecurity lifecycle is provided. This chapter examines current threat modeling methods, intrusion detec-

tion techniques, and ML/DL algorithms used in IIoT environments. It identifies key research gaps, particularly in detecting zero-day attacks, applying ML/DL/Hybrid models, and integrating standardised frameworks into practical systems.

Chapter 3 proposes a threat modeling methodology specific to IIoT, using data flow diagrams, asset identification, and CVSS-based scoring. It integrates MITRE ATT&CK for attack path mapping and prioritisation. Two case studies (in the power and manufacturing sectors) validate the model by highlighting critical assets and potential vulnerabilities.

Chapter 4 Focuses on traditional and known attacks. This chapter introduces ensemble ML-based intrusion detection systems, including SGANs for DDoS, ANN and LSTM like models. Each model is tested using real-world datasets, and results are analysed for detection accuracy, false positives, and computational efficiency.

Chapter 5 tackles the challenge of unknown threats using a CNN-LSTM hybrid model enhanced with zero-shot learning. The method is evaluated on preprocessed datasets, showing improved generalisation and robustness against zero-day attacks.

Chapter 6 Extending previous work, this chapter develops a hybrid IDS that combines ML, clustering, and signature-based methods to detect both zero-day and advanced persistent threats. The system is tested on the CIC-ToN-IoT dataset, and performance metrics validate its efficiency, speed, and adaptability.

Chapter 7 explores how to automate and optimise threat response using ML. It includes model selection, feature reduction, and the design of an Intrusion Prevention System (IPS) that balances speed and accuracy for real-time use in IIoT environments.

Chapter 8 summarises the research contributions, discusses the strengths and limitations of the proposed framework, and outlines future research directions. It suggests extending the framework to include proactive protection and recovery mechanisms, enhancing resilience for broader IIoT applications.

Chapter 2

Literature Review

While navigating the multifaceted terrain of cybersecurity within the dynamic ecosystem of the Industrial Internet of Things (IIoT), a broad spectrum of academic and industry literatures have been reviewed. Several recurring themes emerge, particularly the roles of Cyber-Physical Systems (CPS), Industrial Control Systems (ICS), SCADA, and the broader concept of the Industrial Internet. These concepts appear frequently across academic research papers, white papers, and corporate publications, reflecting their foundational importance in the IIoT discourse.

To provide a comprehensive understanding of how IIoT was evolved, and why it is critical in today's industrial landscape, this review begins with a conceptual foundation rooted in Industry 4.0. By unpacking key concepts, the emergence and development of IIoT within this transformative industrial paradigm can be better contextualised.

2.1 Adoption of Industrial Internet of Things: Overview

The concept of connecting industrial devices via the internet dates back to the early 1990s. Initially, the focus was on integrating sensors and actuators with basic networking capabilities. Over time, advancements in wireless communication, cloud computing, and data analytics transformed IIoT into a strategic pillar of modern

industry. Today, IIoT encompasses a vast ecosystem of connected devices that enable real-time monitoring, predictive maintenance, and autonomous operations across various sectors.

2.1.1 Infusion of IT & OT Convergence

Set within the broader framework of Industry 4.0, the integration of CPS, IACS, and the Industrial Internet signals a significant shift in industrial operations. These interconnected systems form the digital backbone of modern manufacturing, where physical processes are seamlessly interwoven with intelligent computing and networked communication. This technological convergence (Fig. 2.1) is what gives rise to IIoT, positioned at the core of current industrial transformation. Through the interconnection of smart devices, sensors, and control infrastructures, IIoT facilitates real-time insight, system automation, and process optimisation in complex industrial settings.

Although terms such as Industrial Internet, Industry 4.0, CPS, IIoT, IACS, SCADA, and Information and Communication Technology (ICT) are often used interchangeably, their definitions and applications are not always consistent in existing literature [16]. In the remainder of this section, the aim is to synthesise these concepts and highlight their progressive interrelationships. This synthesis reveals a conceptual continuum that underpins the development of IIoT and its integration within industrial ecosystems.

1. **ICT (Information and Communication Technology):** The foundational layer comprises computers, software, networking, and telecommunications used to handle information.
2. **SCADA (Supervisory Control and Data Acquisition)**: A form of ICT specialised for industrial environments to control and monitor systems in

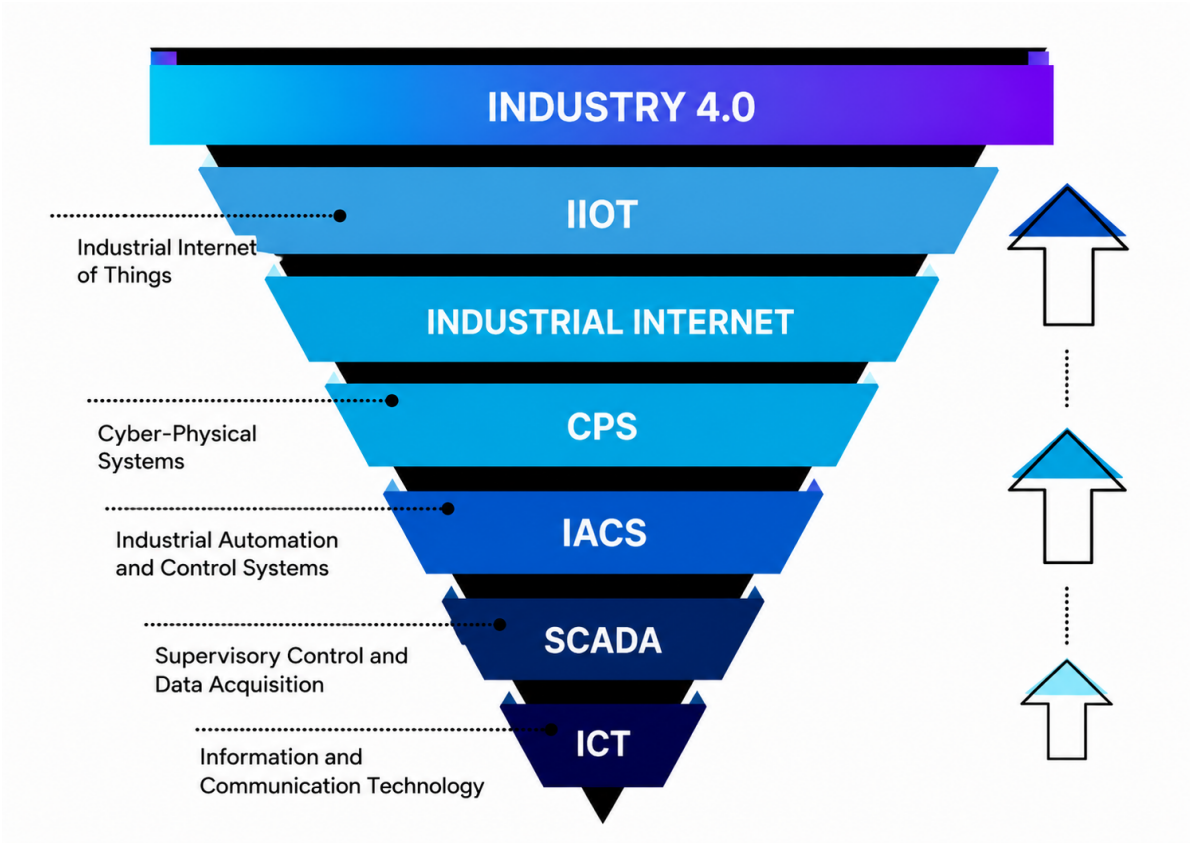


Figure 2.1: "From ICT to Industry 4.0: A Conceptual Continuum"

real-time.

3. **IACS (Industrial Automation and Control Systems):** Broad systems that include SCADA, PLCs (Programmable Logic Controllers), DCS (Distributed Control Systems), and other automation technologies for managing industrial processes.

4. **CPS (Cyber-Physical Systems):** Systems that integrate computing (cyber) with physical processes. IACS evolve into CPS when digital control is tightly integrated with physical operations.

5. **IIoT (Industrial Internet of Things)**: An extension of CPS where interconnected sensors, instruments, and other devices network together for data collection, exchange, and analysis in industrial contexts.

6. **Industrial Internet**: A broader term often synonymous with IIoT, emphasising the integration of big data analytics, machine learning, and cloud computing with industrial equipment.

7. **Industry 4.0**: A strategic vision encompassing all the above ICT, SCADA, IACS, CPS, IIoT, and Industrial Internet, aiming at smart factories and autonomous production through digital transformation.

2.1.2 The Hyperindustrialisation

The Industrial Internet of Things (IIoT) developed from the broader concept of the Internet of Things (IoT), which gained popularity in the early 2000s. Initially, IoT was geared towards consumer applications such as smart homes. However, the industrial sector recognised the potential for operational efficiency and began integrating IoT technologies into industrial processes, resulting in the birth of IIoT in 2010–2012. The "Hyperindustrialisation" refers to the growth or development phases in IIoT, which is detailed below[15]:

Early Stages (2010-2015): Large corporations such as General Electric drove initial adoption, coining the term "Industrial Internet" to describe the integration of smart machines, advanced analytics, and people at work.
Growth Phase (2015–2020): IIoT experienced rapid growth, thanks to advances in sensor technology, cloud computing, and data analytics. Standards such as OPC UA (Unified Architecture) were created to ensure interoperability.
Current State (2020–Present): IIoT has become an essential component in industries such as manufacturing, energy, transportation, and healthcare, with a focus on real-time data analytics, predictive maintenance, and automation.

A clear phase-wise mapping (Ref: Table 2.1) of the evolution of industries (Industry 1.0 to Industry X.0) aligned with the evolution of the Industrial Internet of Things (IIoT) is given below:

Table 2.1: Industry Evolution vs. IIoT Development

Phase	**Era**	**Technology**	**IIoT Evolution**	**Cybersecurity Concerns**
Industry 1.0	1780s	Steam and water-powered mechanization	Pre-IIoT; no digital connectivity	Physical sabotage and equipment damage only.
Industry 2.0	1870–1914	Electricity, assembly lines, mass production	Early industrial control concepts emerge	Insider threats and operational failures dominate.
Industry 3.0	1970s	Computers, PLCs, electronics, automation	Foundation of IIoT; machine-generated data appears	Software flaws, logic manipulation, and system tampering.
Industry 4.0	2010-Present	CPS, IoT, AI, cloud, big data	Industrial Internet (2010-15): sensor/cloud adoption (2015-20): edge AI, digital twins, and predictive maintenance (2020-Present)	IoT vulnerabilities, insecure APIs, ransomware, supply-chain attacks, and ICS-targeted malware.
Industry 5.0	2023+	Human-centric, sustainable, personalized systems	Autonomous and ethical IIoT integration	Privacy risks, biometric theft, AI poisoning, and digital twin attacks.
Industry 6.0	2030+	Quantum AI, swarm robotics, bio-digital convergence	Bio-IoT and quantum-enabled IIoT ecosystems	Quantum attacks, swarm hijacking, synthetic biology misuse.
Industry X.0	Future	XR, AI, blockchain, quantum computing, digital twins	Distributed cognition and autonomous value chains	Hyperconnected attack surfaces, AI compromise, and quantum-enabled breaches.

2.2 CyberSecurity Lifecycle: Before & Beyond IEC 61508

The Cybersecurity Lifecycle, especially in industrial and safety-critical systems, has evolved significantly, particularly with the influence of IEC 61508 [52], a foun-

dational functional safety standard. Here's (Ref: Table 2.3) a structured breakdown of how the cybersecurity lifecycle looks before and after IEC 61508 in terms of principles, integration, and process rigour. Post IEC 61508, the provisions for threat modelling and attack detection to include in the cybersecurity lifecycle were strongly focused.

Table 2.3: Cybersecurity Lifecycle: Pre vs. Post IEC 61508 Era

Key Phases	**Pre IEC 61508**	**Post IEC 61508**
Lifecycle Integration	Ad hoc, often reactive. Cybersecurity was considered after design.	Integrated into the system safety lifecycle from concept to decommissioning.
Design Philosophy	Focused mostly on availability or basic IT security.	Emphasises risk-based design tied to Safety Integrity Levels (SIL).
Threat Modeling	Rare or informal.	Formal threat/risk assessments (e.g., HAZOP, FMEA, or ISO/IEC 62443 based).
Security Objectives	Often generic (e.g., "protect the network").	Tied to functional safety objectives and asset criticality.
Standards/Best Practices	Fragmented; borrowed from general IT standards.	Guided by IEC 61508, IEC 62443, and harmonised standards.
Development Process	Waterfall or non-standardized.	Requires V-model or lifecycle models with verification/validation gates.

Validation & Testing	Mostly functional testing.	Includes failure analysis, robustness testing, and security validation.
Change Management	Ad hoc patches or updates.	Requires structured change control, including risk reassessment.
Documentation	Often insufficient.	Mandatory, detailed documentation throughout the lifecycle.
Responsibility & Governance	Security often isolated from safety teams.	Shared responsibility between safety, security, and engineering teams.
Incident Response Preparedness	Rarely formalized.	Integrated into lifecycle: includes detection, response, and recovery plans.

2.2.1 ISA/IEC 62443

ISA/IEC 62443 is a globally recognised series of standards [53] developed by the International Electrotechnical Commission (IEC) to address cybersecurity within IACS. Its core objective is to protect industrial environments from the growing landscape of cyber threats. The standard establishes a structured and systematic framework for implementing cybersecurity practices specifically to IACS. By providing detailed guidance on identifying and mitigating system vulnerabilities, ISA/IEC 62443 ensures the preservation of critical system integrity, availability, and confidentiality.

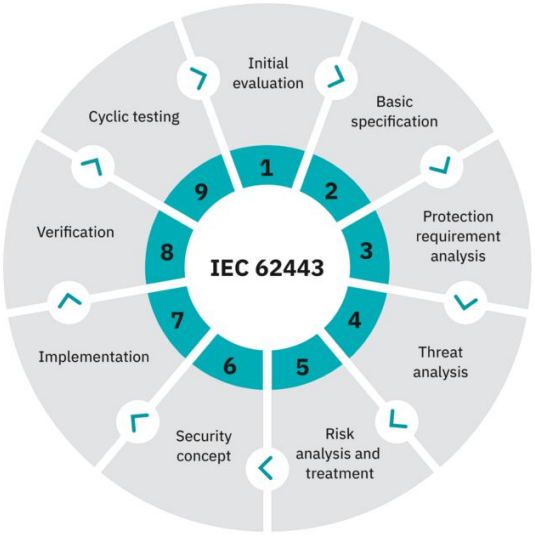


Figure 2.2: IEC 62443 Cybersecurity Lifecycle [109]

2.3 Existing Cybersecurity Frameworks for IIoT Systems

IIoT-adapted businesses are increasingly becoming targets for cyberattacks due to their limited resources, lack of expertise, and often inadequate cybersecurity measures. To address these challenges, IIoT-adapted businesses can adopt cybersecurity frameworks that provide structured guidance for managing risks and improving resilience. This research provides a detailed descriptive analysis of four prominent cybersecurity frameworks—NIS, NIS2, KRITIS, and NIST Cybersecurity Framework (CSF)—focusing on their applicability to IIoT systems.

2.3.1 NIS & NIS2 Directive Characteristics

The "NIS Directive" is a European Union (EU) legislation enacted in 2016 to improve cybersecurity across critical infrastructure sectors [26]. It applies to "Operators of Essential Services (OES)" and "Digital Service Providers (DSPs)". The directive aims to ensure a high level of cybersecurity across the EU by requiring

organizations to implement appropriate security measures and report significant incidents.

Key Features

- "Risk Management": Organizations must identify and manage cybersecurity risks by implementing appropriate technical and organizational measures.
- "Incident Reporting": Significant cybersecurity incidents must be reported to national authorities, ensuring timely response and mitigation.
- "Cross-Border Cooperation": The directive encourages collaboration between EU member states to address cybersecurity threats and share best practices.

Applicability to IIoT systems

While the NIS Directive [26] is primarily designed for large organizations and critical infrastructure, IIoT systems operating in sectors like energy, transport, and healthcare may also fall under its scope. For these IIoT systems, the directive provides a structured approach to cybersecurity, encouraging the adoption of best practices and fostering collaboration with other organizations and authorities. However, compliance can be resource-intensive, requiring significant investment in cybersecurity measures and expertise. IIoT systems outside the scope of critical infrastructure may find the directive less relevant and overly complex for their needs.

NIS2 Overview

NIS2, or the Network and Information Security Directive 2 [26], is the EU's updated cybersecurity framework, replacing the original NIS Directive, introduced in 2020 to address the evolving cybersecurity landscape. It expands the scope of the

original directive and introduces stricter requirements, aiming to ensure a higher level of resilience across more sectors.

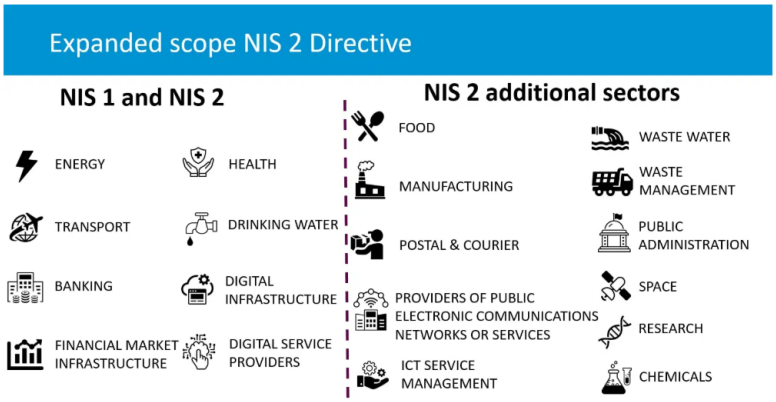


Figure 2.3: NIS2 Directive, an extension of the former NIS Directive that aims to establish a common level of cybersecurity across the European Union (Image Source [154])

Key Features

- "Expanded Scope": NIS2 includes additional sectors such as public administration, space, and postal services, broadening the range of organizations subject to its requirements.
- "Stricter Requirements": The directive introduces more stringent incident reporting and supervisory measures, ensuring that organizations take cybersecurity more seriously.
- "Supply Chain Security": NIS2 places a strong emphasis on securing the supply chain, requiring organizations to assess and mitigate risks associated with third-party vendors and service providers.



Figure 2.4: NIS2 Pillars (Image Ref [51])

Applicability to IIoT systems

For IIoT systems operating in sectors covered by NIS2, the directive provides a comprehensive framework for improving cybersecurity resilience. The focus on supply chain security is particularly relevant for IIoT systems, as they often rely on third-party vendors and may be vulnerable to supply chain attacks. However, the stricter requirements and expanded scope may pose challenges for IIoT systems with limited resources and expertise. Compliance with NIS2 may require significant investment in cybersecurity measures, making it less accessible for smaller organizations.

2.3.2 KRITIS Coverage

"KRITIS" is a German framework focused on protecting critical infrastructure sectors [29]. It mandates that operators of critical infrastructure implement spe-

cific security measures and report incidents to the "Federal Office for Information Security (BSI)". The framework aims to ensure the resilience of critical infrastructure against cyber threats.



Figure 2.5: Critical infrastructures (KRITIS) by BSI (Image Ref [25])

Key Features

- "State-of-the-Art Security Measures": Organizations are required to implement advanced security measures to protect critical infrastructure.
- "Regular Risk Assessments": Regular risk assessments and audits are mandated to identify and mitigate vulnerabilities.
- "Incident Reporting": Cybersecurity incidents must be reported to the BSI, ensuring timely response and mitigation.

Applicability to IIoT systems

KRITIS is primarily designed for critical infrastructure sectors, such as energy, water, and healthcare. IIoT systems operating in these sectors may benefit from

the framework's emphasis on advanced security measures and proactive risk management. However, for IIoT systems outside the scope of critical infrastructure, KRITIS may be less relevant and overly complex. Compliance with KRITIS can be resource-intensive, requiring significant investment in cybersecurity measures and expertise, which may be challenging for smaller organizations.

2.3.3 NIST Cybersecurity Framework Overview

The "NIST Cybersecurity Framework (CSF)" is a voluntary framework developed by the "National Institute of Standards and Technology (NIST)" in the United States [101]. It provides a flexible approach to managing cybersecurity risks and is widely adopted across various industries. The framework is designed to help organizations of all sizes improve their cybersecurity posture.

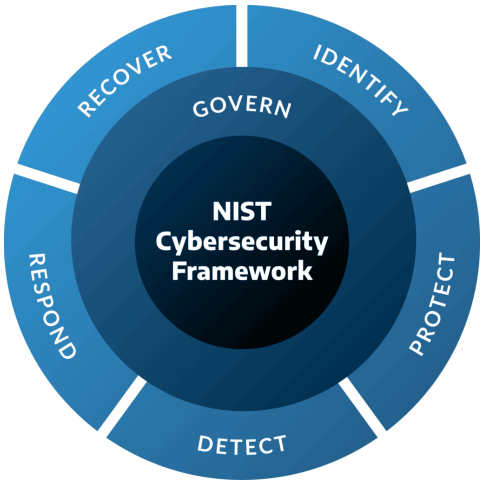


Figure 2.6: NIST V-2.0 CSF (Image Ref [13])

Key Features

- "Core Functions": The framework is organized into six core functions—Govern, Identify, Protect, Detect, Respond, and Recover—providing a comprehensive

approach to managing cybersecurity risks.

- "Flexible and Scalable": The NIST CSF can be to organizations of all sizes and sectors, making it particularly suitable for IIoT systems.
- "Risk-Based Approach": The framework emphasizes a risk-based approach, allowing organizations to prioritize their cybersecurity efforts based on their specific needs and resources.

Applicability to IIoT Systems

The NIST CSF is highly suitable for IIoT systems due to its flexibility, scalability, and risk-based approach. IIoT systems can adopt the framework at their own pace, focusing on the most critical areas of their cybersecurity posture. The framework provides a clear roadmap for improving cybersecurity, making it accessible even for organizations with limited resources and expertise. Additionally, the NIST CSF is widely recognized and supported by industry best practices, ensuring that IIoT systems adopting the framework are aligned with global standards.

2.3.4 Comparative Study of Cybersecurity Frameworks

Table 4.2 provides a comparative study of the four frameworks in terms of their suitability for IIoT environments. The study reveals that the NIST CSF is the most suitable framework for IIoT systems due to its flexibility, scalability, and risk-based approach. In contrast, NIS, NIS2, and KRITIS are primarily designed for critical infrastructure sectors and may be too complex and resource-intensive for IIoT systems.

NIST CSF allows IIoT-systems to implement cybersecurity measures to their specific needs and resources, making them more accessible than the more rigid

Table 2.5: Comparison of Cybersecurity Frameworks for IIoT systems

Framework	Scope	Key Features	Strengths	Challenges
NIS	EU legislation for critical infrastructure	Risk management, incident reporting	Structured approach for critical sectors	Complex, resource-intensive
NIS2	Expanded EU legislation	Stricter incident reporting, supply chain security	Higher resilience for critical sectors	More stringent than NIS
KRITIS	German framework for critical infrastructure	State-of-the-art security measures, risk assessments	High security standards for critical sectors	Resource-intensive
NIST CSF	Voluntary framework for all sectors	Flexible, risk-based approach	Flexible, scalable, widely recognized	Requires expertise

and resource-intensive frameworks like NIS, NIS2, and KRITIS.

2.4 Positioning Threat Modeling, Detection, & Response in the NIST Cybersecurity Lifecycle

The integration of Industrial Internet of Things (IIoT) in critical infrastructure introduces complex cybersecurity challenges. A comprehensive security strategy must include threat modeling, intrusion and attack detection mechanisms, and response mechanisms. These components should align with the NIST Cybersecurity Framework (CSF), which includes five core functions: Identify, Protect, Detect, Respond, and Recover.

2.4.1 Aligning Threat Modeling with NIST CSF

Although Threat Modeling fundamentally belongs to the Identify stage of NIST CSF, but it is not a one-time task; it supports every phase of the NIST framework by enabling proactive risk identification, informed defense planning, and continuous feedback.

Table 2.6: Mapping Threat Modeling to NIST Cybersecurity Framework (CSF 2.0) Stages

NIST CSF Function	**Role of Threat Modeling**	**Details**
Govern	Aligns threat modeling with policy and risk strategy	Ensures threat modeling practices are integrated into governance frameworks, aligning with organizational risk appetite, roles, responsibilities, and compliance requirements.
Identify	Foundation for modeling	Identifies assets, data flows, threat actors, and system boundaries. Supports frameworks like STRIDE and PASTA for structured analysis.
Protect	Design preventive controls	Guides security control implementation by showing what needs protection based on modeled threats. Informs decisions on encryption, access control, and segmentation.
Detect	Threat pattern recognition	Helps predict likely attack vectors and behaviors (e.g., through Attack Trees or Kill Chain), improving detection mechanisms.
Respond	Plan based on modeled scenarios	Enables creation of effective incident response plans based on likely threats and attack paths identified during modeling.
Recover	Feedback for improvement	Post-incident analysis updates threat models, contributing to organisational learning and resilience building.

2.4.2 Aligning Attack Detection & Response with NIST CSF

Similarly, attack detection fundamentally belongs to the "Detect" stage of NIST CSF, and attack response belongs to "Response" stage of NIST CSF 2.0 but it also supports every phase of the NIST framework indirectly. Below Table (2.7) reflects NIST CSF 2.0 phases.

Table 2.7: Intrusion Detection Across NIST CSF for IIoT

NIST CSF Phases	**Role of Detection Mechanisms in IIoT**
Govern	Establishes policies, responsibilities, detection criteria
Identify	Identify detection requirements and potential blind spots across the IIoT ecosystem.
Protect	Deploy proactive detection tools like honeypots and harden detection systems.
Detect	Use anomaly-based and signature-based IDS/IPS systems tailored for IIoT traffic and protocols.
Respond	trigger containment actions or alerts. Interface with SIEM and automate response workflows.
Recover	analyse historical logs post-incident. Feed insights back into detection tuning.

Table 2.8: Response Mechanisms Across NIST CSF for IIoT

NIST CSF Function	**Response Mechanism Role in IIoT**
Govern	Defines response plans, escalation, and safe containment
Identify	Define incident thresholds and classify IIoT-specific incidents.
Protect	Configure fallback operations and isolation procedures in advance.
Detect	Initiate response based on detections and anomaly signals.
Respond	Activate playbooks, isolate compromised zones, coordinate with OT teams.
Recover	Restore services safely, analyze response effectiveness, and update protocols.

2.5 Prevalent Cybersecurity Threats in Industrial IoT

Prevalent Cybersecurity Threats Facing Indian & IIoT systems are particularly exposed to various cybersecurity risks. The following are some of the most frequently encountered threats:

2.5.1 Supply chain attacks

Supply chain attacks [90]are increasingly becoming a significant threat aimed at smart supply chains, manufacturers, and software developers. These attacks take advantage of weaknesses in third-party components, insecure communication protocols, and poor coding practices to compromise reliable software or hardware and take control of entire distribution networks. Given the involvement of numerous vendors and stakeholders, supply chains are particularly susceptible to data breaches. In industrial environments, malicious code can be concealed within hardware or delivered through third-party updates and connected devices, often going undetected for extended periods, which makes these attacks particularly serious and challenging to trace.

2.5.2 Data Breaches

A data breach occurs when unauthorized individuals gain access to confidential company data, including customer information, financial records, or trade secrets. Such breaches can lead to significant financial losses and reputational damage. Implementing robust access controls, encryption protocols, and firewalls can help safeguard sensitive information [86].

2.5.3 Cyber Espionage

Stealing confidential information, such intellectual property, for financial, political, or competitive benefit is known as cyber espionage [90], or cyber spying. Attackers use trojans, spear phishing, and malware to breach networks and target supply chains in industrial settings by taking advantage of flaws in manufacturing systems. Businesses that depend heavily on data, like mining, are particularly at risk. These kinds of attacks frequently fall under the category of Advanced Persistent Threats (APTs), which aim to compromise vital data and interfere with operations.

2.5.4 Malware Infections

Malware, or malicious software, is designed to infiltrate, disrupt, or gain unauthorized access to computer systems and networks. Common forms include viruses, ransomware, worms, and spyware. Attackers leverage malware to steal data, damage infrastructure, or exploit system vulnerabilities. Deploying up-to-date antivirus software, conducting system updates, and training employees on cybersecurity hygiene can help mitigate these risks [86].

2.5.5 Third-Party Vulnerabilities

Collaborating with external vendors, suppliers, or service providers can expose to cybersecurity risks, especially if third parties lack adequate security measures. Cybercriminals exploit these weak links to gain unauthorized access to critical data. Establishing stringent vendor security policies, conducting routine audits, and ensuring compliance with security standards can minimize such risks [90].

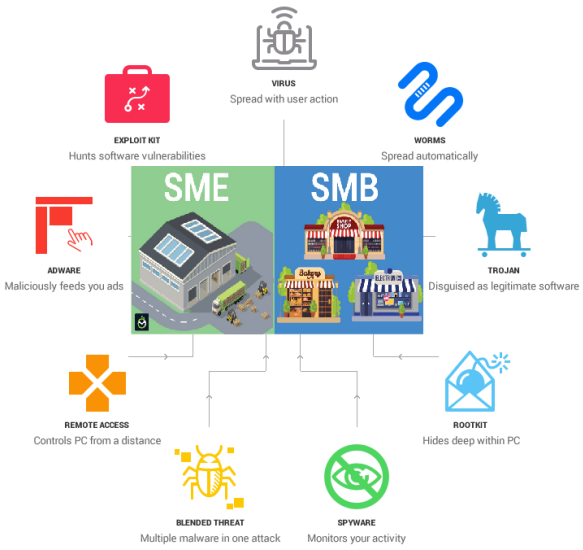


Figure 2.7: Malware Attacks on IIoT systems

2.5.6 Phishing Scams

Phishing [90]is a deceptive technique used by attackers to impersonate legitimate entities and trick employees into disclosing sensitive credentials, such as passwords or payment information. Cybercriminals often use fraudulent emails, messages, or websites to manipulate unsuspecting victims. Businesses can mitigate phishing risks by conducting employee awareness training, implementing multi-factor authentication, and deploying advanced email security solutions.

2.5.7 Identity Fraud

Identity fraud involves attackers gaining access to personally identifiable information (PII), including tax IDs, bank account details, and social security numbers, to engage in fraudulent activities. should implement stringent authentication measures, monitor financial transactions regularly, and use secure payment gateways to prevent unauthorized access to sensitive data [41].

2.5.8 Cryptojacking

Cryptojacking is an emerging cyber threat in which attackers secretly use an organization's computing resources to mine cryptocurrency without authorization. This malware-based attack consumes system resources, leading to reduced performance and increased operational costs. The malware is often introduced through malicious email attachments, compromised websites, or software vulnerabilities. Businesses can mitigate cryptojacking risks by implementing endpoint security solutions, blocking unauthorised scripts, and continuously monitoring system performance [8].

2.5.9 Active or Passive Eavesdropping

In this, an attacker might install spyware and carry out exploits by gaining access to the wired or wireless network across Maximum Transmission Units and sub-Maximum Transmission Units or Remote Telecontrol Units [17].

2.5.10 Man in the Middle (MITM)

It is a kind of attack [90], where the perpetrator observes and intercepts the network traffic before inputting altered data and transmitting it to the target. If the attack succeeds, he seizes control of the current session and maintains the connection from a fictitious IP to remain undetected.

2.5.11 BotNet Attack

IoT devices have limited computational power, storage capacity, and communication abilities because of their resource constraints. Statista reports that there are currently 21 billion IoT devices in operation, with projections suggesting that this number will grow to 50 billion by 2030[104]. With the rapid growth of IoT devices, there is also an increase in IoT-based Distributed Denial of Service (DDoS) attacks, particularly through the use of bots within the network. A botnet refers to a group of compromised internet-connected devices that are controlled by malicious actors. Fig.2.8 illustrates the typical scenario of a botnet attack. The attacker utilizes a command and control server to interact with the compromised IoT devices. Subsequently, these devices are employed as bots to overwhelm web servers, rendering the website inaccessible to legitimate users, which can potentially be mitigated by implementing a botnet attack detection model, as represented in the model layer of Fig. 2.8. A notable example of a botnet attack was the ”Mirai,” which disrupted a significant portion of the Internet in 2016. At the beginning of 2021, organizations recorded a 31% occurrence of botnet activity. By the end of 2021, this is expected to rise to 51%[36].

2.5.12 DoS and DDoS Attacks

Malicious actions designed to disrupt a service, referred to as Denial of Service (DoS), occur when a website becomes inaccessible to legitimate users [117, 68]. Distributed Denial of Service (DDoS) attacks are executed by a collection of compromised devices known as “Bots,” which are automated programs targeting a remote system (typically a server) [110]. A DoS attack transpires when a hacker interferes with the services of a networked host to prevent authorized users from reaching a server or network resource. When multiple sources attempt to com-

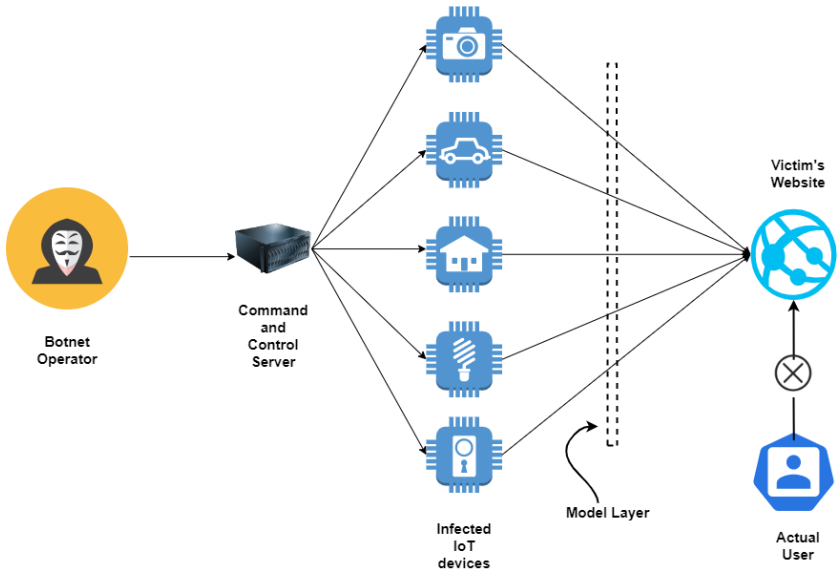


Figure 2.8: Botnet Attack on IoT devices

promise the targeted server, the attack is termed a ”Distributed Denial of Service” (DDoS) attack [36]. The intent behind these assaults is to create a disruption that could be potentially life-threatening. Significant contributors to attacks in IoT are DoS and DDoS. These attacks can either be volumetric or protocol-based [177]. Volumetric attacks impact a large number of individuals. These attacks generate substantial network traffic by either compelling the victim to execute the attacker's orders or causing the system to crash, thus making the service unavailable. Common examples of volumetric DDoS attacks include SYN flood, ICMP flood, and UDP flood attacks [42]. Protocol-based attacks utilize Transport Control Protocol (TCP) / Internet Protocol (IP) or User Datagram Protocol (UDP) / IP mechanisms to deplete the CPU and memory resources, causing the targeted system to be unable to respond to requests. Figure 1 illustrates the DDoS attack scenario.

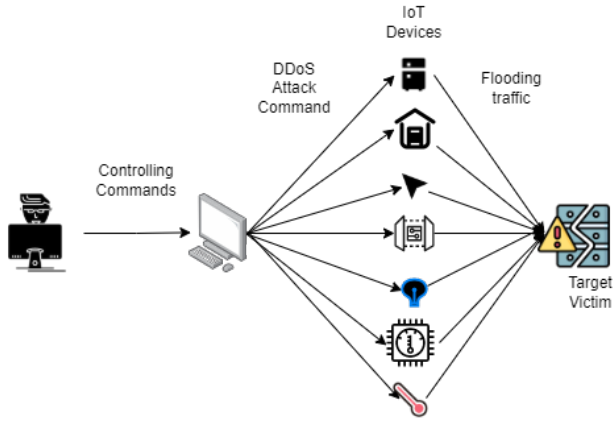


Figure 2.9: DDoS attack scenario diagram

2.5.13 Advanced Persistent Threats (APTs)

Advanced Persistent Threats (APTs) represent highly advanced strategies often used in cyber intrusions. The term "Advanced" refers to the use of complex attack techniques by the attackers. These methods include the use of publicly available exploits that specifically target well-known vulnerabilities, as well as the application of custom payloads and delivery methods, commonly referred to as zero-day exploits [141]. The latter are created to adapt and evolve based on the reactions of the targeted system. "Persistence" indicates the ongoing and uninterrupted connection between the attacker and the compromised systems until the attacker's goals are realized. In summary, APTs involve carefully planned covert attacks that utilize advanced exploitation techniques to focus on specific targets while maintaining long-term access. APT attacks consist of cybercriminals breaching a company's network and sustaining long-term, unauthorized access to extract confidential information. These covert attacks are generally employed for corporate espionage, data theft, or to disrupt business activities. To guard against APT attacks, organizations should implement ongoing network surveillance, carry out frequent security assessments, and deploy advanced threat detection solutions[80].

2.5.14 Zero Day Attacks

A zero-day attack refers to a cyberattack that exploits a software vulnerability that is unknown to both the software developers and the public. Since the flaw has not been identified, there is no available patch or fix, leaving the software exposed to the attack. These attacks pose significant risks as they can cause severe damage before the flaw is recognized and remedied. The term ”zero day” indicates that once a vulnerability is discovered, developers lack the time to create a remedy for it (”zero day”)[128][4].

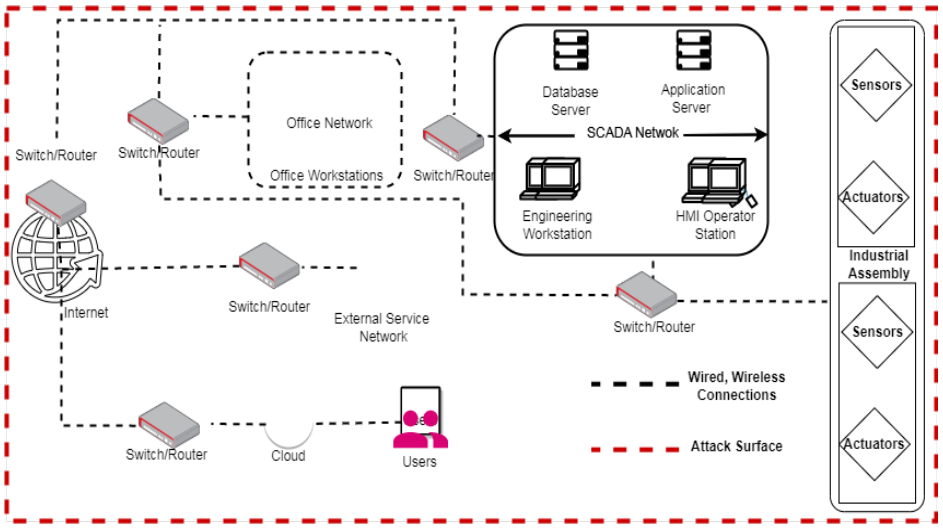


Figure 2.10: Attack Surface in an Industrial IoT

Intrusion detection systems (IDSs) have historically been used to identify zero-day attacks (ZDAs) and advanced persistent threats (APTs) [140]. The field of cybersecurity has experienced advancements through the incorporation of machine learning (ML) techniques, which allow for the extraction of insights from data via statistical algorithms. ML models are skilled at predicting, classifying, and forecasting occurrences and scenarios by recognizing complex data patterns that may escape the notice of experts in the field [59] and can provide self-healing in

response to cyber threats. As a result, many organizations across various sectors have started using ML to enhance their security measures. Traditional "Signature-based IDS" works by scanning for known attacks by recognizing specific Indicators of Compromise (IOCs). Nevertheless, these systems struggle with detecting zero-day attacks, which capitalize on undiscovered vulnerabilities prior to their resolution, creating a significant security threat [176]. In contrast, ML-based IDSs [132, 134, 130, 133] utilize behavioral patterns gleaned from incoming network packets to spot anomalies or malicious elements, extending beyond mere IOCs. These models have a higher capability for identifying zero-day attacks as they learn complex traffic patterns and are not exclusively dependent on established IOCs.

2.6 Datasets and Algorithms: A Technical Landscape

In this research work, majorly IIoT-specific datasets have been used along with various popular ML and DL models for known and unknown threat detection and protection systems, which are explained in the following subsections.

2.6.1 Overview of Experimental Corpora

In order to conduct a thorough and comparative assessment of intrusion detection methods within IIoT settings, this research utilizes a variety of well-recognized cybersecurity datasets. These datasets include standard benchmarks commonly utilized in the research community, such as KDD Cup 99, CIC-IDS 2017, and UNSW-NB15, in addition to more recent, IIoT-specific datasets like Bot-IoT, CIC-ToN-IoT, and NF-ToN-IoT-v2. Each dataset features a wide range of attack types, network traffic behaviors, and protocol attributes, providing a well-rounded plat-

form to evaluate the efficacy and applicability of the proposed detection models.

2.6.2 UNSW-NB15

Since 2015, this dataset has been widely utilized for assessing Network Intrusion Detection Systems (NIDS), developed by the Cyber Range Lab at UNSW-Canberra [127, 95, 94, 96, 97]. The dataset comprises 49 features (excluding class labels) and contains a total of 2,540,044 records divided among 4 CSV files. It includes 10 class labels, with 9 representing different varieties of Botnet attack vectors, and 1 representing a Normal class. The different types of attack traffic encompass analysis, backdoor, generic, reconnaissance, Denial of Service (DoS), shellcode, fuzzers, exploits and worms. In the study, 13 significant features were identified from the 49 available, which include the source IP address, duration, destination IP address, source port, destination port, source bytes, destination bytes, destination TTL, source TTL, destination load, source load, destination packets, and source packets.

2.6.3 Bot-Iot Dataset

Developed by the Cyber Range Lab at UNSW-Canberra in 2019, this dataset features network and botnet traffic integrated within a network setting. It comprises 47 attributes, including the class label, attack category, and subcategory [70, 73, 71, 72, 75, 76]. The dataset comprises 72 million records, representing both normal and attack traffic. A curated training and testing dataset was created that includes the 10 most important features, which have been utilized for training and validation purposes. The combined total of records in the training and testing dataset is 3.6 million. It encompasses various attack vectors including DDoS, DoS, OS and service scans, keylogging, and data exfiltration attacks. Based on

the protocols used, the categories of DDoS and DoS are further classified into different types.

2.6.4 CIC-ToN-IoT Dataset

The CIC-ToN-IoT dataset, introduced in [126], serves as a valuable resource for studying IoT network traffic. With the aid of the CICFlowMeter tool, researchers can extract features from pcap files within the ToN-IoT dataset, offering a rich dataset for analyzing and evaluating various methodologies IoT network security field. By leveraging the capabilities of the CICFlowMeter-v4 tool, researchers can extract 83 distinct attributes, providing a comprehensive and detailed characterization of the data. The dataset may be augmented[1] using data augmentation techniques [85] to broaden and diversify the training set, which will further improve the performance metrics. However, the training set of the dataset used in this manuscript is also good enough to train the proposed model, hence the idea of augmenting the training set is proposed as a future scope of this research work.

This comprehensive array of features enhances the ability to comprehend and analyze network traffic within IoT settings, facilitating the creation of robust security measures.

The research employed the CIC-ToN-IoT dataset, which contains a total of 5,351,760 data samples. Of these samples, 2,836,524 (53.00%) are identified as attacks, whereas 2,515,236 (47.00%) are labeled as benign instances (Fig. 2.11). This nearly even distribution between attack and non-attack data enables a thorough representation of both categories, supporting efficient analysis and modeling of their unique characteristics and patterns. It is essential to highlight that the

[1]Data augmentation is a method that uses pre-existing data to create modified copies of a dataset, thereby artificially expanding the training set.

Attack Distribution	
Attack type	Number of Samples
Xss	2149308
Password	340208
Injection	277696
Scanning	36205
Backdoor	27145
Ransomware	5098
MITM	517
DDoS	202
Dos	145

Table 2.9: Distribution of different types of attacks in the dataset.

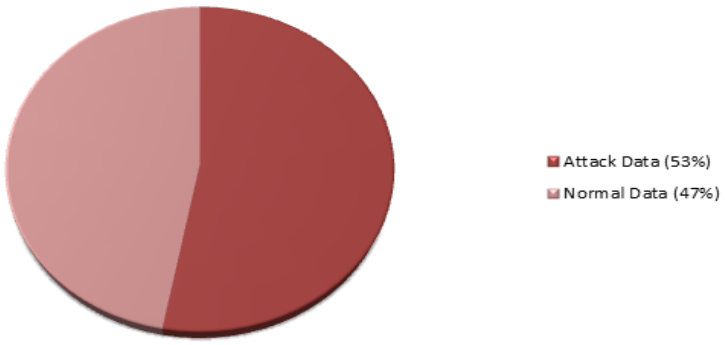


Figure 2.11: CIC-ToN-IoT-Dataset Distribution

dataset was specifically developed for this research study. Researchers looking to access and use this dataset can consult the provided source [126], which includes comprehensive details on how to obtain the dataset for further analysis and investigation.

2.6.5 KDD-99 cup Dataset

The KDD-99 cup dataset[156] is widely regarded as the primary dataset for training algorithms. It is a portion of the DARPA-98 dataset. The KDD-99 dataset is a dataset that contains several varieties of data. The dataset comprises 4.8 million occurrences. The attributes utilised in the dataset possess category and integer properties. The object possesses a total of forty-two characteristics. TheKDD-99 cup dataset include the following:

- This dataset consists of TCP dump data from simulated network traffic that was gathered in 1998 at Lincoln Labs. After seven weeks, a total of five million connection data points were collected and utilised as a training set. An additional two weeks of network traffic resulted in the creation of a test set including two million instances. KDD-99 is a refined version of this dataset.
- The KDD-99 dataset consists of five distinct kinds of patterns, namely Normal, DoS (Denial of Service), U2R (User to Root), R2L (Remote to Local), and Probe (Probing Attack). Each intrusion category is further categorised based on the specific method employed to carry out the attacks. The attacks are tabulated in Table [2.10]

2.6.6 NSL-KDD Dataset

In order to address the problems present in the KDD-99 cup dataset, researchers have introduced a new dataset called NSL-KDD. This dataset is composed of carefully chosen records from the original KDD dataset and is free from any of the aforementioned concerns. NSL-KDD is a project initiated by Tavallaee et al. [156] to address the issues of KDD-99 and overcome its limitations. Nevertheless, the authors acknowledge that the dataset

Table 2.10: Class Distribution in Training and Test Sets

Class	Training Set		Test Set	
	Count	Percentage	Count	Percentage
Normal	812,814	75.611%	60,593	19.481%
DoS	247,267	23.002%	229,853	73.901%
Probe	13,860	1.289%	4,166	1.339%
R2L	999	0.093%	16,189	5.205%
U2R	52	0.005%	228	0.073%
Total	1,074,992	100%	311,029	100%

still suffers from some issues, including its failure to accurately reflect low-footprint assaults.

Table 2.11: NSL-KDD Dataset Distribution

Class	**Training Set**	**Percentage**	**Test Set**	**Percentage**
Normal	67,342	53.458%	9,710	43.075%
DoS	45,927	36.458%	7,457	33.080%
Probe	11,656	9.253%	2,421	10.740%
R2L	995	0.790%	2,754	12.217%
U2R	52	0.041%	200	0.887%
Total	**125,972**	**100%**	**22,542**	**100%**

2.6.7 CIC IDS 2017

The CIC IDS 2017 dataset [142], created by the Canadian Institute for Cybersecurity, contains realistic, labeled network traffic for evaluating intrusion detection systems. It includes both benign and various attack types (e.g., DDoS, brute-force, infiltration, XSS, botnet) generated over a 5-day period. Each flow is described by over 80 features (Table 2.12), making it ideal for training and testing machine learning models for cybersecurity research.

Table 2.12: Daily Label of CICIDS 2017 Dataset

Days	Labels
Monday	Benign
Tuesday	SFTP, BForce and SSH
Wednesday	Heartbleed and DoS Attacks
	slowloris, Hulk, , Slowhttptest and GoldenEye
Thursday	Infiltration and Web Attacks
	Sql Inject, XSS, and Web BForce
	Infiltration Dropbox Download and Cool disk
Friday	Botnet ARES, DDoS LOIT
	PortScans (sF, sA, sL, sV, sR, sS, sO, sP, sX, sN, sU, sW, sT, B)

2.6.8 NF-ToN-IoT-v2

The NF-ToN-IoT-v2 dataset, a version of the ToN-IoT dataset that is based on NetFlow, is a publicly available heterogeneous collection of network traffic traces from a realistic IoT network [10]. The ToN-IoT dataset is a freely accessible heterogeneous dataset. The NF-ToN-IoT dataset was generated by processing pcap files from the ToN-IoT dataset to obtain NetFlow records, which encapsulate network traffic details such as source and destination IP addresses, port numbers, protocol types, as well as the number of packets and bytes (Table 2.13). This dataset, which is derived from NetFlow offer a more structured representation of network traffic data, making it valuable for network monitoring, anomaly detection, and machine learning classification tasks. The adaptability of this dataset enables its application in several areas, including the development of intrusion detection systems, evaluation of security solutions, and training machine learning models for classifying network traffic.

Table 2.13: The attacks with the distributions found in the NF-ToN-IoT v2 dataset :

S.No.	Class	Count	Description
1	Benign	6099469	Normal unmalicious flows
2	Backdoor	16809	A method for attacking remote systems by responding to specially created client programs
3	DoS	712609	A deliberate attempt to overtax a computer system resources to restrict access to or make data unavailable
4	DDoS	2026234	A DoS-like attack that uses a variety of dispersed sources
5	Injection	684465	many attacks try to change how code is executed by providing untrusted inputs; two of the most common are SQL injections and code injections.
6	MITM	7723	A technique known as "Man in the Middle" involves inserting an attacker between a victim and the host with the victim is attempting to contact in order to eavesdrop on traffic and communications.
7	Password	1153323	Includes several attacks that use brute force or sniffer to recover passwords.
8	Ransomware	3425	A cyberattack that encrypts the data kept on a host and demands payment in exchange for the method or key used to decrypt it.
9	Scanning	3781419	An assortment of methods together referred to as probing that seek to learn details about hosts and networks
10	XSS	2455020	A cross-site scripting attack using online applications to send harmful scripts to end users

2.6.9 Machine Learning & Ensemble Learning Models: An Overview

Various ML and Ensemble Learning models have been used in the proposed methodologies of different research contributions, algorithms which include Logistic Regression (LR), Naive Bayes (NB), Support Vector Machines (SVM), Random Forest (RF), Extra Trees (ET), Extreme Gradient Boosting (XGBoost), CatBoost, etc. which are briefly explained below.

2.6.10 Logistic Regression (LR)

Binary categorization uses logistic regression. The logistic function estimates the binary outcome probability based on input characteristics. The logistic

function, also known as the sigmoid function, is a key component of logistic regression and converts every real-valued input to a probability of the positive class between 0 and 1. The sigmoid function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where z represents a linear combination of the input variables and their associated weights:

$$z = w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n$$

Here:

- w_0 is the bias term,
- $w_1, w_2, \ldots, w_n$ are the weights associated with the input features $x_1, x_2, \ldots, x_n$.

2.6.11 Naive Bayes (NB)

NB is a classification algorithm that relies on probability theory principles, especially Bayes' theorem. The features within the dataset are assumed to be conditionally independent, meaning that the occurrence of one feature does not influence the likelihood of another feature existing or not.

The Naive Bayes classifier calculates the probability of a class C_k based on the input features using Bayes' theorem:

$$P(C_k|X) = \frac{P(X|C_k) \cdot P(C_k)}{P(X)}$$

In this equation:

- $P(C_k|X)$ represents the posterior probability of class C_k given the features X.

- $P(X|C_k)$ denotes the likelihood of observing the features X for class C_k.
- $P(C_k)$ signifies the prior probability of class C_k.
- $P(X)$ refers to the evidence, or the probability of witnessing the features X.

The "naive" aspect of Naive Bayes lies in the assumption that the features are conditionally independent given the class label. This can be mathematically represented as:

$$P(X|C_k) = P(x_1|C_k) \cdot P(x_2|C_k) \cdot \ldots \cdot P(x_n|C_k)$$

With this assumption, the Naive Bayes classifier can be simplified to:

$$P(C_k|X) = \frac{P(C_k) \cdot \prod_{i=1}^{n} P(x_i|C_k)}{P(X)}$$

The class with the highest posterior probability is ultimately chosen as the predicted output.

2.6.12 K-Nearest Neighbors (KNN)

KNN [128] method is a frequent choice for classification problems because to its non-parametric nature. It assigns a new data point to the feature space class with the most KNNs. For classification tasks, let x be a new data point, and $N_k(x)$ represent the set of k-nearest neighbors of x in the training set. The predicted class y for x can be obtained by a majority vote:

$$y = \text{argmax}_{c_i} \sum_{x_i \in N_k(x)} I(y_i = c_i)$$

Here:

- c_i is a class label,
- y_i is the class label of the i-th neighbor,
- $I(\cdot)$ is the indicator function that returns 1 if the condition is true and 0 otherwise.

For regression tasks, the predicted value is often the average (mean) of the values of the k-nearest neighbors:

$$\hat{y} = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$

2.6.13 Support Vector Machine (SVM)

Support Vector Machine (SVM) [77] is a supervised machine learning algorithm widely used for binary classification tasks. It aims to find the optimal hyperplane that separates data points of different classes with the maximum margin. SVM is effective in high-dimensional spaces and uses kernel functions to handle non-linearly separable data.

Mathematical Formulation

Given a training dataset (x_i, y_i), where $x_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$, the SVM optimization problem is formulated as:

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^{n} \xi_i$$

subject to:

$$y_i(\mathbf{w}^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Here, $\mathbf{w}$ is the weight vector, b is the bias, ξ_i are slack variables, and C is the regularization parameter.

The decision function is:

$$f(x) = \text{sign}(\mathbf{w}^\top x + b)$$

2.6.14 One-Class SVM (OCSVM)

One-Class SVM [77]is an unsupervised variant of SVM used for anomaly detection. It is designed to learn a boundary around normal data and identify points that fall outside this boundary as anomalies.

Mathematical Formulation

Given data points $x_1, x_2, \ldots, x_n \in \mathbb{R}^d$, OCSVM solves the following optimization problem:

$$\min_{\mathbf{w},\rho,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + \frac{1}{\nu n}\sum_{i=1}^{n}\xi_i - \rho$$

subject to:

$$\mathbf{w}^\top \phi(x_i) \geq \rho - \xi_i, \quad \xi_i \geq 0$$

Here, $\phi(x)$ is a kernel function, ρ is an offset parameter, and $\nu \in (0, 1]$ determines the trade-off between the fraction of outliers and support vectors.

The decision rule is:

$$f(x) = \text{sign}(\mathbf{w}^\top \phi(x) - \rho)$$

OCSVM is particularly effective in scenarios where only normal data is available during training, making it suitable for intrusion or anomaly detection in cybersecurity.

2.6.15 Decision Tree (DT)

The DT algorithm is a widely used method in machine learning that employs a tree-like hierarchical structure to analyze and classify data. Through the use of recursive partitioning techniques on the input data, the algorithm produces a set of decision rules that can be applied for making predictions[170]. It is crucial to finely tune various hyperparameters when utilizing DT algorithms to attain peak performance. Examples of these hyperparameters include the maximum number of leaf nodes, the minimum proportion of total sample weights required for a leaf node, the minimum number of samples necessary to split an internal node, and the minimum number of samples needed to create a leaf node[170]. The decision tree is a commonly employed model in machine learning that utilizes a tree structure based on the divide and conquer approach to perform classification tasks. The systematic construction of this structure involves the careful selection of the best feature for partitioning at each node, using criteria such as entropy or information gain. The Decision Tree is a key element of the individual trees that are combined in the ensemble. The preceding algorithms are based on the principles underlying decision trees.

2.6.16 Random Forest

Random forest (RF) Ensemble learning classifiers, such as [157], employ the majority voting mechanism. This process involves generating several decision trees, with the ultimate classification outcome determined by the

consensus of all the decision trees. The rationale behind utilizing multiple decision trees is to enhance both the accuracy and reliability of the system's predictions. In a random forest algorithm, each decision tree is trained on a randomly selected subset of the training dataset as well as a random selection of features. The predictions from each tree are merged to produce the overall prediction.

2.6.17 Extra Trees

Extra Trees [40] represents an ensemble model akin to Random Forests since it also leverages several decision trees for making predictions. However, Extra Trees differs in its methodology for constructing these trees. In Extra Trees, the thresholds for splitting each feature are chosen at random rather than selecting the optimal split point, which introduces additional randomness and diminishes the model's variance. This approach proves effective for high-dimensional datasets containing noisy features and is generally quicker to train.

2.6.18 XGBoost

XGBoost [21] is another ensemble learning algorithm designed to enhance both the speed and effectiveness of the model by applying gradient descent techniques. This algorithm fuses multiple decision trees to form an efficient ensemble model. The XGBoost algorithm uses gradient boosting to refine the accuracy of the decision trees. The process entails creating a sequence of decision trees, whereby each new tree is designed to reduce the loss function related to the predictions made by the previous tree. The minimization is carried out via a gradient descent strategy. The complexity of XGBoost is $O(Kd\|\mathbf{x}\|\log n)$, with d representing the maximum tree depth, K denoting

the number of trees employed, n indicating the data size, and $\|\mathbf{x}\|$ referring to the count of non-missing samples.

2.6.19 CatBoost

CatBoost [113] is a potent, open-source gradient boosting algorithm centered on decision trees and is specifically tailored for managing categorical data. It has demonstrated remarkable performance across various machine learning challenges, particularly with high-dimensional categorical datasets. The algorithm employs an ordered boosting methodology to enhance decision tree accuracy by training them in a predetermined sequence. Additionally, it includes features designed to avert overfitting, such as L2 regularization. The complexity of the CatBoost model is $O(SN)$, where N denotes the count of base decision tree models and S signifies the number of potential permutations of the subsets used to train each base model.

For a dataset with n instances, f features, and t decision trees, the time complexities for Extra Trees, Random Forest, and Decision Trees are $O(nft)$, $O\left(n^2\sqrt{f}t\right)$, and $O\left(n^2 f\right)$, respectively [172].

2.6.20 Performance Metrics

For research and model evaluation, it is imperative to acknowledge that a singular metric cannot comprehensively gauge the excellence of a specific model. Therefore, multiple metrics [130] are employed to assess and analyze the performance of the models.

In the context of proposed research, a set of metrics has been employed to assess the performance of the proposed predictive model. True positives (TP) represent instances where our model correctly identifies and predicts attack

samples within the dataset. Conversely, false positives (FP) indicate cases where the proposed model incorrectly predicts an attack when the sample is, in fact, normal. True negatives (TN) signify the accurate prediction of normal samples, while false negatives (FN) are instances where the model wrongly predicts a sample as normal when it is, in reality, an attack. These metrics are instrumental in evaluating the effectiveness and reliability of our predictive algorithms within the scope of this research work.

1. **Accuracy:** Ratio of correctly predicted samples to total samples.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{2.1}$$

2. **Precision:** Ratio of correctly predicted attack samples to total predicted attack samples.

$$PR = \frac{TP}{TP + FP} \tag{2.2}$$

3. **Recall:** Ratio of correctly predicted attack samples to total number of attack samples

$$RE = \frac{TP}{TP + FN} \tag{2.3}$$

4. **F1 Score:** Weighted mean of precision and recall.

$$F1Score = \frac{2 * Recall * Precision}{Recall + Precision} \tag{2.4}$$

5. **Weighted avg:** The data points which have higher frequency contribute more than others.

2.7 Literature: Gaps, Problems, and Key Contributions

This section presents a critical review of existing intrusion detection approaches in IIoT environments, highlighting their limitations in addressing evolving threat landscapes. It identifies key research gaps, formulates the core problem statements, and outlines the specific contributions made to advance the state of IIoT security through novel solutions.

2.7.1 Threat Modeling and Attack Path Analysis

To conduct a review of the state-of-the-art methodologies and to explore how cybersecurity and threat modelling are relevant to industrial IoT threat modeling and attack path analysis, a literature study has been performed. The review specifically examines cybersecurity standards and the latest methods for modeling threats in ICS.

Threat assessment in industrial IoT is a growing research topic with several notable works. The closest and most recent is a study by Jbair et al.[56] which looked at threat analysis in the smart manufacturing industry. The study proposed a method of threat modeling and risk analysis in systems specifically for the smart manufacturing industry. Their threat modeling process included noteworthy elements such as a factor of asset importance, albeit with a non-standard scoring system. Lorenzo et al.[35] surveyed all available protocols in industrial IoT networks and used a CVSS to quantify the threat each protocol poses. The study also presents a framework to characterize the severity of each protocol in the IIoT environment.

To ensure that this study covered all possible cyber threat avenues when modeling for industrial IoT systems, a thorough analysis of all possible cyber threats in IIoT was essential. Hassija et al.[46] achieved the former require-

ment by describing various security issues present in an IIoT application at different levels, including the sensing layer, network layer, middleware layer, gateways, and application layer. The study also presents an overview of upcoming cybersecurity threats in fields such as blockchain and cloud computing. A literature survey into cybersecurity issues in industrial IoT by Jayalakshmi et al.[55] took a deeper look into the architecture of an industrial IoT system and the solutions proposed for attack detection. These studies served as a catalog of current security issues and not as a threat model for IoT systems.

The modeling done by Falco et al.[27] was a specific approach focusing exclusively on SCADA systems to draw a data-driven organizational schema that could help researchers to recognize security breaches that are crucial to our operations, so they presented a framework to prioritize the modeled threats using the CVSS scoring system. To ensure that researchers choose the necessary threat modeling approach, Nataliya Shevchenko et al. [146] comprehensively explained the twelve threat modeling methods, namely STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service (DoS), and Elevation of Privilege) and their associated derivations. PASTA (Process for Attack Simulation and Threat Analysis), LINDDUN (Linkability, Identifiability, Non-Repudiation, Detectability, Disclosure of Information, Unawareness, and Non-Compliance), CVSS (Common Vulnerability Scoring System), Attack Trees, Persona Non-Grata, Security Cards, HTMM (Hybrid Threat Modeling Method), Quantitative Threat Modeling Method, Trike, VAST (Visual, Agile, and Simple Threat) Modeling, OCTAVE (Operationally Critical Threat, Asset, and Vulnerability Evaluation), that came from various articles, came from various articles, and they focus on diverse functions of the process. Each threat modeling method is special and is not suggested over others; the selection of which methods to use should be based

on the project's requirements and its considerations[146]. These studies have been found useful as a checklist for covering all security issues for creating a threat model for an industrial IoT system.

The comprehensive analysis encompasses a diverse array of methodologies and frameworks employed in the field of threat modeling and risk analysis within the context of industrial Internet of Things (IoT) systems. Certain studies have been dedicated to examining particular industries or protocols within the realm of industrial Internet of Things (IoT), aiming to gain a deeper understanding of cybersecurity challenges within these specific contexts. Conversely, other research endeavors adopt a broader perspective, seeking to offer a comprehensive overview of the various cybersecurity issues that arise in the industrial IoT landscape as a whole. The scoring systems employed in various studies exhibit considerable variation, with certain studies adopting the Common Vulnerability Scoring System (CVSS), while others opt for non-standard scoring systems predicated on the significance of assets. The aforementioned studies provide significant insights pertaining to the establishment of a comprehensive threat model for an industrial Internet of Things (IoT) system. The process involves the identification of security concerns across various layers, the prioritization of risks that have been modeled, and the selection of the most suitable threat modeling methodologies in accordance with the project's requirements and considerations.

Research Gaps

Foolowing are the key research gaps based on the literature survey:

Table 2.14: Summary of Literature Survey on Threat Assessment in Industrial IoT Systems

Study	Focus	Methodology	Frame-work	Scoring-System	Key Findings
Jbair et al. [56]	Threat analysis in smart manufacturing industry	Threat modeling and risk analysis	N/A	Non-standard scoring system based on asset importance	Proposed a method for threat modeling and risk analysis specifically for smart manufacturing industry
Lorenzo et al. [35]	Survey of available protocols in industrial IoT networks	Common vulnerability scoring system (CVSS)	Framework to characterize the severity of each protocol	CVSS	Framework to characterize the severity of each protocol in the IIoT environment
Hassija et al. [46]	Identification of security issues present in IoT applications at different levels	N/A	N/A	N/A	Identified various security issues present in an IoT application at different levels
Jayalakshmi et al. [55]	Literature survey into cybersecurity issues in industrial IoT	N/A	N/A	N/A	Cataloged current security issues in industrial IoT systems
Falco et al. [27]	Modeling of threats in SCADA systems	Data-driven organizational schema	Framework to prioritize modeled threats	CVSS	Proposed a framework to prioritize modeled threats in SCADA systems
Nataliya-Shevchenko et al. [146]	Comprehensive explanation of twelve threat modeling methods	N/A	N/A	N/A	Comprehensive explanation of twelve threat modeling methods and their associated derivations
Proposed Work (TMAP)	Asset-centric versatile and generalized approach	Quantitative Threat Modeling and Attack Path Analysis using Purdue Model	Framework for cyber threats specific to industrial equipment.	CVSS	Proposes a robust framework for quantitative threat modeling and path analysis, validated through insightful case studies.

Lack of Standardized and Industry-Agnostic Threat Modeling for IIoT

While existing research extensively covers threat modeling and detection, there remains a significant gap in the form of a standardized, step-by-step methodology for identifying, assessing, and prioritizing threats in Industrial Control and Cyber-Physical Systems (ICPS) [135]. Current approaches suffer from three major limitations:

End-to-End Framework

IIoT environments are increasingly exposed to cyber threats, yet existing approaches lack early-stage threat modeling and asset criticality analysis. This research proposes a systematic integration of Purdue, STRIDE, MITRE ATT&CK, and CVSS for IIoT cybersecurity assessment that enables asset classification and comprehensive threat modeling from the design phase. While not performing risk assessment directly, the framework can integrate with it, serving as an end-to-end foundation for securing ICPS.

Lack of Standardization

Most threat modeling techniques are fragmented, with no unified framework that integrates the STRIDE based threat modeling [135] with the Common Vulnerability Scoring System (CVSS) [135] based on risk prioritisation.

Industry-Specific Limitations

Existing models are often tailored to niche sectors (e.g., smart grids or manufacturing), limiting their applicability across diverse IIoT environments. This absence of a systematic, industry-agnostic methodology hinders organiza-

tions from efficiently allocating security resources, leaving critical vulnerabilities unaddressed.

Research Problem

Can a standardized, quantitative threat modeling approach be developed to analyze and prioritize attack vectors across diverse industrial IoT environments such as manufacturing (IoM) and power generation?

Research Contribution

TMAP presents a systematic integration of Purdue model, STRIDE, MITRE ATT&CK, and CVSS for IIoT cybersecurity assessment threat modeling. The method is validated using "Internet of Manufacturing (IoM)" and "Internet of Production (IoP)" case studies. It enables organizations to map vulnerabilities and prioritize mitigation using a unified approach.

2.7.2 Detection of known Cyber Threats Leveraging Ensemble ML Techniques

The article [172], discusses cyber-attacks on autonomous and connected vehicles and proposes an IDS for securing vehicular networks. The IDS uses tree-based ML models to detect threats, with the SMOTE oversampling method applied for class imbalance and a tree-based averaging feature selection approach for irrelevant features. The IDS was tested on CAN-intrusion and CICIDS2017 datasets, achieving 99.98% and 99.86% accuracy respectively. In article [37],the authors propose an IDS for vehicular networks using ToN-IoT dataset, addressing missing values and class imbalance issues. They use Chi2 and SMOTE for feature selection and class imbalance, and

XGBoost outperforms other models for binary and multi-class classification with 97% and 98% accuracy, respectively. The article [171] discusses the development of a method called LCCDE (Leader Class and Confidence Decision Ensemble) for improving security in the Internet of Vehicles (IoV). The method identifies the best machine learning models for attack detection and combines them into an ensemble model. The model achieved an F1 score of 99. 8% in the CICIDS2017 data set.

The UNSW-NB15 dataset was examined utilizing ensemble learning techniques and the stacking approach in article [3]. Random forest, Naive Bayes, SVM, and logistic regression were employed as meta-classifiers, and the results showed a 95% accuracy rate. In article [169], the multi-tiered hybrid intrusion detection system (MTH-IDS) model, proposed by Yang et al., identifies zero-day cyber-attacks and known attacks on vehicular networks. The system consists of four learning models: tree-based supervised learners, Bayesian optimization with a Tree-structured Parzen estimator, CL-K-means model, and Bayesian optimization with Gaussian Processes. The combination of these tiers improves the model's performance, achieving 99.8% and 99.9% accuracy on CICIDS2017 and CAN-intrusion, respectively.

Research Gap

Current ML-based NIDS systems show varying accuracy due to inconsistent feature sets in available datasets, making results difficult to compare or reproduce. Existing models struggle to maintain high detection rates across diverse IIoT network topologies. Additionally, ensemble learning models for NIDS are underexplored, particularly in the context of industrial networks where real-time and scalable detection is needed. Additionally, the prevalent use of datasets like CICIDS2017 and CAN-intrusion does not fully capture the complexity, scale, and unique threat landscape of IIoT systems.

Research Problem

How to develop a reliable NIDS that provides consistent, high accuracy across diverse IIoT scenarios using standardized features and ensemble methods, while overcoming challenges of false positives and dataset variability.

Research Contribution

ELA-NIDS uses the NF-ToN-IoT-v2 dataset and compares multiple ensemble classifiers against traditional ML models. XGBoost achieved 96% detection accuracy. NF-ToN-IoT-v2 is a flow-based dataset specifically designed for IIoT environments, featuring labeled bidirectional NetFlow records generated from diverse industrial protocols (e.g., Modbus, MQTT). It enables efficient training of ML/DL-based IDS models by providing high-dimensional features and multi-class attack scenarios reflective of real-world IIoT threats

2.7.3 Machine learning Approach for Botnet Attack Detection

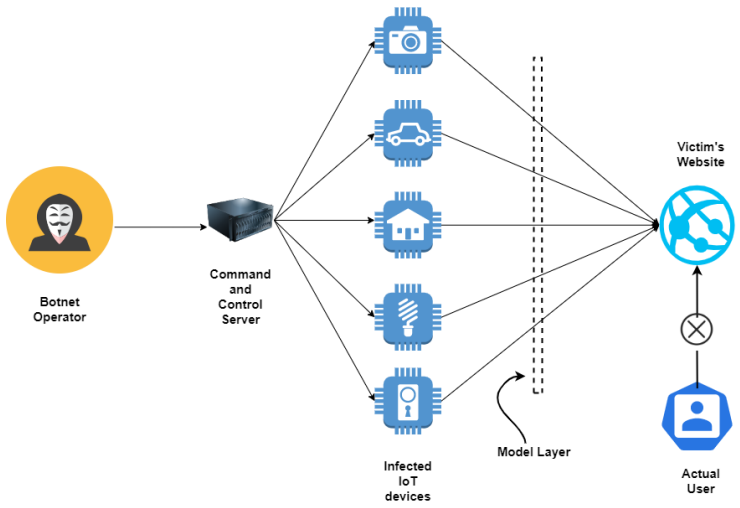


Figure 2.12: Botnet Attack on IoT devices

Fig. 2.12 depicts the general botnet attack scenario. The attacker uses the

command and control server to communicate with the infected IoT devices. The attacker then uses these IoT devices as bots to overload the web servers, which makes the website inaccessible to the actual use,r which may be prevented by installing a botnet attack detection model as shown in the model layer of Fig. 2.12. One of the notable botnet attacks was the one "Mirai" which took down a large part of the Internet in 2016. At the beginning of 2021, organizations have recorded 31% botnet activity. By 2021, it is predicted to increase to 51% [36]. Thus, there is a growing concern to detect botnet attacks. The increase in IoT devices also results in increasing of efforts in the labeling process of the data generated all around the world. This thesis work investigates a semi-supervised learning technique using Generative Adversarial Networks (GAN) to minimize the efforts in the labeling process while successfully detecting botnet attacks. Semi-supervised learning is a machine learning technique in which a small amount of labeled data is combined with a large amount of unlabeled data to train a model. Semi supervised learning uses additional unlabeled data to learn the features. Unsupervised learning algorithms do not require any labeled data, while supervised learning algorithms are trained on a large set of labeled data. The semi supervised algorithms lies in between these two techniques [168].

Existing Intrusion Detection Systems (IDS) based on Machine Learning (ML) and Deep Learning (DL) techniques are described in the following: Ensemble and ML techniques have been used for anomaly detection [115] on UNSW-NB15 [97] and CICIDS2017 [142] datasets for multi-class classification. Random Forest (RF) performed better than others i.e., 95.45% and 99.7% accuracy on UNSW-NB15 and CICIDS2017 datasets respectively in ML techniques while the stacking algorithm achieved 96.83% and 99.99% respectively in ensemble approach, hence, the ensemble approach performed better than ML techniques.

A DL-based intrusion detection system at the fog layer has been proposed in [33] where the Convolutional Neural Network (CNN), Deep Neural Network (DNN) and Recurrent Neural Network (RNN) architectures were implemented on CIC-DDOS2019 [143] as well as ToN_IoT [93] dataset for the multi-class classification. It could be concluded that CNN outperformed the state of art DL based IDS system. CNN achieved 99.95% accuracy on binary classification while 99.92% on multi-class classification without any mitigation strategy.

ML and DL techniques on N-BAIoT [89] dataset were investigated in [65]. Decision Tree (DT) and Random Forest (RF) achieved the F1 score of 1 in both multi-class and binary classification. DL models like CNN, RNN and Long Short Term Memory (LSTM) achieved F1 scores of 0.87, 0.44 and 0.46 on an average on dataset.

A Deep Learning framework at the fog layer was proposed[123] where six DL algorithms were analyzed on five different datasets with the best accuracy of 99.85% achieved in binary classification. It was concluded that LSTM outperformed to all other supervised DL models. Anomaly detection using feature selection was studied in [1]. The authors investigated the logistic regression and Artificial Neural Network on N-BAIoT [89] dataset having an accuracy of 99.98% performed better than the proposed ANN with 96.4% accuracy.

In [116], Distributed attack detection framework based on semi-supervised learning achieved 86% accuracy on NSL-KDD [156] dataset. The NSL-KDD dataset lacks the botnet attack. It was suggested to improve the performance using the DL techniques.

Research Gap

The existing literature predominantly emphasizes supervised learning approaches for intrusion detection, which depend heavily on manually labeled datasets. This manual labeling process is time-consuming, resource-intensive, and not scalable to real-time, high-volume IoT environments. Furthermore, most current models are limited to binary classification tasks, often neglecting the detection and classification of specific threats such as botnet attacks. Botnet communication often uses stealthy protocols, making labeled sample collection more challenging. Despite the growing prevalence of botnets in IoT ecosystems, there is a noticeable lack of research addressing this challenge using more scalable methods. Therefore, there exists a significant gap in developing semi-supervised learning models that can effectively detect botnet attacks while minimizing the reliance on labeled data, thus enabling more efficient and real-time threat detection in IoT networks.

Research Problem

How to detect IoT botnets such as Mirai and Bashlite using models that can learn from both labeled and unlabeled traffic, thus reducing the manual labeling burden.

Research Contribution

GANIBOT proposes a semi-supervised GAN model trained on the N-BaIoT dataset. It achieves 99.89% accuracy in binary classification and 59% in multi-class classification. The model learns latent patterns in unlabeled traffic, enabling reliable botnet detection despite data labeling challenges. The approach is validated against state-of-the-art supervised models, showing superior efficiency.

2.7.4 Deep Learning based IDS for IIoT systems

In paper [14], IDS was developed to learn the behavior of normal network traffic. Dataset used was UNSW-NB 15 for communication of external networks. The methods being experimented here were Artificial Immune System (AIS), Filtered-based SVM (FSVM), Euclidean distance Map (EDM), and Geometric Area Analysis (GAA) which performed 85%, 92%, 90% and 93% respectively. In [72], BoT-IoT and UNSW-NB15 datasets are considered in the experiment. After the data standardization is done, the MLP (Multi-Layered Perceptron) model is used followed by adapting the hyperparamters. The comparison was based on the BoT-IoT dataset while the classifier was built on UNSW-NB15 dataset. ARM (Association Rule Mining) and Naïve Bayes only had 85% and 72% accuracy respectively. The normal perceptron model was basic and gave 63% accuracy but when converted to the prima facie 13- feature DNN model, it gave 99% accuracy. Among the ML techniques, Decision tree gave the highest accuracy (93%). The process of training the models consumes a lot of time and memory [72]. In paper [74], the authors explore ML based approach. On carrying out an experiment it was seen that the ML methods along with flow identifiers were effective in detecting botnet attacks. Four ML algorithms were used namely, ARM, Artificial Neural Network (ANN), Naïve Bayes and Decision Tree. Decision Tree gave the best results with the highest accuracy of 93.23% and the lowest False Alarm Rate of 6.77% whereas ANN was the least accurate with an accuracy of 63.97% and False Alarm Rates of 36.03%. In [97], the authors have addressed the issue of lack of a data set which can appropriately show the modern network traffic and attacks. So this research looks into the creation of a UNSW-NB15 dataset. It has 49 features developed using Argus and Bro-IDS tools. In paper [149], the authors proposed a hybrid IDS for IoT networks by combining CNNs (Convolutional Neural Network) and

LSTM. They used UNSW_NB-15 dataset and compared the performances of the hybrid IDS and RNN for binary classification, and the accuracy achieves was 95.7% and 98.7% respectively. In [180], the authors compared the performances of DL methods like RNN, GRU and Text-CNN with some of the traditional ML methods on the KD99 and ADFA-LD datasets. The DL methods were able to outperform the traditional ML techniques. In paper [124], the authors implemented Linear Discriminant Analysis (LDA), Classification and Regression Tree (CART) and Random Forest (RF) on KD99 dataset. The accuracy achieved were 98.1%, 98% and 99.65% respectively. In [39], the authors implemented a Feed-Forward Neural Network for binary and multi-class classification on the Bot-IoT dataset for normal and three different attack categories (DDoS/DoS, Reconnaissance, Information Theft). They were able to achieve 98%, 99.4%, 98.4% and 88.9% accuracy on the normal and the three attack categories respectively. In paper [84], the authors proposed an IDS based on blockchain and deep learning. The DNN model was able to achieve 98% accurclassification and 97% accuracy for multi-class classification on the NSL-KDD dataset.

Research Gap

Traditional intrusion detection models in IoT environments have shown limitations in effectively handling multi-class attack detection, particularly in the presence of heterogeneous, high-dimensional, and time-dependent network traffic. Classical machine learning algorithms such as Decision Trees, SVMs, and k-NNs suffer from scalability issues, lack robustness against zero-day attacks, and often yield high false positives when faced with evolving botnet traffic.

Although deep learning (DL) techniques offer improvements in learning complex patterns, basic DL models (e.g., ANN, CNN) struggle to preserve tem-

poral dependencies and are insufficiently optimized for sequential IIoT traffic. More critically, prior IDS implementations often operate under binary classification settings, which fail to distinguish between multiple attack types (e.g., Exploits, Worms, Reconnaissance, Theft). Furthermore, existing models rarely leverage optimized feature spaces, resulting in computational overhead and overfitting risks in large-scale datasets like BoT-IoT and UNSW-NB15.

Existing models often rely on outdated datasets and fail in multi-class classification with high false positives. There is a lack of IDS advanced sequence-aware models like LSTM variants that can effectively learn temporal patterns for IIoT systems.

Research Problem

To build a robust IDS using LSTM to detect complex and new attack patterns in IIoT systems using deep learning models that can effectively learn temporal patterns in IIoT network traffic to improve the detection accuracy while maintaining low false positive rates.

Research Contribution

LBDMIDS presents variants of LSTM (stacked and bidirectional) trained on UNSW-NB15 and BoT-IoT datasets. The models outperform classical approaches in detecting complex attack categories including DoS, worms, and backdoors. The research highlights LSTM's ability to retain long-term dependencies in time-series traffic, offering both accuracy and robustness.

2.7.5 DDoS Attack Detection Leveraging ML & DL Approaches

In [167], the goal of the suggested research was to create an intelligent model that may aid in the security of the IoT framework. In [19], The authors conducted a review of current cybersecurity research and discovered two significant trends. First is the adoption of old, established approaches that are still in use in many applications. Another is Machine learning (ML) based techniques, which are becoming increasingly popular. It also mentions that ML is now being utilised in both malware and security. In [61], the author proposed a solution of traditional intrusion detection which gathers and uses huge data containing irrelevant information, leads to an increase the detection time and a decrease the accuracy. The solution uses a combination of Particle Swarm Optimization (PSO) and Grey Wolf Optimization (GWO). Features obtained were given to Random Forest (RF) for classification, where these models were able to achieve 99.66% accuracy for multiclass classification. In [162], the author showed ML techniques are not able to detect attacks for unpredictable network technologies, buta Deep Learning (DL) model was able to solve this problem. The DL models used were Convolution Neural Networks (CNN) along with Transfer Learning, which was used for multiclass and binary classification on the Bot-IoT dataset. In [7], the author proposed a novel framework using the IoTID20 dataset. The proposed framework used a CNN, an LSTM and a hybrid CNN-LSTM model. The dimensionality of the network dataset was lowered, and the PSO approach was used to optimise the suggested system. These models were able to obtain 96.60%, 99.82% and 98.80% accuracy, respectively. In [28], Fan et. al. proposed an alternate model as a single detection model for heterogeneous data using Federated Transfer Learning. The model provided improved detection of unknown attacks with 91.93% of detection accuracy as compared to traditional methods. The model also generated a lower False Positive rate than the single cen-

tralised model.

Several improvements have been made to enhance the accuracy of intrusion detection for IoT. However, simple DL-based IDSs face several inherent challenges during attack classification in an intrusion detection system for IoT. A dataset with a high number of features and a high amount of data increases the time required to train a DL model. Some features can negatively impact the training process of the model, which reduces the model's attack detection capability. Table 2.15 shows the summary of the most recent research with special emphasis on the use of feature selection and data imbalance issues. The proposed approach involves using feature selection techniques and also handles the data imbalances.

Table 2.15: Different intrusion detection approaches

S.No.	References	Handling data imbalance	Feature Selection	Classification Technique	Attack model for Prediction	Dataset
1	Roddaet. al. [118]	Not used	Not used	Multi layer Perceptron	N/w Intrusion Detection System	NSL –KDD
2	Yanmiao et. al. [82]	Not used	Not used	Multi-CNN fusion model	NIDS	NSL –KDD
3	Mohamed et. al. [34]	Three RDTIDS mode	Not used	Rules and Decision Tree	NIDS	Bot-IoT
4	Torres et al. [160]	Not used	Not used	RNN	NIDS	NSL –KDD
5	Keserwani et. al. [60]	Not used	GWO –CSA	DSAE	NIDS	UNSW NB-15
6	Tama et al. [155]	Not used	PSO+ ACO+ GA	Rotation forest	Binary classification	UNSW NB-15
7	Alkahtani et al. [7]	Not used	PSO	LSTM	Binary classification	IoTID 20

Research Gap

Despite the growing application of deep learning (DL) in intrusion detection for IIoT environments, existing approaches suffer from many limitations.

IIoT networks are vulnerable to DDoS attacks, but current DL-based IDS systems are often too heavy to be deployed in these environments. There is a lack of computationally efficient models that balance speed, accuracy, and computational efficiency.

Research Problem

To create a computationally efficient deep learning-based IDS capable of accurately detecting DDoS attacks in IoT networks without sacrificing speed or performance.

Research Contribution

NFDLM proposes four ANN/LSTM-based architectures combined with correlation and mutual information feature selection. The models achieve nearly 99% accuracy on DDoS detection tasks using minimal computing resources. The approach allows fast training and scalability across real-time environments.

2.7.6 Zero Day Attack Detection in IIoT

This subsection reviews key studies on zero-day attack detection. Zhang et al. [179] evaluated ML-based NIDS for zero-day detection using zero-shot/open-set learning and a sparse autoencoder to map known attack features into semantic space. Using NSL-KDD, their method achieved an average accuracy of 88.3%, showing the potential of semantic mapping for detecting unseen attacks. Hindy et al. [47] proposed an autoencoder-based approach evaluated on CICIDS2017 and NSL-KDD, achieving 90.01% and 92.96% accuracy, respectively; however, performance depended heavily on

fixed threshold values, which may limit adaptability. Sarhan [125] introduced a ZSL/OSL framework using UNSW-NB15 and CICIDS2017, reporting that some zero-day attacks remain difficult to detect due to low zero-day detection rates.

Several deep learning approaches have also been explored. Vigneswaran et al. [163] showed that DNNs outperform shallow networks for intrusion detection on KDDCup-99. Naseer et al. [100] improved anomaly detection using DNNs on NSL-KDD, achieving high accuracy with low false positives. Rustam et al. [147] used a ResNet-based IDS to address vanishing gradients, achieving 84.7% and 90.5% accuracy on NSL-KDD groups with fast detection times. Halbouni et al. [45] proposed a CNN-LSTM hybrid model, achieving 99.64%, 94.53%, and 99.67% binary accuracy on UNSW-NB15, CICIDS2017, and WSN-DS, although detection remained weak for some attack types.

Other studies focused on specialized zero-day and anomaly detection methods. Li et al. [83] integrated attribute learning with IDS using feature selection and spatial clustering, outperforming decision trees on NSL-KDD for selected DoS and probe attacks. Kumar et al. [78] and Sabeel et al. [120] developed models for high-volume zero-day attacks using heavy-hitter and graph-based techniques, achieving around 88.98%–91.62% accuracy across real-time and CICIDS18 datasets. Mokhtari et al. [92] proposed MIDS for SCADA systems, reaching 99.76% fault detection accuracy using random forest. Khan et al. [62] converted network flows into spectrograms and applied CNNs, achieving 99.35% binary and 98.76% multi-class accuracy.

Recent works also explored early detection and IoT security. Hwang et al. [49] introduced D-PACK, combining sampling, CNN-based traffic profiling, and autoencoders to achieve nearly 100% accuracy with low false positives while reducing detection delay. Feng et al. [31] developed a plug-and-play

DL-based device for detecting XSS, DoS, and SQL attacks in ad-hoc networks, though more malicious samples and adaptive security designs are needed. Abdalgawad et al. [2] applied BiGAN and AAE models to IoT-23, showing that GAN-based models can effectively classify IoT attacks and detect anomalous scenarios.

Overall, ML- and DL-based IDSs show strong potential for detecting zero-day and unknown attacks. However, many studies still rely on outdated datasets such as NSL-KDD and KDDCup-99, which may not reflect modern network traffic or evolving threats. Future research should prioritize recent, diverse, and realistic datasets, broader attack coverage, adaptive thresholds, and stronger real-world evaluation to improve the reliability and generalizability of zero-day intrusion detection systems.

Research Gap

Despite significant advancements in IDS leveraging ML techniques, there remains a **notable research gap** in effectively detecting **zero-day attacks**. Existing works predominantly focus on identifying known threats or use generalised anomaly detection techniques that often suffer from high false alarm rates and lack behavioral context.

Specifically, **key gaps** include:

1. **Limited Focus on Zero-Day Scenarios**: Most ML-based IDS research emphasizes known attacks. Only a subset adopts approaches like zero-shot learning (ZSL) or generative models to address truly unseen threats.
2. **Over-Reliance on Outdated Datasets**: A significant portion of studies use datasets like NSL-KDD or CICIDS2017, which do not capture the complexity of modern traffic or sophisticated attack vectors, limiting real-world applicability.

3. **Inadequate Real-World Evaluation**: Many promising methods (e.g., autoencoders, attribute learning, graph-based volume signatures) lack validation across diverse, live network environments, making them vulnerable to practical limitations such as evolving attack behaviours and data imbalance.
4. **Neglect of Behavioural Semantics**: Several anomaly detection systems ignore deeper behavioural patterns of attacks, leading to alert fatigue due to misclassification of benign anomalies as threats.
5. **Scarce Use of Hybrid or Semi-Supervised Models**: There is limited exploration of **semi-supervised**, **transfer learning**, or **adversarially robust frameworks** that could generalize better to unknown threats.

Research Problem

1. To design a zero-shot deep learning model capable of identifying zero-day threats by learning attack characteristics indirectly from similar known classes.
2. To design a hybrid IDS that integrates threat intelligence and combines supervised and unsupervised learning for the detection of both known and unknown (zero-day/APT) attacks in IIoT environments.

Research Contribution

1. A hybrid CNN-LSTM architecture enhanced with zero-shot learning to detect unseen classes was proposed. The model is trained using CICIoT2023 with one class omitted during training. Results show superior performance over ensemble and traditional ML models. It reduces false positives and captures unseen behavioral patterns.

2. Proposes HMS-IDS, a multi-stage system using CIC-ToN-IoT data, achieving 99.49% accuracy for known and 98.94% for unknown threats. It employs a two-tier detection system with anomaly detection and k-means clustering for zero-day threats. Threat intelligence is incorporated through preprocessing, semantic analysis, and behavior profiling. The framework is validated across diverse datasets, making it generalizable and robust.

2.7.7 Intrusion Response Mechanisms in IIoT

Authors in paper [111] propose an IPS with reCAPTCHA Controller using information-based metrics for DDoS [134] attack on VANET prohibits bots from launching consistent automated attacks as well as many automated DDoS attacks. This method examines the randomness of user requests using information theory-based measures, such as entropy. Both frequency and entropy are utilised to determine the attack's vulnerability. DDoS attacks are carried out using AODV, Firecol and reCAPTCHA. It has been proven that reCAPTCHA performs well during DDoS attacks.

In paper [9], Intrusion Prevention Systems (IPSs) are focused on big data. The author presented a huge data-based hierarchical deep learning system to improve efficiency. It captures both network traffic and content information using behavioural and content-functional capability. This model enhances the accuracy of fake attack categorization. SVM and DT were the algorithms used for implementation in this article. The independent test and the 52 cross-validation F test are used to assess IDS performance. The function extraction and selection method may consume too much storage space for large datasets. Storage for sparks For an efficient approach to role collecting, the MapReduce architecture is used.

In paper [139], according to the Hybrid Machine Learning Network Intrusion Prevention System in Real-Time was capable of high performance and real-time categorization. This system employs two classifiers. Only if the Level 1 classifier can categorise packets as normal or attack with precision can it receive them at line speed. Level 2 classifier has no real-time processing limits; therefore, it can take any leftover packets that were not classified as an attack in level 1 and categorise them slowly but accurately. The dataset is the same in both.

In paper [5], Improving Cyber Security in advanced Internet of things (IoT) Using Intrusion Prevention Algorithm for IoT (IPAI) was discussed. The suggested technique focuses on the investigation and research of bandwidth attacks with a special focus on DDoS, which is a difficult to detect problem that degrades network performance. DDoS is a sort of assault in which a collection of attacker nodes attacks the target, preventing legitimate users from accessing the resources and network services. The proposed technique is based on the report generated by the IDS after analysing the forensic analysis report, which further leads to the prevention of DDoS assaults in the network.

In paper [38], an IDS/IPS (Intrusion Prevention/Detection System) named Dique that can identify and stop denial of service (DoS) assaults was developed using a deep learning algorithm. The information of recorded and identified packets can be graphically and textually displayed in real time via Dique's Graphical User Interface (GUI). Deep Feed Forward neural network with multiple layers is used in the proposed model for classifying DoS attacks. For training, the CICDDoS2019 Dataset was utilized. The Deep Feed-forward technique has a 99.4% accuracy rate. To verify the functionality of Dique, the online application Diluvio was created and constructed.

In paper [12], the author proposed a cutting-edge unsupervised intrusion prevention system (IPS) for vehicle Controller Area Networks that can identify

and prevent assaults without altering the design of the electronic control units (ECUs) or requiring data that is only available to automakers. They have assessed which machine learning (ML) method can identify fuzzy and spoof attacks with the greatest degree of precision with the fewest amount of data bytes. It is the only approach that can reject attacking frames before damage is done and may be utilised on a low-cost Raspberry Pi. They have used three datasets from the Hacking and Countermeasure Research Lab [43]. Detection results on these datasets are fuzzy dataset 99.29%, spoof dataset 99.24%.

Table 2.16: Different intrusion prevention and detection approaches

S.No.	Ref.	Feature Selection	Classification techniques	Attack Model	Dataset
1	[111] Poongodi M	Not used	Information theory based metric	reCAPTCHA technique	Realtime
2	[9] Abdulrahman Saad Alqahtani	Not used	Hierarchical Deep Learning System	HIDS	CICIDS2017
3	[139] Wooseok Seo , Wooguil Pak	Uses each byte value without one-hot encoding	RF and DT	NIDS	UNSW-NB15 and CICIDS2017
4	[5] Abdulaziz Aldaej	Not used	Forensic analysis of IDS report	Not used	DDos Attacks
5	[38] Canola Garcia, J. F., Blandon, G. E. T.	CICFlowMeter tool	Multilayered Deep Feed Forward neural network	Not used	CICDDoS2019
6	[12] Paulo Freitas de Araujo-Filho	Not Used	One-Class Support Vector Machine (SVM) and Isolation Forest	Not used	Hacking and Countermeasure Research Lab datasets [16]

Research Gap

Most ML-based IDS systems detect intrusions but do not include automated mechanisms to prevent them. There's a lack of computationally efficient and adaptive IPS solutions designed specifically for real-time IoT environments. Also, prevention techniques may involve packet drop, which leads to data loss. Models with higher accuracy require more computational memory, which makes real-time prevention and detection less significant in terms of cost. Whereas signature-based methods have low computational cost but lower accuracy, which is not ideal for real-world applications [139].

Research Problem

To design a computationally efficient, real-time machine learning-based Intrusion Prevention System (IPS) that can proactively prevent attacks on resource-constrained IoT networks.

Research Contribution

MLAPS uses a two-phase approach: at Level 1, DT, for fast filtering, and at Level 2, RF, for deep inspection. It calculates a threshold to dynamically split traffic, and adapts to changing attack behavior. Validated on UNSW-NB15, the system reduces training time and maintains high accuracy, showcasing real-time decision-making capability.

Chapter 3

Threat Modeling and Attack Path Analysis

In this proposed work, a quantitative threat modeling and attack path analysis method for industrial IoT systems has been proposed. In this work, two case studies have been taken for the implementation of the proposed model, one being the power industry (Internet of Production: IoP) and the other being the manufacturing industry (Internet of Manufacturing: IoM). The advantage of taking a generalized approach is that it gives you the flexibility to apply it to several different industries. This study offers significant contributions to researchers and organizations involved in the IIoT systems.

3.1 Threat Modeling: Characteristics & Issues

The process of identifying, enumerating, classifying, and mitigating possible dangers is referred to as "threat modeling." It is a preventative method that is utilized to gain an understanding of how various dangers and assaults could be carried out[165] . The objective of threat modeling is to supply security teams with a methodical study of what countermeasures need to be developed, taking into account the characteristics of the asset, the attack vectors

that are most likely to be used, and the assets that are most desired by an adversary. Threat modeling provides answers to questions such as:

- "Where is it most vulnerable to attack?"
- "What threats, if carried out, may result in a greater impact?"
- "What countermeasures are required to safeguard against these threats?"

3.1.1 Types of Threat Modeling Approaches

In general, there are three basic approaches to threat modeling: software-centric, attacker-centric, and asset-centric.

1. Software Centric Approach: This approach prioritizes the examination of the software system in order to detect possible security risks and vulnerabilities. It requires having an understanding of the architecture, design, and specifics of the execution of the system. The objective of this methodology is to identify vulnerabilities present in the software code, system configuration, and the overarching security stance of the system. Commonly, it encompasses tasks such as scrutinizing code, conducting security assessments, and evaluating the constituents, connections, and information transmissions of the system.

2. Attacker Centric Approach: This approach is centered on attackers and takes into account the viewpoint of possible attackers and their tactics. The process entails the identification of possible threats through the examination of the motives, abilities, and potential avenues of attack of adversaries. This methodology facilitates comprehension of the methodologies that malevolent actors may employ to capitalize on security weaknesses. The approach of considering the attacker's perspective entails directing attention toward the identification of possible vulnerabilities that may be leveraged to undermine the security of the system. This methodology frequently encompasses

methodologies such as the acquisition of threat intelligence, conducting penetration testing, and emulating plausible attack scenarios.

3. Asset-Centric Approach: The proposed work is based on an asset-centric approach, which is centred on the identification and safeguarding of essential assets within a given system. The process entails the classification and ranking of assets according to their significance, sensitivity, and worth to the enterprise. The objective is to recognise plausible hazards that may have an impact on said resources and subsequently rank the measures taken to alleviate them in order of importance. This methodology facilitates knowledge of the possible effects of a security breach on the vital assets of the enterprise. The process frequently encompasses tasks such as managing an inventory of assets, evaluating potential risks, and selecting suitable security measures to protect the assets that have been identified.

3.1.2 Threat Modeling Methodologies

Organisations frequently opt for a blend of threat modeling methodologies, taking into account their particular needs and the characteristics of the system or application under inspection. Various threat modeling methodologies are available for organizations to methodically recognize and tackle potential threats. STRIDE, PASTA, OCTAVE, and DREAD are some popular methods for threat modeling and are structured approaches to identify assets and mitigates threats in the system, specially in cybersecurity context which are briefly explained below:

1. STRIDE

The STRIDE [66] approach was created by Microsoft and is a commonly employed methodology that offers a systematic framework for conducting threat

modeling. The STRIDE acronym is utilized to denote six distinct categories of threats, namely Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege. By taking into account the following categorizations, the STRIDE framework offers a systematic approach to identifying credible threats.

2. DREAD

Microsoft also developed the widely-used DREAD technique [66], which provides a standardized framework for performing threat modeling. Damage potential, Reproducibility, Exploitability, Affected users, and Discoverability are the five unique risk aspects represented by the DREAD mnemonic. Each component is given a score between 1 and 10 that reflects the degree of risk it poses. Potential risks associated with known threats can be evaluated and prioritized with the help of the DREAD framework.

3. OCTAVE

The Operationally Critical Threat, Asset, and Vulnerability Evaluation (OCTAVE) [66] technique is a systematic approach to identifying and evaluating vulnerabilities and threats to an organization's most important assets. People from different departments or roles within an organization work together in this method. The study methodology consists of three main steps: developing asset-based threat profiles; identifying potential threats; and conducting vulnerability assessments. The OCTAVE methodology takes a methodical approach to evaluating risks, with a spotlight on recognizing and appreciating the value of a business's assets.

4. PASTA

The Process for Attack Simulation and Threat Analysis (PASTA) [66] is a risk-centric approach to threat modeling that combines elements from other approaches like STRIDE, attack trees, and risk analysis. There are seven distinct phases in total: planning, detecting, evaluating, reacting, and checking in on progress. The PASTA methodology prioritizes an iterative and adaptable approach, enabling organizations to consistently enhance their threat models by incorporating new information. The primary aim of this endeavor is to recognize plausible dangers, evaluate their possible outcomes and probability, and offer pragmatic suggestions for reducing linked hazards. The aforementioned methodologies function as systematic frameworks that provide direction to entities in executing thorough threat modeling exercises.

These methodologies serve as structured frameworks that offer guidance to organizations in conducting comprehensive threat modeling exercises. It is noteworthy that entities have the capacity to modify and tailor these methodologies in accordance with their individualized necessities, industry prerequisites, and the intricacy of their systems.

Summary

It is imperative to note that threat modeling is a continuous process that necessitates periodic review and modification to accommodate system changes, emerging threats, and evolving security needs. For IIoT systems, the threat modeling methodologies PASTA and STRIDE are particularly more suitable because of the complexity and cyber-physical nature of IIoT environments. PASTA integrates risk assessment with system architecture and attack surface to help in prioritising the attack surfaces that could impact safety, uptime, and data integrity, whereas STRIDE has distinct advantages, being easy to apply during system design and architecture phases. Stride also highlights threats

special to device authentication and data integrity. It is recommended in literature to primarily start with STRIDE for system level threat categorization and followed by PASTA for more detailed business driven attack simulation. In article [69], the authors also compared various threat modeling methodologies and concluded that the STRIDE method is the one used by the vast majority of people.

3.2 Proposed Methodology

In Fig. 3.1, an attack path analysis work has been proposed that takes into account the position of the IIoT device from various viewpoints that are applicable in terms of the device's vulnerabilities towards threats from both physical and cyber perspectives. This procedure is further categorised into the following steps:

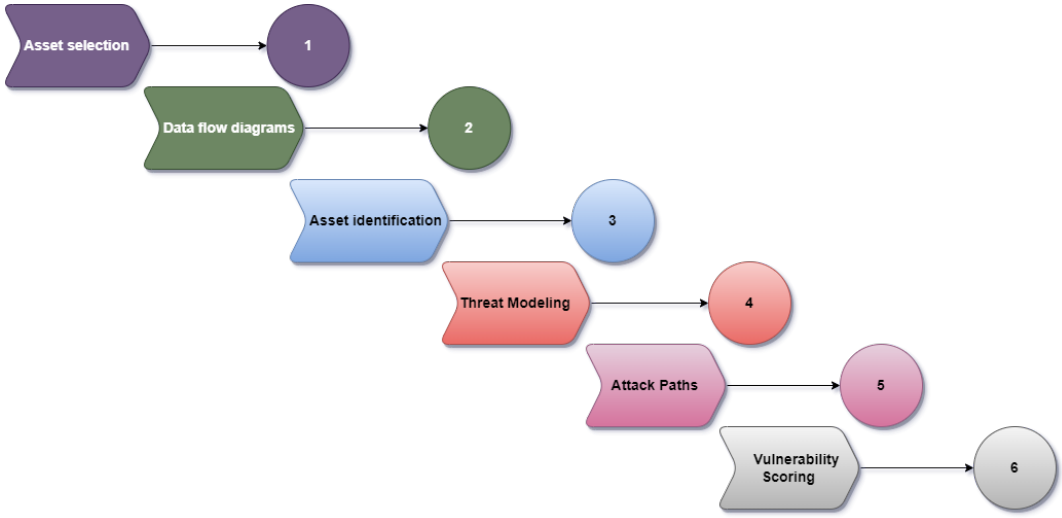


Figure 3.1: Proposed Methodology

3.2.1 Architectural Setup & Asset Selection

The first step in the threat modeling approach is setting up an architectural setup for a selected set of industries and by setting up the right architectural setup and analyzing potential attack vectors, one can greatly reduce the risk of any potential threats. In the proposed approach, this is done with the help of the Purdue model. The Purdue Model fits better when compared with other popular architectural setup methodologies in Table I. All popular models share similarities with the Purdue Model in that they define a hierarchical structure for organizing enterprise information systems. However, they differ in their focus and level of detail, depending on the specific needs of their respective industries. While the Purdue Model is specific to the manufacturing industry, other industries have developed similar reference architecture models to organize and standardize their enterprise information systems.

When it comes to whether or not the Purdue model is still accurate and relevant today, that depends on the assets being used. Although it may not be an exact fit for an OT blueprint in an IIoT environment, a three-component architecture may be needed instead of sensors and devices, field/cloud gateways, and a services backend, it's still a great place to start for securing an OT network [138].

3.2.2 Data Flow Diagrams

To assess the threats to an asset, its entries, events, and boundaries could be recognized in the system. It could be achieved by building data flow diagrams (DFDs) that map out the flow of information in the system. It uses varied symbols and routes between each destination. Generally, STRIDE evaluates the system's detailed design, and an accurate DFD shows how successful our STRIDE modeling can be, which is done in the next step. Each connection

between entities represents the flow of data, and based on this connection, the threats associated with it are also considered.

3.2.3 Asset Identification

The process of asset identification in IIoT, as depicted in the Data Flow Diagram (DFD), entails the identification of various assets that participate in the data flow within the IIoT system. Assessing the potential security risks associated with each asset and implementing appropriate measures to secure them is a critical step. To gain insights into the assets present in the Industrial Internet of Things (IIoT) through the utilization of Data Flow Diagrams (DFD), a comprehensive analysis of the DFD is imperative. The comprehensive analysis should encompass the identification of the discrete processes, data stores, and external entities that are involved in the transfer of data. Each and every constituent within a given context can be regarded as a valuable asset, warranting a comprehensive evaluation of the associated risks. In the context of IIoT, the identification of assets via DFD assumes a pivotal role in ensuring the security of IIoT systems. By engaging in the systematic procedure of risk identification and subsequent evaluation for every asset within the IIoT system, it becomes feasible to implement appropriate measures aimed at mitigating the identified risks. This proactive approach serves to bolster the overall security of the IIoT system, thereby minimizing potential vulnerabilities and enhancing its resilience. The identification of assets from a Data Flow Diagram (DFD) entails a systematic approach that can be undertaken by following a series of recommended steps. These steps serve as a guideline to ensure a comprehensive and accurate identification process:

1. To begin, examine the system's context-level diagram to determine the external actors in the scenario. They could be individuals, other systems,

or even whole corporations.

2. Find out what steps are taken inside the system. Each process is a representation of a unique system function.
3. Follow the trail of information as it moves from one procedure to another and beyond. This will aid in determining the nature of the information being sent and the appropriate security measures to take.
4. Look for data stores within the system. These are places where data is stored and can be considered assets.
5. Identify any connections or interfaces between the system and other systems or networks. These connections could be potential attack vectors and should be secured appropriately.

By following these steps, one should be able to identify the assets within a system and take appropriate steps to protect them.

3.2.4 Threat Modeling

STRIDE is a popular threat modeling paradigm that is an acronym for the various threat categories it encompasses: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege. STRIDE is considered the most mature threat modeling methodology. The reasoning behind choosing Stride in place of other alternatives is due to its low false positive rate and comprehensive aggregation of all possible attacks (12 methods). The downside of using the STRIDE method is that a large number of modeled threats are obtained. This is later taken care of by threat rating, which facilitates the prioritization of the attack vectors with a higher rating. Table 3.1 shows what each attack category means.

"Microsoft Threat Modeling Tool" has been used to identify potential security threats to software applications, systems, or networks. It is used to

Table 3.1: Elements of STRIDE

Threat	Threat Definition
Spoofing Identity	Disguising the real identity to appear as trusted source
Tampering with data	Modifying data without permission
Repudiation	Denying taking part in a transaction falsely
Information Disclosure	Revealing sensitive data to unauthorized entities
Denial of Service	Denying access to resource or data
Elevation of privilege	Gaining unauthorised access to elevated rights

design, analyze, and refine security models for applications and industrial networks. This tool uses a systematic approach (STRIDE) to model threats, which has common categories of threats that can be used to analyze and mitigate security risks. Using the Microsoft Threat Modeling Tool, DFD is created, and this tool in turn, models and generates a report of potential security vulnerabilities[57].

3.2.5 Using Attack Vectors to create Attack Paths

Attack paths are a useful way to diagrammatically model potential attack vectors. It depicts attacks on a system in path form. The root of the path is represented by a goal for the attack, and its branches are ways to achieve that goal. When the threat analysis of a system is performed, it has several threat-based goals that are represented by different attack paths[112]. To create accurate and effective attack paths for ICS environments, the use of taxonomies and ATT&CK characteristics defined by MITRE[159]have been employed. ATT&CK by MITRE is primarily a knowledge library of adversarial strategies as well as a breakdown and classification of offensively focused activities that may be utilized against multiple systems. It focuses

more on how attackers interact with systems throughout an operation than on the tools and viruses they deploy.

ATT&CK puts these strategies together into a set of tactics to help explain and put the technique in its proper context. Tactics are important contextual categories for particular approaches since they encompass conventional, higher-level notations for what attackers do during an operation, such as persist, find information, move laterally, execute files, and exfiltrate data. The ATT&CK Matrix represented in Table 3.2 and Table 3.3 illustrates the link between tactics and methods. For example, the strategy of persistence (the adversary's purpose to persist in the target environment) includes several approaches such as AppInit DLLs, New Services, and Scheduled Tasks. Each of them is a single tactic that opponents may employ to attain the aim of persistence.

As it provides a matrix with multiple possible actions that can be taken by the attacker at each stage in the attack process, enabling us to assess a challenging multi-level attack. It is of particular interest to us as it is the latest standard for industrial control systems and has been proven to be practical in real-world attacks. Attack paths are easy to adopt and understand when the system and its security concerns are well understood. In the past few years, this method has often been used in combination with other threat modeling techniques like STRIDE, which presents a much better understanding of the modeled threats.

3.2.6 Vulnerability Scoring

The vulnerability score measurement is done using the Common Vulnerability Scoring System (CVSS) at step 6 of the process. Since scoring with CVSS requires inputs regarding the specifics of attack vectors and therefore

it is positioned at the end of the process.

This score acts as a prioritisation mechanism for security teams in the organisation. This information is also crucial for Chief Information Security Officers (CISOs) and Chief Security Officers so that they can be equipped to mitigate the threats assessed through the process.

The CVSS is a framework for assessing and communicating the severity of security vulnerabilities. CVSS provides a score that helps organisations prioritise and address vulnerabilities. CVSS metrics are divided into three groups: Base, Temporal, and Environmental [23].

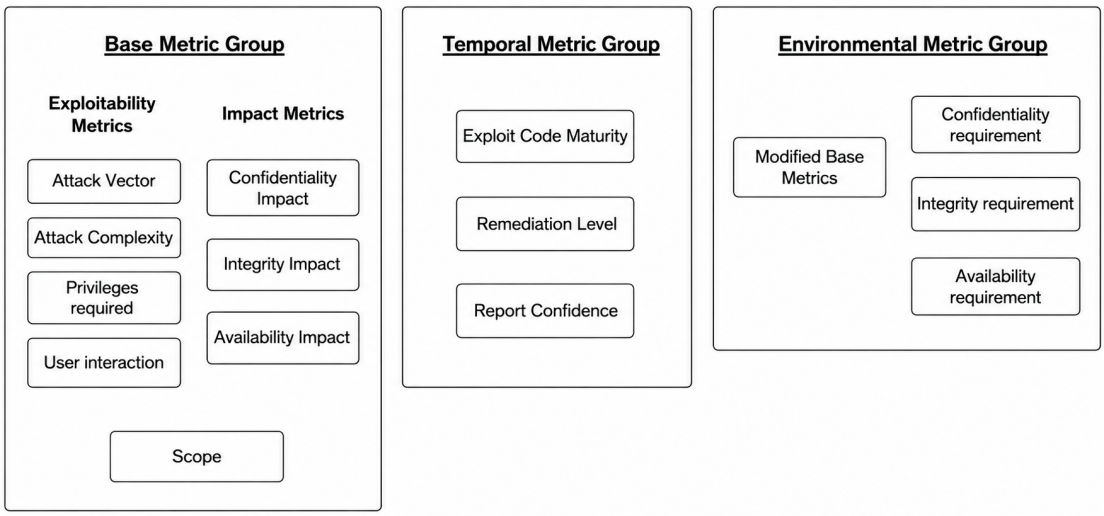


Figure 3.2: CVSS Metrics

3.2.7 Base Metrics

These metrics capture the intrinsic qualities of a vulnerability and do not change over time or in different environments. They are used as the foundation for calculating the overall CVSS score. Generally, base metrics include:

1. **Access Vector (AV):** Describes how the vulnerability is accessed. It can be Local, Adjacent Network, or Network.

2. **Access Complexity (AC):** Reflects the level of complexity required to exploit the vulnerability. It can be Low, Medium, or High.
3. **Authentication (Au):** Indicates the level of authentication required to exploit the vulnerability. It can be None, Single, or Multiple.
4. **Confidentiality (C), Integrity (I), and Availability (A):** Represent the impact of the vulnerability on these three security attributes. They can be None(0), Partial(0.5), or Complete(1.0).
5. **Exploitability (E):** Measures how likely it is that the vulnerability will be exploited. It can be Unproven(0.85), Proof-of-Concept(0.9), Functional(0.95), or High(1.0).
6. **Impact (I):** Reflects the overall impact on the system if the vulnerability is exploited. It can be None(0), Low(0.22), Medium(0.56), High(0.7), or Critical(0.85).

AV = Network, AC = Low, Au = Single, C = Complete, I = Complete, A = Complete, E = Functional, I = High

3.2.8 Temporal Metrics

These metrics reflect the characteristics of a vulnerability that change over time but are not specific to a particular environment. They provide a dynamic view of the vulnerability risk. Generally, temporal metrics include:

1. **Exploit Code Maturity (E):** Indicates the current state of exploit development. It can be Not Defined(0.0), Unproven(0.9), Proof-of-Concept(0.95), Functional(1.0), or High(1.0).
2. **Remediation Level (RL):** Represents the official response to the vulnerability. It can be Official Fix(0.0), Temporary Fix(0.25), Workaround(0.75), or Unavailable(1.0).

3. **Report Confidence (RC):** Reflects the confidence in the accuracy of the reported vulnerability information. It can be Unconfirmed(0.0), Uncorroborated(0.5), or Confirmed(1.0).

E = Functional, RL = Official Fix, RC = Confirmed

3.2.9 Environmental Metrics

These metrics consider the unique characteristics of the environment in which the vulnerability exists. They allow organizations to tailor the CVSS score to their specific circumstances. Generally, environmental metrics include:

1. **Collateral Damage Potential (CDP):** Measures the potential impact on systems beyond the one directly affected. It can be None(0.0), Low(0.1), Low-Medium(0.3), Medium-High(0.4), or High(0.5).
2. **Target Distribution (TD):** Reflects the proportion of systems in the environment that are affected. It can be None(0.0), Low(0.25), Medium(0.75), or High(1.0).
3. **Confidentiality Requirement (CR), Integrity Requirement (IR), and Availability Requirement (AR):** Represent the importance of these security attributes in the environment. They can be Not Defined(0.0), Low(0.5), Medium(1.0), or High(1.51).

CDP = Low, TD = Medium, CR = High, IR = Medium, AR = Low

To calculate the overall CVSS score, the following formula could be used:

$$\text{CVSS Score} = \left(\frac{(1 - (1 - C) \cdot (1 - I) \cdot (1 - A)) \cdot (E \cdot RL \cdot RC)}{(0.6 \cdot (1 - C) + 0.4 \cdot (1 - CDP)) \cdot \text{Impact} + 0.6 \cdot CDP} \right) \cdot \text{Exploitability} \tag{3.1}$$

3.3 Case Study: Threat Modeling of an IIoT System

In this work, two case studies have been taken for the implementation of the proposed model, one being the Internet of Production (IoP) [1][108] like power industry and the other being the Internet of Manufacturing (IoM)[2] like manufacturing plants. The advantage of taking a generalised approach is that it gives you the flexibility to apply the framework to several different industries.

3.3.1 Case Study I: Electric Power Systems as an Internet of Production (IoP)

An electric power system refers to a complex network of electrical components that are designed to facilitate the generation, transmission, and distribution of electricity for various applications. The power system, exemplified by the electrical grid, serves as a means of distributing electricity to both residential and commercial consumers across a vast geographical area. The electrical grid comprises three primary components, namely the power generators responsible for electricity production, the transmission system that facilitates the transportation of power from the production centres to the load centres, and the distribution system that disseminates the power to the neighbouring households and businesses.

The electric power industry is widely acknowledged as a crucial sector, given that its impairment or annihilation would result in a severe impact on security, national economic stability, public health, and safety, or a combination

[1]The definition of production is "the process through which raw materials are transformed into completed goods"[18]

[2]What differentiates "Production" from "Manufacturing" and vice versa, the Manufacturing is defined as a sequence of interrelated activities and processes comprising the design, material selection, planning, production, quality assurance, management, and marketing of discrete consumer and durable goods[18]

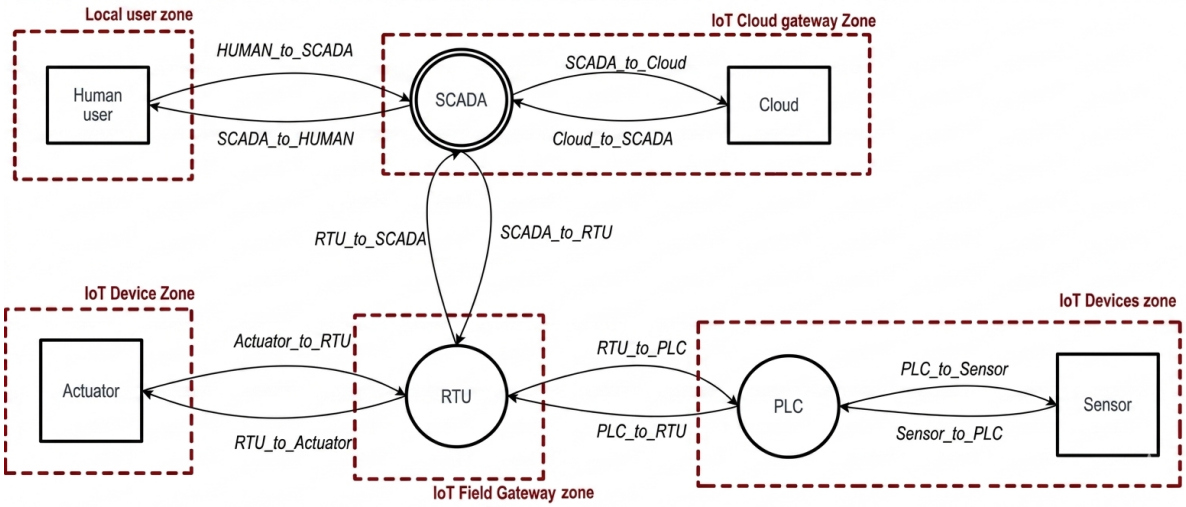


Figure 3.3: DFD for SCADA in Electric power system

of these factors. Consequently, the process of threat modeling holds significant importance for the aforementioned industry as it aids in the identification of potential threats aimed at a specific target and provides a rationale for the implementation of security measures. And, for this study, the industry standards of the electric power industry have been considered from [137].

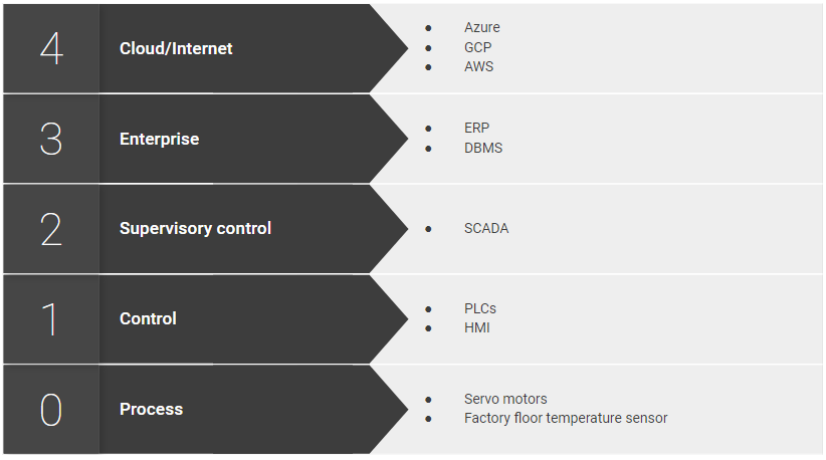


Figure 3.4: Purdue model for Electric power systems

Threat modeling: The first step in threat modeling is architecture setup using the Purdue model. Fig. 3.4 shows the Purdue model for electric power

systems. For this analysis, the Supervisory Control and Data Acquisition (SCADA) systems have been chosen. SCADA is placed in the grid control centre, which has under its control many remote substations and gathers data from them using the Remote Terminal Unit (RTU). SCADA enables human managers to operate these substations remotely. It also compiles the data being gathered in real-time to generate backlogs and provide insights about the safety of devices.

Fig. 3.3 shows the data flow diagram for a SCADA device in the electric power system. Using the data flows, the possible attack vectors were derived using the STRIDE modeling technique with Microsoft Threat Modeling Tool, through which some modeled attacks and threat counts per asset could be found mentioned in DFD[3]. The devices denoted with circles are considered IoT gateways, meaning they interface with multiple IoT devices to gather or pass data, and rectangles represent IoT devices. Table 3.4 gives a summary of the number of modeled attacks on electrical power systems.

Table 3.4: Threat summary for Electric power systems

Category	Number of modeled attacks
Denial Of Service	20
Elevation Of Privilege	45
Information Disclosure	12
Repudiation	22
Spoofing	26
Tampering	47

After gathering all the attacks, attack paths could be drawn using the MITRE[159] notation. Attack paths enable us to understand the flow of the attack from origin to destination and the CWEs exploited in the process. Fig. 3.5 shows the attack paths for 3 such attacks modeled for SCADA.

[3]Computation using the Microsoft Threat Modelling Tool (MTMT) was done on a computer running Windows 10 Pro with an Intel 9300H CPU and an NVIDIA GTX 1650 GPU.

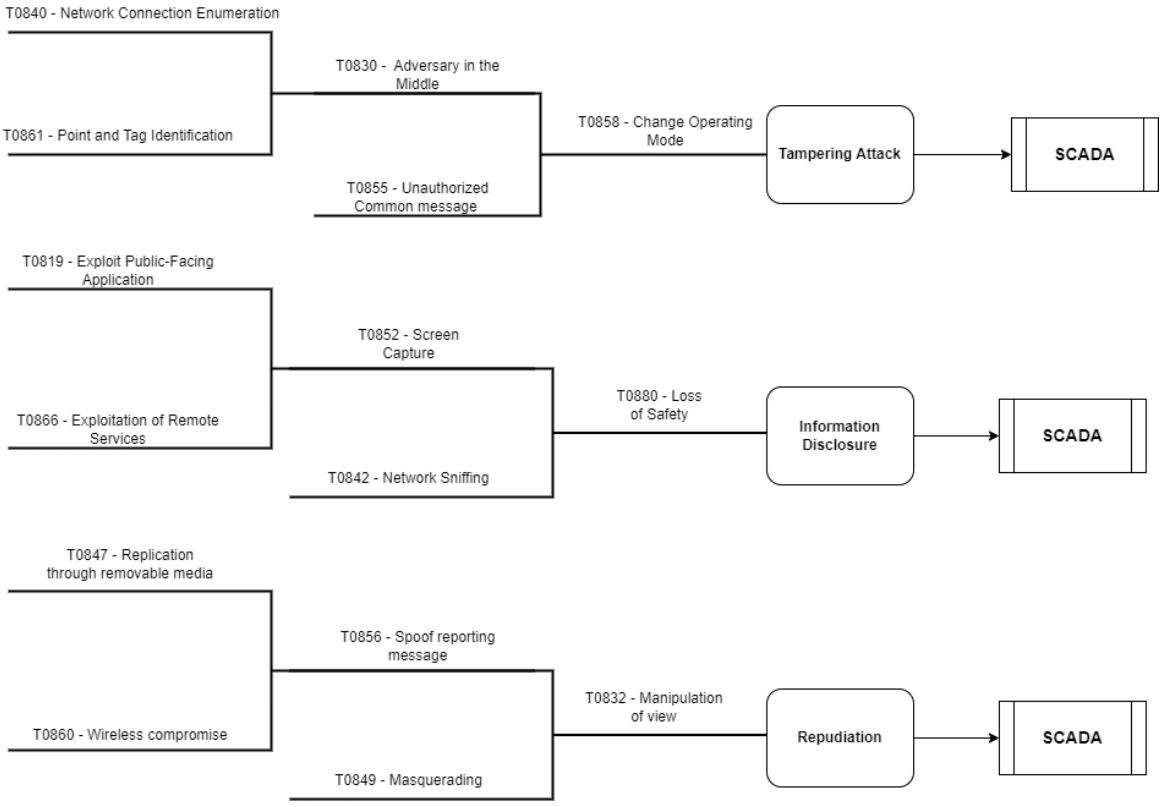


Figure 3.5: Attack paths for SCADA in Electric power system

3.3.2 Case Study II: Manufacturing Industry (IoM)

The manufacturing industry comprises establishments engaged in the mechanical, physical, or chemical transformation of materials, substances, or components into new products. Examples of manufacturing companies include automotive companies, petroleum-based companies, food production, and processing companies etc.

The manufacturing industry is considered a critical industry as its incapacitation or destruction would have a debilitating effect on security, national economic security, national public health or safety, or any combination thereof. As a result, threat modeling is critical in this industry for identifying a list of threats against a target and justifying security efforts.

In this section, for our security analysis, human resources such as operators, contractors, and engineers have been excluded, as the scope of our study, did not cover how human behavior impacts the operational security of a smart manufacturing system (Fig. 3.7). This outlines how the factory floor network is segregated from other networks such as the Internet and enterprise network by firewalls. In this configuration, the red nodes represent potential entry points that an attacker could exploit [56].

Threat modeling: The first step in threat modeling is asset selection using the Purdue model. Fig. 3.9 shows the Purdue model for the manufacturing industry. The Human Machine Interface (HMI) has been used for this analysis. HMI is a software and hardware combination that allows the operator to interact with systems and machines. The HMI system is unavoidable; in most manufacturing industries, the automation system would be incomplete without an HMI.

An HMI allows an operator to monitor the floor's activity in real-time. If there is a problem, the HMI will notify the operator. In essence, the HMI

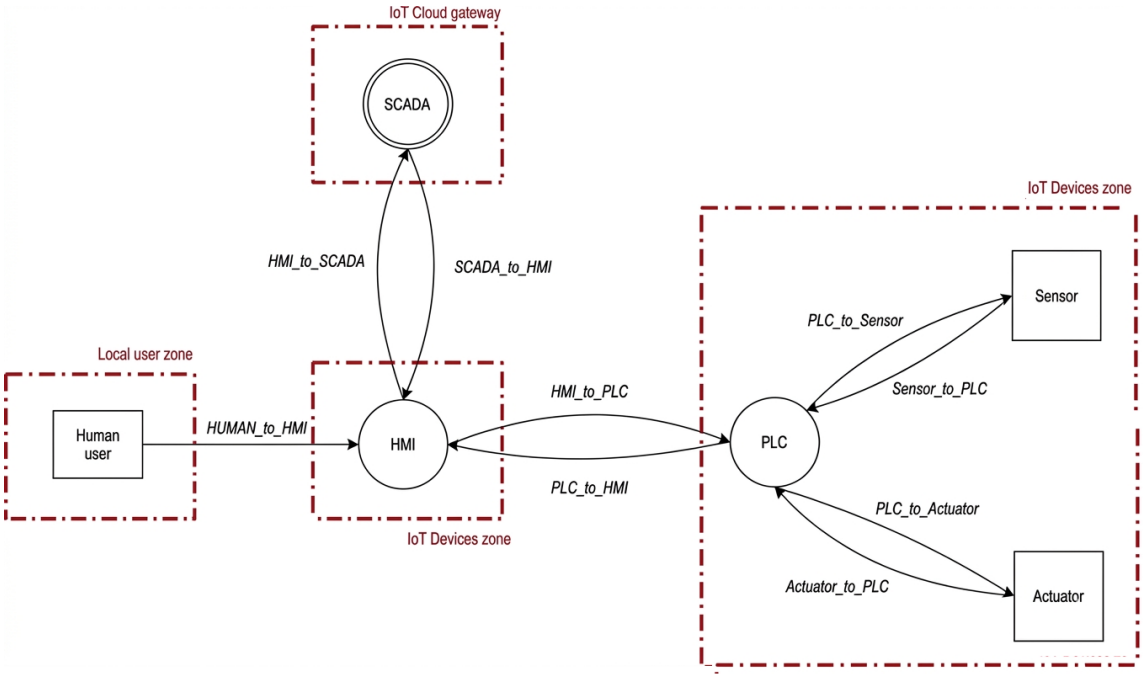


Figure 3.6: DFD for HMI in a Manufacturing Industry

serves as the user interface in a manufacturing or process control system. The HMI also provides a graphical representation of the industrial control and monitoring system.

Fig. 3.6 shows the data flow diagram for an HMI device in the manufacturing industry. Similar to Electric power system case study, by using the data flows, all possible attack vectors could be drawn using the STRIDE modeling technique through the Microsoft Threat Modelling Tool. The devices denoted with circles are considered IoT gateways, meaning they interface with multiple IoT devices to gather or pass data, and rectangles represent IoT devices. Table 3.5 gives a summary of the number of modeled attacks on the manufacturing industry.

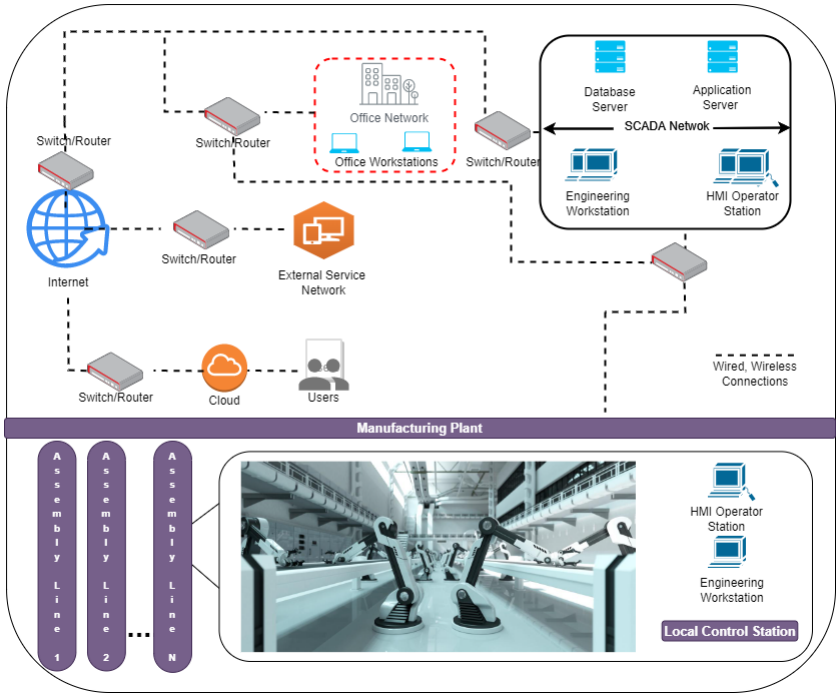


Figure 3.7: The Architecture of A Manufacturing Plant

Table 3.5: Threat summary for Manufacturing Industry

Category	Number of modeled attacks
Denial Of Service	10
Elevation Of Privilege	32
Information Disclosure	9
Repudiation	9
Spoofing	11
Tampering	39

After gathering all the attacks, attack paths could be drawn using the MITRE [159] notation. Attack paths enable us to understand the flow of the attack from origin to destination and the CWEs exploited in the process. Fig. 3.8 shows the attack paths for 3 such attacks modeled for HMI.

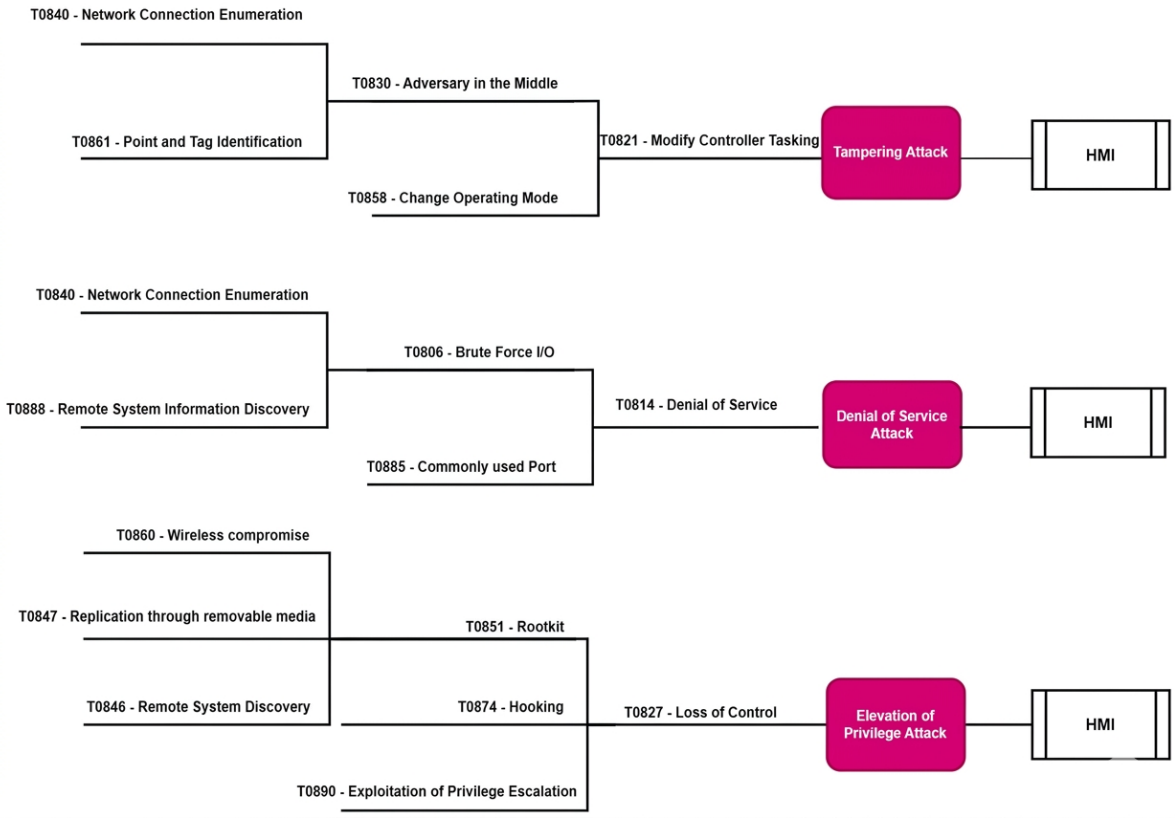


Figure 3.8: Attack paths for HMI in a Manufacturing Industry

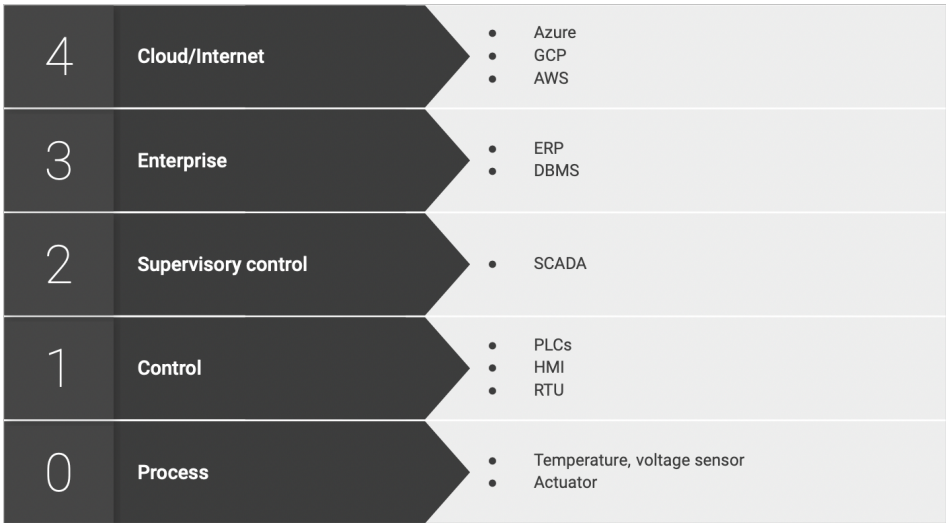


Figure 3.9: Purdue model for a Manufacturing Industry

3.4 Identification of Key Assets with Predominant Threats

As discussed above, the CVSS is an industry-standard scoring system for detected vulnerabilities. In the below case studies, the same standard from the National Vulnerability Database (NVD) has been utilized to quantitatively rank the vulnerabilities in the order in which they need to be prioritized.

Case Study I: Electric Power Systems as an Internet of Production(IoP): The number of attacks for each device is calculated and tabulated in Table 3.6.

Table 3.6: Threat count per asset

Asset	Threat count
SCADA	41
Cloud	13
PLC	36
RTU	57
Actuator	12
Sensor	9

CVSS scores can be calculated using the available CVSS calculator. This calculation must be precise and done by experts. So, to get the reality of CVSS scores, consider the example of Schneider Electric, whose vulnerabil-

ities are scored and are available at the National Vulnerability Database [102]. Schneider Electric SE is a European corporation that has expertise in the context of energy management. Its focus is on providing solutions for a variety of sectors, including homes, buildings, data centres, infrastructure, and industries. Table 3.7 shows the top 5 threats identified by using this method from the National Vulnerability Database.

Table 3.7: Top 5 threats

Category	Interaction	CVSS Score
Elevation Of Privilege	plc_to_rtu	9.8
Spoofing	human_to_plc	9.1
Spoofing	human_to_scada	8.1
Tampering	human_to_scada	7.5
Denial Of Service	rtu_to_plc	7.5

Case Study II: Manufacturing Industry (IoM): Similar to the first case study, here also the same standard from the National Vulnerability Database (NVD) has been utilised to quantitatively rank the vulnerabilities in the order in which they need to be prioritised. The number of attacks for each device is calculated and tabulated in Table 3.8.

Table 3.8: Threat count per asset

Asset	Threat count
PLC	47
SCADA	11
HMI	34
Actuator	9
Sensor	9

CVSS scores can be calculated using the available CVSS calculator. This calculation must be precise and done by experts. So, to get the reality of CVSS scores, the example of Johnson & Johnson is considered, whose vulnerabilities are scored and are available at NVD. Johnson & Johnson is a large American multinational corporation that specializes in the development, manufac-

turing, and distribution of a broad range of healthcare products. Johnson & Johnson's products include medical devices, pharmaceuticals, and consumer healthcare products such as Band-Aids, Tylenol, and Neutrogena. Table 3.9 shows the top 5 threats identified by using this method from the National Vulnerability Database [102].

Table 3.9: Top 5 threats

Category	Interaction	CVSS Score
Spoofing	sensor_to_plc	9.8
Tampering	plc_to_sensor	9.8
Spoofing	actuator_to_plc	7.5
Information Disclosure	plc_to_hmi	7.5
Spoofing	plc_to_hmi	4.8

3.5 Chapter Summary

This chapter presents a structured, industry-agnostic framework for threat modeling and attack path analysis in Industrial IoT systems to enhance the security of ICPS in smart factories. Using standardised methods such as STRIDE and CVSS, the six-step approach addresses the gap in existing literature by offering a clear, adaptable process for identifying, assessing, and prioritising cyber threats. Two case studies, industrial manufacturing and power sectors, demonstrate the method's broad applicability and effectiveness in guiding cybersecurity decisions and investments across diverse industrial domains.

Table 3.2: Tactics and Techniques Representing The MITRE ATT&CK® Matrix for Industrial Control Systems (ICSs) [159]

Initial Access	**Execution**	**Persistence**	**Privilege Escalation**	**Evasion**	**Discovery**	**Lateral Movement**	**Collection**	**Command and Control**	**Inhibit Response Function**	**Impair Process Control**	**Impact**
Drive-by Compromise	Change Operating Mode	Hardcoded Credentials	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Adversary in the Middle	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Exploit Public-Facing Application	Command-Line Interface	Modify Program	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Automated Collection	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Exploitation of Remote Services	Execution through API	Module Firmware		Indicator Removal on Host	Remote System Discovery	Hardcoded Credentials	Data from Info Repositories	Standard App Layer Protocol	Block Command Message	Module Firmware	Denial of View
External Remote Services	GUI	Project File Infection		Masquerading	Remote Sys Info Discovery	Lateral Tool Transfer	Data from Local System		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Internet Accessible Device	Hooking	System Firmware		Rootkit	Wireless Sniffing	Program Download	Detect Operating Mode		Block Serial COM	Unauthorized Cmd Message	Loss of Control
Remote Services	Modify Controller Tasking	Valid Accounts		Spoof Reporting Message		Remote Services	I/O Image		Change Credential		Loss of Revenue
Replication via Removable Media	Native API					Valid Accounts	Monitor Process State		Data Destruction		Loss of Protection
Rogue Master	Scripting						Point & Tag ID		Denial of Service		Loss of Safety

Table 3.3: Tactics and Techniques Representing The MITRE ATT&CK® Matrix for Industrial Control Systems (ICSs) (Continued) [159]

Initial Access	**Execution**	**Persistence**	**Privilege Escalation**	**Evasion**	**Discovery**	**Lateral Movement**	**Collection**	**Command and Control**	**Inhibit Response Function**	**Impair Process Control**	**Impact**
Spearphishing Attachment	User Execution						Program Upload		Device Restart/Shutdown		Loss of View
Supply Chain Compromise							Screen Capture		Manipulate I/O Image		Manipulation of Control
Transient Cyber Asset							Wireless Sniffing		Modify Alarm Settings		Manipulation of View
Wireless Compromise									Rootkit		Theft of Operational Information
									Service Stop		
									System Firmware		

Chapter 4

Detection Mechanisms for known Cyber Threats Leveraging ML Techniques

Network Intrusion Detection Systems (NIDS) are used to detect cyber attacks and malicious activities like Denial of Service (DoS), Distributed DoS (DDoS), Worms, Backdoor, etc. Network traffic is monitored, and potential threats are identified. Signature-based NIDS is good at detecting known attacks but fails at detecting attacks which have not been seen before. Anomaly-based NIDS is good at detecting new attacks

4.1 Significance of NIDS in Second Line of Defence

Intrusion Detection Systems (IDSs), Intrusion Prevention Systems (IPSs) and Intrusion Response Systems (IRSs) are referred to as defensive security systems for invasions [133]. Any software or hardware that identifies intrusions is referred as an Intrusion Detection System (IDS). Intrusion Prevention Systems (IPSs), sometimes known as inline IDS are an extension of IDS that aim to detect intrusions and then protect the system in real time. Intrusion Prevention Systems (IPS) and Intrusion Detection Systems (IDS) are worthless without Intrusion Response Systems (IRSs), which utilise pro-

tection mechanisms to evaluate system performance and even identify and address possible intrusions. Two categories of IDS are signature-based IDS and behaviour-based IDS, also known as anomaly-based attack detection systems.

4.1.1 Intrusion Detection System

Monitoring, detection and response are the three general components of a conventional IDS [64]. The traffic flow behaviour is analysed by the monitoring component. Any questionable behaviour is noticed by the detection component, which then tells the response component. The response component sends a message to the network administrator or raises an alert. Two categories of IDS are signature-based IDS, and the second is behavioral based, also known as anomaly-based attack detection systems.

Two Categories are Signature-Based IDS and the second is behavioral-based, also known as Anomaly-based Attack detection systems.

4.1.2 Signature-Based intrusion detection system

The IDS accesses a database of known vulnerabilities and fingerprints when using the signature-based technique [114]. Each intrusion assault has a fingerprint that comprises all attack's information and is used to identify and stop similar attempts in the future. The primary disadvantage of this approach is the requirement for regular database updates to classify new attacks.

4.1.3 Anomaly based intrusion detection system

The anomaly-based IDS, also known as behaviour-based, exploits baseline patterns to train from in order to identify fresh intrusion attempts. Any de-

parture from the established baseline patterns is considered as an assault, and alarms are set off [114].

4.2 ELA-NIDS: An Ensemble Learning base

In recent years, there has been a notable surge in the utilization of machine learning methodologies[88] to improve the performance of NIDS. Researchers have investigated the application of machine learning algorithms to inspect network traffic patterns and detect possible intrusions. Nevertheless, there exists a need for research in enhancing the precision and effectiveness of NIDS through the utilisation of machine learning methodologies. A promising way involves the implementation of ensemble learning, a technique that combines numerous machine learning models to attain superior detection performance of NIDS. Despite the better performance of ensemble learning approaches, further research is required to optimize ensemble techniques and develop novel ensemble architectures to build reliable NIDSs.

The lack of labeled and diverse datasets with updated attack features in Industrial Control System (ICS) environments poses a significant research challenge for training and evaluating machine learning-based network intrusion detection systems (NIDS). Addressing these gaps will lead to more robust and effective systems using ensemble machine learning approaches.

The study presents IDS [129] a NIDS using ensemble learning algorithms to detect cyber-attacks on IIoT devices. Tree-based models like Extra Trees, Random Forest, CatBoost, and XGBoost were compared with base machine learning models like Decision Tree, K-nearest neighbor, and support vector machine. The study demonstrates the effectiveness of machine learning in enhancing IIoT security, offering real-time detection capabilities and effective feature selection using tree-based models. This manuscript utilizes a

publicly available dataset containing network traffic traces from a realistic IoT network environment.

4.2.1 Proposed Methodology

After splitting the dataset into "Training Set" and "Testing Set", data transformation is the first step that deals with the class imbalance problem, which helps to increase the classification accuracy, followed by label encoding, data normalisation, data augmentation, and feature engineering to improve training efficiency. The steps involved in this methodology are explained below.

4.2.2 Dataset

Many publicly available datasets, such as ToN-IoT, UNSW- NB15, CICIDS 2017, CICIDS 2018, CICIDS2021, NSL- KDD, KDD Cup, etc., have long been utilized by researchers to create ML-based NIDS. This work utilizes the NF-ToN- IoT-v2 dataset, a NetFlow-based variation of the ToN-IoT dataset, which is a publicly accessible heterogeneous dataset containing network traffic traces from a realistic IoT network environment [10] Table 4.1 lists the popular datasets used for NIDS.

The NF-ToN-IoT-v2 dataset's success in industrial environments relies on its representation of diverse protocols, attack scenarios, traffic patterns, legacy systems, and scalability. It covers various IIoT protocols, attack types, and industry-specific traffic patterns while also incorporating legacy systems and adapting to different network architectures and data volumes.

The NF-ToN-IoT v2 dataset comprises approx16 million data records. Out of the total data records, 63.99% or approx 10 million samples are classified as attack data, whereas the remaining 36.01% or approx 6 million samples are benign.

Table 4.1: Comparison of Datasets for Attack Detection in Industrial IoT

Dataset	**Description**	**Features**	**Size**	**Attacks**	**Year**	**Samples (Millions)**
NF ToN-IoT V2	Comprehensive IIoT dataset capturing realistic ICS network traffic.	43	Large	8	2021	4.5
ToN-IoT	IIoT dataset capturing network traffic for intrusion detection research.	30+	Large	5	2018	2.5
UNSW-NB15	Network traffic dataset from a controlled environment.	42	Large	9	2015	2.5
CICIDS2017	Real network traffic dataset from an enterprise environment.	80+	Large	16	2017	2.8
CICIDS2018	Network traffic dataset with synthetic attacks in an enterprise environment.	80+	Large	23	2018	2.8
CICIDS2021	Network traffic dataset capturing real and simulated attacks in an enterprise environment.	78	Large	15	2021	4.4
NSL-KDD	Network traffic dataset derived from the original KDD Cup 1999 dataset.	41	Moderate	4	2009	1.5
KDD Cup	Network traffic dataset for intrusion detection in computer networks.	41	Large	22	1999	4.9

4.2.3 Data Preprocessing

Effective data preprocessing is crucial for enhancing the accuracy, efficiency, and robustness of machine learning models, particularly in network traffic analysis. This section outlines the key preprocessing steps employed, including data transformation, normalization, augmentation, and feature engineering.

4.2.4 Data transformation

Data transformation helps reduce computational complexity and prepares raw input into a more suitable format for model training.

4.2.5 Data transformation using clustering

Training machine learning models on network traffic datasets is a challenging task that demands substantial time and computational resources[169]. The process of selecting a smaller subset of data from a larger dataset, known as sampling, is a commonly employed technique to reduce the complexity of model training [169]. The proposed methodology utilizes a clustering-based approach employing the k-means algorithm to extract a subset of data that exhibits high representativeness. Cluster sampling is a frequently employed method for the purpose of selecting a subset of data for analysis[169].

This technique involves grouping the initial data points into multiple clusters and then randomly selecting a proportion of data from each cluster to generate a representative subset. K-means is a commonly used unsupervised learning algorithm in machine learning, which involves clustering a group of data points based on their similarities [99]. The k-means algorithm initiates by randomly selecting K centroids and assigning each data point to the centroid nearer to it. The centroids are then shifted to the mean of all the data

points assigned to that particular cluster. This procedure is repeated until either the centroids stop moving or a predefined maximum number of iterations is reached. The primary goal of the k-means algorithm is to reduce the total sum of squared distances between each data point and its corresponding cluster centroid, denoted by [169]:

$$\sum_{i=0}^{n_k} \min_{u_j \in C_k} (\mathbf{x}_i - u_j)^2$$

where $(\mathbf{x}_1, \cdots, \mathbf{x}_n)$ is the data matrix; The centroid or mean of the cluster C_K is denoted by u_j and is computed as the average of all samples in C_k where n_k is the total number of sample points in C_k. The time complexity of k-means is $O(nkt)$, where n is the size of the data, k is the no. of clusters, and t is the total iterations [99].

To divide the data points into k clusters based on their distance, a k-means clustering method is used. The distance is computed using Manhattan Euclidean or Mahalanobis metrics. The process involves applying k-means clustering to the original dataset to group the data samples into k clusters, followed by random sampling of a certain percentage of data from each cluster to create a representative subset. The percentage of data selected for sampling depends on the resources available.

4.2.6 Label Encoding

In machine learning, encoding is a technique that is used to transform categorical data into numerical data. This is done to enable algorithms to better interpret and process the data. Since numerous machine learning algorithms only accept numerical input data, categorical data must be transformed for the algorithms to make predictions. Label encoding is one such technique,

where each distinct category is assigned a numerical label, and each label denotes a unique category. The labels are allocated in an arbitrary fashion with no inherent order or ranking between them.

4.2.7 Data Normalization

Network traffic data usually consists of features with significantly varying ranges. To enhance the performance of machine learning models trained on such data, it is recommended to normalize the dataset [6]. If the feature scales are largely different and the dataset is not normalized, a biased model may result, prioritizing the significance of large-scale features. To address this, the Z-score technique is used to normalize the features. This technique transforms the data into a distribution with a mean of 0 and a standard deviation of 1 by subtracting the mean value of each feature from its respective data points and dividing it by the feature's standard deviation. This process guarantees that the data is on a uniform scale and can be used effectively in machine learning models. When using the Z-score method to normalize features, each normalized feature value, denoted by x_n, is determined as follows:

$$x_n = \frac{x - \mu}{\sigma} \tag{4.1}$$

where x is the value of feature taken, μ is the mean and σ is the standard deviation respectively [169].

4.2.8 Data Augmentation

Imbalanced classes are a common problem in network traffic data, which can lead to biased models [22]. Resampling methods are often used to address this issue. One such technique is SMOTETomek, which is a hybrid approach

that combines two sampling techniques, SMOTE and Tomek links, to balance the minority and majority classes in the dataset. SMOTE is used to oversample the minority class by creating synthetic samples that are similar to existing samples in that class, thereby avoiding a bias towards the majority class. On the other hand, Tomek links are pairs of samples in different classes that are close to each other but are classified differently. Removing these pairs can help to improve the separation between the classes and reduce the overlap. The SMOTETomek technique first oversamples the minority class using SMOTE and then removes majority class samples identified as Tomek links. This approach helps to improve the balance between the classes and reduces the risk of overfitting or underfitting the minority class. The impact of this is improved class balance, reduced overfitting, and better class separation.

4.2.9 Feature Engineering

For training, Information Gain (IG), a feature selection method, is used in machine learning models to remove irrelevant features while retaining essential ones. IG measures the amount of information a feature can bring to the target variable by calculating changes in entropy. The proposed system uses IG due to its fast speed and low computational complexity. Mutual Information, also known as information gain, evaluates the statistical dependence between two variables by measuring the reduction in uncertainty. A higher score indicates a stronger association between the variables, always equal to or greater than zero. The Information Gain value associated with a particular feature X and target variable T is represented by $IG(T|X)$.

Information Gain value using feature X denoted by [169]:

$$IG(T \mid X) = H(T) - H(T \mid X), \tag{4.2}$$

Table 4.2: Comparison of the Proposed NIDS Performance Metrics with Recent Works

	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
ELA-NIDS	NF-ToN-IoT	96.01	96.51	96.10	96.32
	NSL-KDD	94.30	94.10	94.50	94.30
	CICIDS2017	95.60	95.20	95.90	95.50
Thi-Thu-Huong Le et al. (2022)[121]	NF-ToN-IoT	99.00	-	-	99.00
Hussain et al. (2023)[10]	NF-ToN-IoT	˜97.00	-	-	˜97.00
	CICIDS2017	˜96-98	-	-	˜96-98

where $H(T)$ is entropy T, $H(T|X)$ is conditional entropy of T given X. If the value of $IG(T|X)$ is higher than the value of $IG(T|Y)$, then feature X is considered more significant to the target T than feature Y [169].

4.2.10 Model Selection Criteria:

The system's proposed methodology involves the selection of tree-based Machine Learning algorithms which include Random forest(RF), Extra Trees(ET), Extreme gradient boosting (XGBoost), and CatBoost. Apart from the proposed algorithms, performance evaluations are also being performed using the common classification models such as Decision Tree, K-nearest neighbor (KNN), and support vector machine (SVM).

The reasons for selecting these algorithms are outlined as follows [172]:

1. Ensemble algorithms are significantly more efficient for non-linear and complex datasets, such as those involving network traffic, which makes them more effective than traditional machine learning models like linear models.
2. These models determine feature importance during training, aiding in the feature engineering process.
3. Tree-based algorithms utilize a random construction methodology, which leads to the formation of diverse decision trees. This diversity contributes to the development of a robust ensemble model that can generalize more effectively compared to other machine learning methods.

In order to enhance the performance of the proposed model, it is essential to identify the most effective hyper parameters. A grid search was utilized to determine the optimal hyper parameters [147] for the proposed models. This approach involves establishing a range of hyperparameters organized in the form of a grid, which encapsulates all potential combinations of these hyperparameters. Next, for each combination within the grid, the classifier is trained and evaluated using cross-validation. The best hyperparameters for the model are chosen based on which combination yields the highest score from cross-validation [148]. This method provides advantages in pinpointing the ideal hyperparameters for the classifier, which in turn improves the model's accuracy and generalizability. Additionally, it simplifies the hyperparameter tuning process, a task that can be laborious and complex when handled manually.

4.2.11 Results and Performance Analysis

To evaluate the performance of the models, 5-fold cross-validation is also used to measure reliable accuracy, precision, recall, and F1 scores. Results tabulated in Table 4.3 show that the XGBoost model achieved the highest accuracy of 96%, other models such as Extra Trees, Random Forest, and CatBoost also performed well, achieving an accuracy of 95%, 94 %, and 93% respectively. Apart from the ensemble models, other models such as DT, SVM, and KNN also produced good results with an accuracy of 94%, 94%, and 93% respectively. For XGBoost, the multi-classification results are also tabulated in Table 4.4. It is visible that the model is good at detecting all attacks but struggles with Dos attack detection.

The proposed study is also performed on different datasets. The results with their corresponding datasets NSL-KDD and CICIDS2017 are listed in

4.2 with two other recent works by other researchers. The proposed work achieves competitive metrics across datasets with a simpler XGBoost approach. Despite other studies having slightly better accuracy, the work also contributes by enhancing the generalizability and scalability of NIDS in IIoT systems, making it highly valuable for industrial applications. A combination of higher F1 scores, balanced metrics, and diverse, real-world applicability sets the proposed research apart as a competitive and practical solution for IIoT cybersecurity. Overall, the results suggest that Random Forest and XGBoost are the most suitable models for predicting the target variable on the NF-Ton-Iot-v2 dataset. However, it is important to note that further optimisation and tuning of the models may lead to even better performance.

Table 4.3: Model performance on evaluation metrics

Model	Accuracy	Precision	Recall	F1-score
XGBoost	**96.01**	**96.51**	**96.10**	**96.32**
Extra Trees	95.23	96.56	95.23	95.80
Random Forest	95.07	96.76	95.07	95.72
CatBoost	93.77	96.35	93.77	94.78
Decision Tree	0.75	0.76	0.71	0.73
SVM	0.89	0.91	0.87	0.88
KNN	93.23	0.79	0.74	0.76

4.3 GANIBOT: A Semi-Supervised Generative Adversarial Network Approach for DDoS Attack Detection

At the beginning of 2021, organisations recorded 31% botnet activity. By 2021, it is predicted to increase to 51% [36]. Thus, there is a growing concern about detecting botnet attacks. The increase in IoT devices also increases efforts in the labeling process of the data generated all around the world. This research investigates [130] a semi-supervised learning technique using Generative Adversarial Networks (GAN) to minimise the efforts in the la-

Table 4.4: XG-Boost Multiclassification performance

S.No.	Attacks	Precision	Recall	F1- score
1	Benign	.95	.95	.95
2	Backdoor	1.00	1.00	1.00
3	DDos	.96	97	.96
4	Dos	.16	.26	.20
5	Injection	.65	0.74	0.69
6	MITM	0.96	0.92	0.93
7	Password	.90	0.96	0.94
8	Ransomware	1.00	1.00	1.00
9	Scanning	.93	0.92	0.93
10	Xss	.77	0.88	0.82
	Macro Avg.	0.83	0.86	0.84
	Weighted Avg.	0.97	0.96	0.96

beling process while successfully detecting botnet attacks. Semi-supervised learning is a machine learning technique in which a small amount of labeled data is combined with a large amount of unlabeled data to train a model. Semi-supervised learning uses additional unlabeled data to learn the features. Unsupervised learning algorithms do not require any labeled data, while supervised learning algorithms are trained on a large set of labeled data. The semi-supervised algorithms lie in between these two techniques [168].

The performance of a Semi-supervised GAN (SGAN) is approximately similar to the logistic regression [1] and other deep learning techniques[123], but it minimizes human efforts to label the dataset. The major contribution of this work is that it can successfully detect a botnet attack in IoT devices since it achieved 99.88% accuracy in binary classification. Also, in multi-class classification, the accuracy of SGAN is 59%.

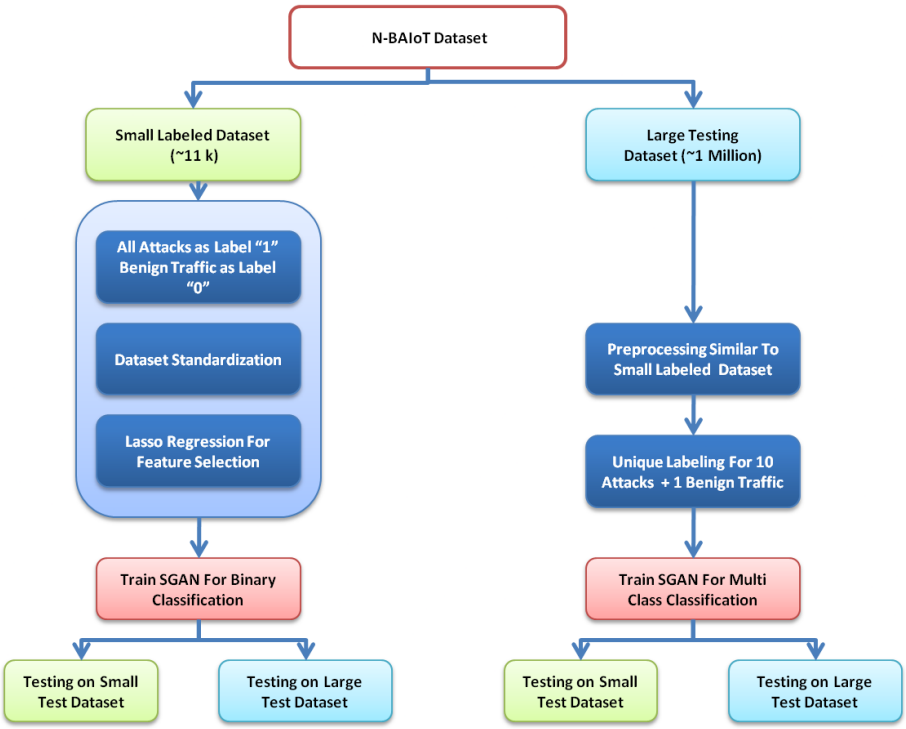


Figure 4.1: Workflow Diagram of the Proposed Methodology

4.3.1 Formulated approach

The workflow diagram of the proposed methodology of GANIBOT is shown in Fig. 4.1. Further details of the methodology are described in the subsequent subsections.

4.3.2 Dataset: N-BaIoT

N-BaIoT[89] suggests real-time traffic information of Mirai and Bashlite infected nine enterprise IoT devices. Mirai botnet and Bashlite botnet have 5 sub-attack types each (Table 4.14). Apart from these 10 class labels, there will be another class label, the benign (normal) label. N-BaIoT is multivariate and sequential data with a total of approximately 7 million. For the current study, the Danmini doorbell's data is used, which has around 1 million data samples and 115 features. 23 features are extracted at 5 time stamps (100 ms,

500ms, 1.5s, 10s and 1 min) which are listed in Table 4.18.

Table 4.5: Dataset details

Class Label	Botnet	Sub-Attack	Total Number
0	Benign	-	49548
1	Bashlite	Scan	29849
2	Bashlite	Combo	59718
3	Bashlite	Junk	29068
4	Bashlite	TCP	92141
5	Bashlite	UDP	105874
6	Mirai	Scan	107685
7	Mirai	Ack	102195
8	Mirai	Syn	122573
9	Mirai	UDP	237665
10	Mirai	Plain UDP	81982

4.3.3 Feature Selection

Most often, regularisation techniques are employed to avoid overfitting of the model and to reduce model complexity. Lasso (Least Absolute Shrinkage and Selection Operator) Regression is one of the regularised linear regression methods for feature selection [1] [98] as it can lead to zero coefficients [158].

$$\sum_{i=1}^{a} (y_{\mathrm{i}} - \hat{y}_{\mathrm{i}})^2 = \sum_{i=1}^{a} (y_{\mathrm{i}} - \sum_{j=0}^{n} (w_{\mathrm{j}} \times x_{\mathrm{ij}})^2 \tag{4.3}$$

Table 4.6: Features of N-BaIoT dataset

Aggregation by	Value	Properties	No.of Features
Source IP	Packet size (outbound)	Mean, Variance	2
Source IP	Packet count	Integer	1
Source MAC-IP	Packet size (outbound)	Mean, Variance	2
Source MAC-IP	Packet count	Integer	1
Channel	Packet size (only outbound)	Mean, Variance	2
Channel	Packet count	Integer	1
Channel	Amount of time b/w packet arrivals	Mean, Variance, Integer	3
Channel	Packet size (inbound and outbound)	Magnitude, Radius, Covariance, Correlation Coefficient	4
Socket	Packet size (outbound)	Mean, Variance	2
Socket	Packet count	Integer	1
Socket	Packet size (inbound and outbound)	Magnitude, Radius, Covariance, Correlation Coefficient	4
Total			23

Eq. 1 defines the cost function of the simplest linear model. Where "n" is the number of features and "a" is the number of data instances. Here, the cost function is altered by adding a penalty equal to the sum of the absolute values of the coefficients. Eq. 2 is the cost function after alteration. It is applied to the training dataset and comprises the two balanced classes, namely the benign class and the attack class.

$$\sum_{i=1}^{a} (y_{\mathrm{i}} - \hat{y}_{\mathrm{i}})^2 = \sum_{i=1}^{a} (y_{\mathrm{i}} - \sum_{j=0}^{n} (w_{\mathrm{j}} \times x_{\mathrm{ij}})^{\ 2} + \lambda \sum_{j=0}^{n} |w_{\mathrm{j}}| \tag{4.4}$$

The SGAN is then trained on the selected features for both binary classifica-

tion and multi-class classification to detect botnet attacks.

4.3.4 The SGAN Model

The main concern with supervised learning is the need for a large amount of training data. Also, since the IoT devices are constantly generating an immense amount of data [48], labeling them is onerous in real-world scenarios. In Semi-Supervised learning, both the unlabeled as well as labeled data are used to train the classifier.

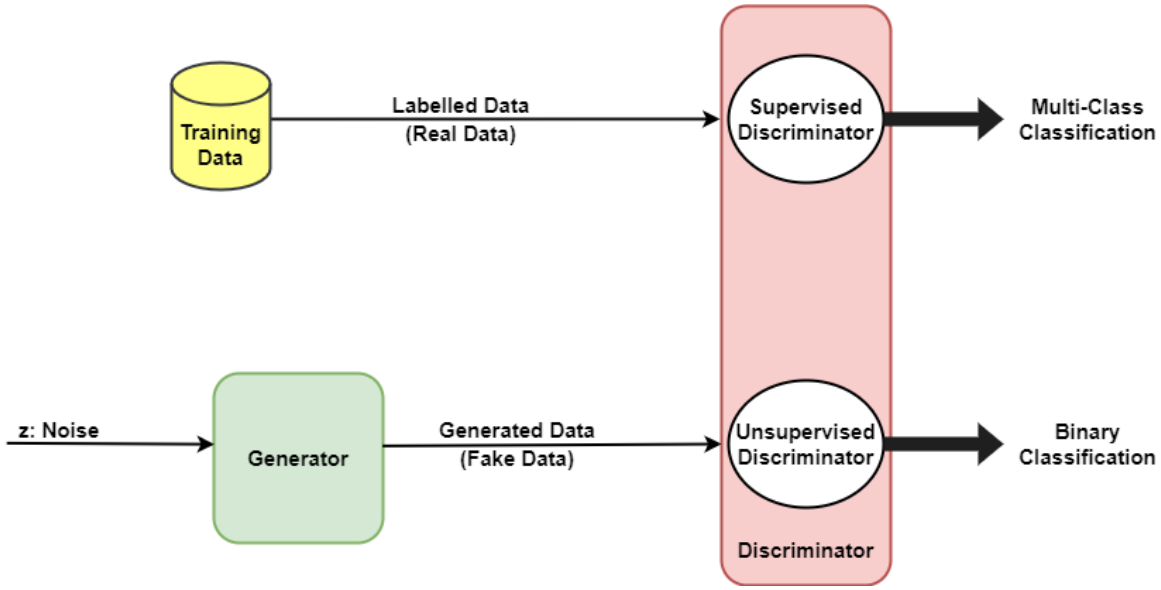


Figure 4.2: SGAN Block diagram

Semi-supervised GAN (SGAN) is a revised Generative Adversarial Network (GAN) architecture. GAN is a class of methods for learning generative models based on game theory. Two neural networks compete against one another in the form of a zero-sum game, i.e., one network's loss is the other network's gain. The goal of GAN is to train a generator model G to generate fake data as close as the real (input) data and train a discriminator model D which classifies input as either real (the user input) or fake (generated by the generator). Generator model G tries to fool the discriminator model D into

classifying the fake data as real data, while D tries to accurately classify them as real or fake. GANs can be described as a two-player min-max game with the objective function defined as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(D,G) = \mathbb{E}_{x\sim P_{data}(x)}[\log D(x)] + \mathbb{E}_{z\sim P_z(z)}[\log(1-D(G(z)))]$$

where G is the generator, D is the discriminator, P_{data} is the real data distribution, and P_z is the noise. G(z) means G tries to learn the P_G distribution from P_z.

GAN based semi-supervised models have achieved success in health-IoT [173], [174], securing edge-based IoT networks [105]. Augustus Odena, in his paper [105] showed how SGAN performs better than CNN on a small set of labeled data. He concluded that for 100 samples of each class in the MNIST handwritten digit dataset, CNN achieved 89.5% accuracy while the SGAN achieved 92.8%.

Semi-Supervised GAN (SGAN) comprises the training of a supervised discriminator (performing multi-class classification) an unsupervised discriminator (performing binary classification), and a generator model. In the SGAN, the supervised discriminator model is trained to predict N+1 classes, where N denotes the total number of classes. One extra class label is required to label the "fake" class. An unsupervised discriminator allows the model to learn more useful features from a large set of unlabeled data, while a supervised discriminator predicts the class labels. As shown in the block diagram in Fig. 3, the training dataset, i.e., the real data and the fake data generated by the generator, is fed to the discriminator.

Supervised and Unsupervised Discriminators are stacked on one another as described in [122]. The supervised discriminator model is responsible for the attack detection. Just before the softmax layer, the unsupervised model uses the output of the supervised model. The normalised sum of both the

Table 4.7: Results for Multi-class classification without feature selection

S.No	Classification Model	Samples	Train Size	Test Size	Test Acc (Small dataset)	Training Time	Test Acc (large dataset)
1	SGAN	1000	8800	2200	79%	660.9s	45%
2	ANN	1000	8800	2200	90%	202.2s	43%
3	CNN	1000	8800	2200	68%	2063.02s	41%

exponential output values mentioned by equations 3 and 4 is then computed.

$$D(x) = \frac{Z(x)}{Z(x)+1} \tag{4.5}$$

$$Z(x) = \sum_{i=0}^{n} exp[l_{\mathrm{i}}(x)]. \tag{4.6}$$

4.3.5 Results

To simulate the current study, a Google Colab GPU with Python programming language 3.7 and Keras version 2.8.0 was used. CPU specifications are Intel i5, 6th generation at 2.30 GHz clock speed and 64 64-bit processor with 8 GB of memory and Windows 10 Operating System. The Generator and Discriminator of the SGAN model are initialised using the uniform distribution with Adam optimiser with an initial learning rate of 0.0002 and beta_1 decay rate of 0.5.

Results for the Multi Class Classification without Feature Selection

For the multi-class classification purpose, the 5 sub-attack classes of Mirai Botnet, 5 sub-attack classes of Bashlite Botnet and one normal/benign class

are classified. Models are trained on 1000 samples of each class for 500 epochs. An average of five training/testing rounds is considered. From Table III, it is evident that the SGAN performed better with an average accuracy of 45%, while the ANN and CNN have 43% and 41%, respectively, on the large testing dataset ($\approx$ 1 million). The small testing dataset accuracy and training time of SGAN is in-between ANN and CNN.

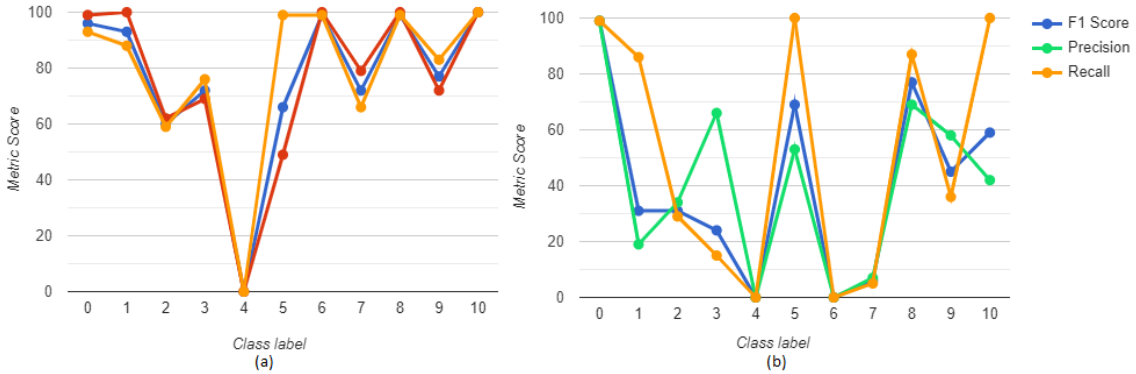


Figure 4.3: SGAN evaluation metrics for multi-class classification on small (Fig. a) and Large testing (Fig. b) data without feature selection

Fig. 4.3 shows the precision, recall and f1 score on small testing data and Fig. 4 4.3 for large testing data, respectively, for multi-class classification without feature selection in one of the testing results.

Results for the Multi Class Classification with Feature Selection:

For the feature selection part, LASSO is used on the small training dataset. The small training data for the LASSO comprises the same training dataset as earlier, but now all the attack classes are given the same label as "1" and the benign class is given "0". The selected features are then applied to the large testing dataset as well for multi-class classification. The SGAN model is also fine-tuned with leaky Relu as the activation function in both the generator and discriminator models.

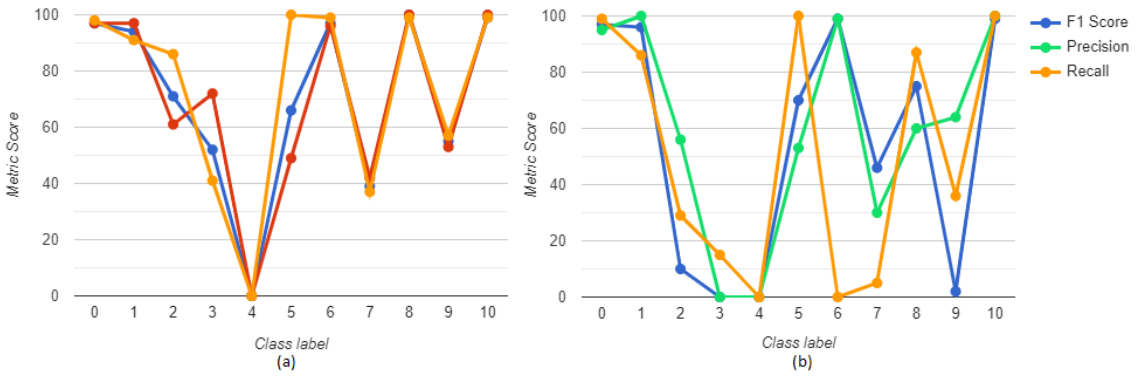


Figure 4.4: SGAN evaluation metrics for multi-class classification on small and Large testing datasets with feature selection

Table 4.8: Results for Multi-class classification with feature selection

S.No	Classification Model	Samples	Train Size	Test Size	Test Acc (Small dataset)	Training Time	Test Acc (large dataset)
1	SGAN	1000	8800	2200	74%	1053.9s	59%
2	ANN	1000	8800	2200	90%	142.86s	48%
3	CNN	1000	8800	2200	66%	1042.02s	30%

Again, models are trained on 1000 samples of each class for 500 epochs. An average of five training/testing rounds is considered. From Table IV, the SGAN performed better with an average accuracy of 59%, while the ANN and CNN had 48% and 30%, respectively, on the large testing dataset ($\approx$ 1 million). The average training accuracy of SGAN is 74%. Fig.4.4(a) shows the F1 score, precision and recall for the small and Fig.4.4(b) shows a large testing dataset with feature selection.

The reported 59% multiclass accuracy indicates that, although the SGAN model outperformed the baseline ANN and CNN models in the considered experimental setting, its performance is not sufficient for direct operational deployment in high-stakes security environments. In particular, such a per-

formance level may lead to missed attacks or excessive false alerts when fine-grained attack categories are required. One possible reason for the limited multiclass performance is that the generator may not have produced sufficiently diverse and class-representative synthetic samples for all granular attack categories. As a result, the discriminator may not have learned sufficiently discriminative boundaries among closely related attack classes.

SGAN Results for the Binary Class Classification with LASSO Feature Selection

Small dataset of size 20,000 (20k) with 10k from benign class and 10k from the attack is used to train the SGAN model. All the sub-attacks are given same labels for the binary classification. 80% of the 20k data is used to train the SGAN and remaining 20% data is used as testing on the small set. Approximately 1 million data is used as testing on a large dataset. On the training dataset, SGAN achieved a precision, recall and F1-score of 1.00. The SGAN model can successfully detect the Botnet attack. Testing accuracy is noted to be 99.88%. From Table V, it can be concluded that SGAN performance is approximately similar to Logistic regression and DNN, while it performs better than other deep learning models.

Table 4.9: Effectiveness Comparison for Binary Classification on N-BaIot Dataset

S.No.	References	Classification Techniques	Accuracy (%)
1	A. Samy et. al. [123]	DNN	98.9
2	A. Samy et. al. [123]	LSTM	99.85
3	A. Samy et. al. [123]	Bi-LSTM	99.81
4	A. Samy et. al. [123]	GRU	99.57
5	A. Samy et. al. [123]	CNN	99.41
6	A. Samy et. al. [123]	CNN-LSTM	99.39
7	F. Abbasi et. al. [1]	ANN	96.4
8	F. Abbasi et. al. [1]	LR	99.98
9	**GANIBOT**	**SGAN**	**99.88**

4.4 LBDMIDS: LSTM based Deep Learning Approach for Network Intrusion Detection System

In this study [131], the literature study shows that the Classic ML techniques have been used in this field for 20+ years while keeping the KDD99 dataset in consideration. But, with the onset of technological boom, the attack categories are increasing spontaneously. So, the potential of any classic ML model is far smaller than the reach of the intrusion. So, methods like SVM (Support Vector Machine), DT (Decision Tree), KNN (K-Nearest Neighbor), etc. have been ruled out long before when considering for industrial applications. Also, the hybrid method using these models has worked well for a long time, but eventually falls short in front of recent DL models. The basis for working of an ML model is basically supervised and unsupervised learn-

ing techniques other than the reinforcement learning part. ML extensively depends upon how rich the dataset is available to us. Also, the inability to scale the data accordingly also poses a large limitation to the extensive use of any ML model.

To handle such issues, DL methods like ANN (Artificial Neural Networks) came into picture. Here the model is built on neurons which are coordinated by parameters and hyper-parameters. To scale the input and use it on extensive scale, the number of layers are kept accordingly to get maximum efficiency. DL methods have proved far better than the ML models in terms of accuracy, precision, etc. with the ability to handle large amounts of data. ANN, CNN (Convolutional Neural Networks), RNN (Recurrent Neural Networks), FDN (Feed Forward Deep Networks) are many examples of DL architectures. DL techniques in general have outperformed ML techniques.

As research progressed, it was seen that basic DL architectures lacked the ability to detect unknown attacks and even if they did, the false positives and false negatives were very high. To tackle this problem, LSTM (Long Short-Term Memory) was introduced. This methodology is a special form of RNN. The most distinguishing feature of LSTM is to keep the information/parameter for later use in the system. Thus, they can handle the data which is time series and could be variable with respect to time.

The primary motivation of using LSTM lies in its approach where is is not restricted to the limitations of conventional DL (neural networks) methods. Here, the input sequences and output sequences between layers are variable which could accordingly work efficiently to detect known as well as unknown attacks.

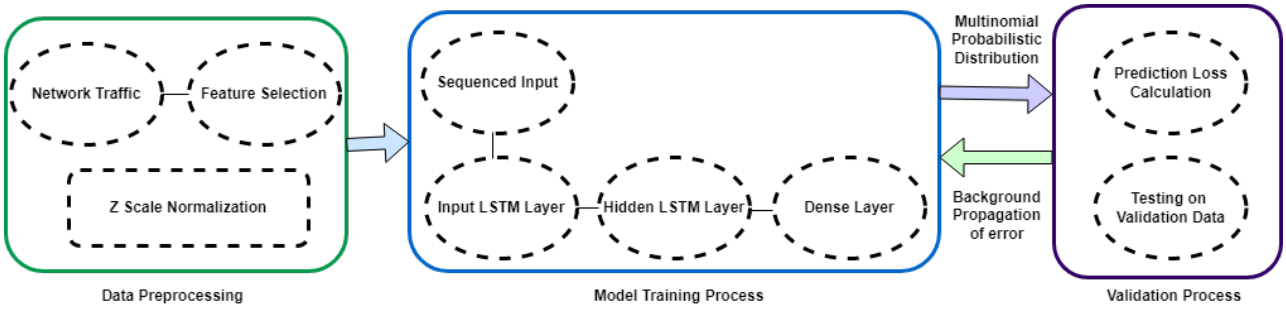


Figure 4.5: Proposed LBDMIDS architecture

4.4.1 Dataset & Outlined method

In order to train our models and to determine their reliability in testing phase, Data selection is necessary. In the past datasets like KDD98, KDDCUP99 and NSLKDD were used as comprehensive datasets for Network Intrusion Detection System(NIDS). In the recent times the researches have shown that the these datasets don't reflect the modern network traffic (normal and attack vectors). Hence, a few datasets have been selected that were made publicly available by researchers in the last few years. These datasets contained labeled network data that were generated in labs with the help of virtual network setup. The datasets are a hybrid of Normal network traffic and synthetic botnet attack traffic. The datasets selected are UNSW NB15 and Bot-Iot Datasets.

In this proposed work, two variants of LSTM are being used in this research, Stacked and Bidirectional.

4.4.2 Proposed Architecture

NIDS is proposed based on two variants of LSTM - Stacked LSTM and Bi-directional LSTM to help filter out normal and attack vectors in IoT network

traffic. The workflow of the NIDS architecture can be divided into three phases, namely the data Preprocessing, Training of the Model and Model Validation, as shown in Fig.1.

4.4.3 Data Preprocessing

In Data Preprocessing stage, the raw data extracted from IoT networks in form of PCAP/CSV files is processed in a suitable form to be fed to the LSTM Model. The data is filtered to remove any redundancies and get rid of Null values. Afterwards the most important features are selected to be fed into the proposed model, which is then followed by z-score Normalization of feature which ensures similar distributions for each feature.

4.4.4 Feature Extraction

The Dimensionality of the raw data has been reduced by selecting the most important features, which made the data suitable for processing. Very often, large Datasets have a lot of redundant and correlated data that can be filtered out without losing important or relevant information. In case of USNW_NB15, the following most important 13 features have been selected: For Bot-IoT dataset, the 10 features that were pre-selected in the Reduced training and testing set were:

4.4.5 Z-scale Normalization

The process of normalization is done to transform the data in a way they are distributed similarly. It helps the model to treat each feature with similar importance as it provides similar weights to each feature. Assuming the feature subspace has N rows and M columns i.e., X (feature subspace) = $\mathbb{R}^{N \times M}$, the

Table 4.10: Network Flow Feature Descriptions

Feature	**Description**
Source packets	Total packets transmitted from the attacker's system to the target
Destination TTL	Time-to-live value for packets returning from destination to source
Destination port	Port number assigned on the target (victim) machine
Source TTL	TTL value for packets sent from the attacker to the target
Source IP address	The IP address initiating the network communication (attacker)
Destination load	Data reception rate (in bps) at the victim's end
Source bytes	Data volume (in bytes) transferred from attacker to victim
Source load	Data transmission speed from source, in bits per second
Destination bytes	Bytes sent from the target back to the attacker
Duration	Total time taken to complete the connection/session
Destination IP address	The target system's IP involved in the network transaction
Source port	Communication port used on the attacker's system
Destination packets	Total number of packets returned from the victim to the attacker

z-scale normalization can be implemented as follows:

$$\mu_m = \frac{\sum_{i=0}^{N-1} x_{im}}{N} \tag{4.7}$$

$$\sigma_m = \frac{\sum_{i=0}^{N-1} (x_{im} - \mu_m)^2}{N} \tag{4.8}$$

z-scale normalized feature vector can be obtained as

$$z_m = \frac{X_m - \mu_m}{\sigma_m} \tag{4.9}$$

Table 4.11: Aggregated Flow Feature Descriptions

Feature	**Description**
mean	The average duration across all combined network records
drate	Packets received per second from destination to source
flgs_number	Encoded numerical value representing flag states
rate	Total number of packets transferred per second in the session
seq	The Argus-generated sequence number for the connection
std_dev	Statistical deviation of durations in the grouped records
state_number	Encoded value reflecting the connection state condition
min	Shortest observed duration in aggregated flows
max	Longest observed duration in aggregated flows
srate	Number of packets sent from source to destination per second

Here, μ_m is the mean of the entries of the m'th column and σ_m is the standard deviation of the entries of the m'th column.

Algorithm 1 Z-SCALE NORMALIZATION

for each col m in X(0,1,2...M-1)
$\mu_m \leftarrow compute(1)$
$\sigma_m \leftarrow compute(2)$
$z_m \leftarrow compute(3)$
end for

4.4.6 Training and Validation Process

Data Processing stage transforms the data into a more suitable form to be processed by the model which will lead to more accurate predictions. It is followed by changing the shape of the data to be processed by the LSTM layers. A suitable Timesteps parameter is selected and the dimensionality of

the dataset is changed(samples, timesteps, features). Timesteps is the number of past samples on which the LSTM model looks back at. The training and validation period of the model consists of feeding the time-series sequential data to the LSTM layers.

4.4.7 Stacked LSTM

In case of stacked LSTM, there are multiple layers of LSTM stacked on top of one another. The LSTM layers help in uncovering the patterns and dependence of features to their class labels because they can learn at higher levels of abstractions. The input LSTM layer is followed by a batch of hidden LSTM layers that process sequenced input and combine learning patterns from the previous layers, to produce learning representations at higher levels of abstraction. The Dense layer is the final layer with the number of nodes equal to the number of categories in the output label. In the decision phase, the soft-max activation function[144] is used by the dense layer to select the most probable of output classes, and the prediction error is calculated with the help of "Sparse Categorical Crossentropy" which is then backpropogated to adjust the weights of the neural network. The model hyperparameters are shown in table 4.12.

Table 4.12: Model hyper-parameters for Stacked LSTM

Dataset	**No. of Layers**	**LSTM Cells/Layer**	**No. of Epochs**	**Learning Rate**
UNSW-NB15	4	40 128 128 64	50	0.002
Bot-IoT	2	32 32	5	0.002

4.4.8 Bi-Directional LSTM

In case of Bi-Directional LSTMs, the recurrent network layer is replicated and it works along side the first layer. The first layer processes the input sequence, while the reversed copy of the input sequence is fed to the second layer. Learning from the past instances and the future instances provides more context to the network and results in better learning. The input layer is a Bi-Directional LSTM layer which feeds forward a non-sequential output to a Dense Layer. The Dense layer is the final layer with the number of nodes equal to the number of categories in the output label. In the decision phase the soft-max activation function is used by the dense layer to select the most probable of output classes, and the prediction error is calculated with the help of 'Sparse Categorical Crossentropy' which is then backpropogated to adjust the weights of the neural network. The model hyperparameters are shown in table 4.12.

4.4.9 Model Structure and Parameters

The model is trained over a number of epochs and training and validation loss decreases gradually with time. The learning process is stopped when the number of epochs (Table 4.13) cross the maximum limit or the model starts overfitting on the training dataset.

Table 4.13: Model hyper-parameters for Bi-LSTM

Dataset	**No.s of Layers**	**LSTM Cells/Layer**	**No.s of Epochs**	**Learning Rate**
UNSW-NB15	1	64	50	0.0015
Bot-IoT	1	12	5	0.001

Table 4.14: Stacked LSTM

Attack	Precision	Recall	F1
Normal	0.98	1.00	0.99
Exploits	0.56	0.86	0.67
Reconnaissance	0.79	0.51	0.62
DoS	0.87	0.01	0.01
Generic	1.00	0.98	0.99
Shellcode	0.62	0.41	0.49
Fuzzers	0.52	0.29	0.37
Worms	1.00	0.00	0.00
Backdoor	1.00	0.00	0.00
Analysis	1.00	0.00	0.00
Weighted avg	0.97	0.96	0.96

4.4.10 Results and Performance Analysis

For this research work, the experimental setup was done where Google Colab was being used for the GPU work. The specifications were Intel(R) Xeon(R) CPU with 2 cores@2.20 GHz, 12.7 GB of RAM and 78 Gb of Hard Disk space. The version of python installed was 3.7.12 and Tensorflow was 2.7.0.

Performance Analysis on UNSW_NB-15 Dataset:

The dataset was splitted into 75% for training and 25% for validation purpose. The training phase consisted of 50 epochs that lasted over 5 hours. In case of Stacked LSTM, the time taken for the validation phase was 128 seconds with the processing speed of 0.2 ms/sample and for Bidirectional LSTM, the validation phase took 90 seconds with the processing speed of 0.14 ms/sample. The accuracy achieved by Stacked LSTM was 96.60% and the accuracy achieved by Bi-directional LSTM was 96.41%

Table 4.15: Bi-directional LSTM

Attack	Precision	Recall	F1
Normal	0.99	0.99	0.99
Exploits	0.54	0.79	0.64
Reconnaissance	0.53	0.59	0.56
DoS	0.47	0.03	0.06
Generic	1.00	0.98	0.99
Shellcode	0.68	0.59	0.64
Fuzzers	0.56	0.35	0.43
Worms	0.53	0.16	0.25
Backdoor	0.67	0.00	0.01
Analysis	1.00	0.00	0.00
Weighted avg	0.96	0.96	0.96

Performance Analysis on Bot-IoT Dataset:

In case of Bot_IoT, the training dataset had 2,934,817 samples and testing dataset had 733,705 samples. The time taken by the training phase was 20 minutes which consisted of 5 epochs for both the stacked and bi-directional LSTM models. In case of Stacked LSTM, the validation phase lasted for 48 seconds with the processing speed of 0.06 ms/sample. For Bidirectional LSTM, the model took 156 seconds with the processing speed of 0.195 ms/sample.

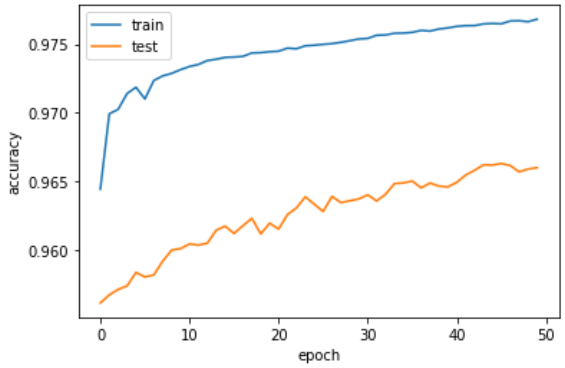


Figure 4.6: Training vs. Validation accuracy (STACKED LSTM)

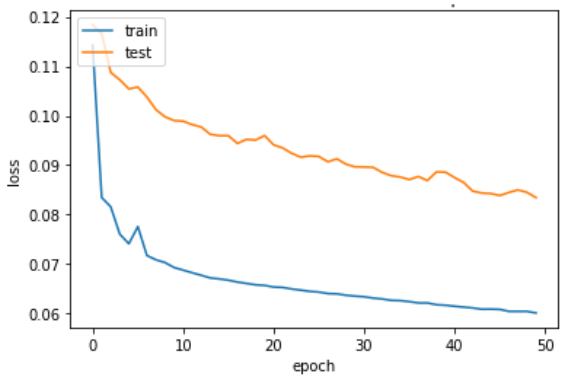


Figure 4.7: Training vs. Validation loss (STACKED LSTM)

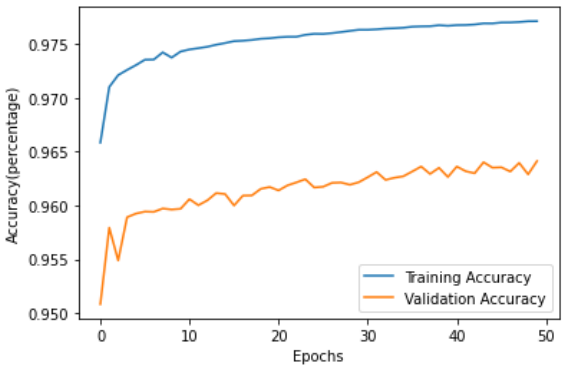


Figure 4.8: Training vs. Validation accuracy (BI-LSTM)

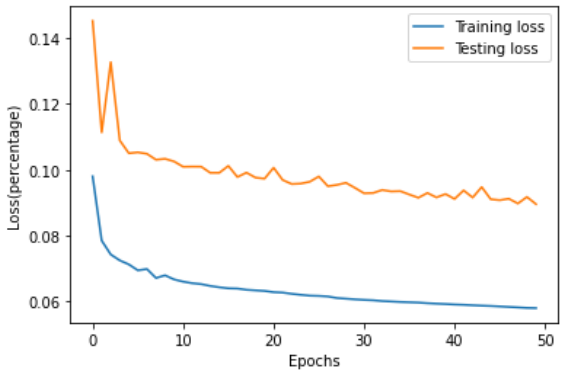


Figure 4.9: Training vs. Validation loss (BI-LSTM)

Table 4.16: Stacked LSTM & Bi-directional LSTM

Attack	**Precision**	**Recall**	**F1-Score**	**Precision**	**Recall**	**F1**
Normal	0.98	0.83	0.89	1.00	1.00	1.00
DDoS	1.00	1.00	1.00	1.00	1.00	1.00
DoS	1.00	1.00	1.00	1.00	0.79	0.88
Reconnaissance	1.00	0.93	0.96	1.00	0.93	0.96
Theft	1.00	0.36	0.53	1.00	0.36	0.53
Weighted avg	1.00	1.00	1.00	1.00	1.00	1.00

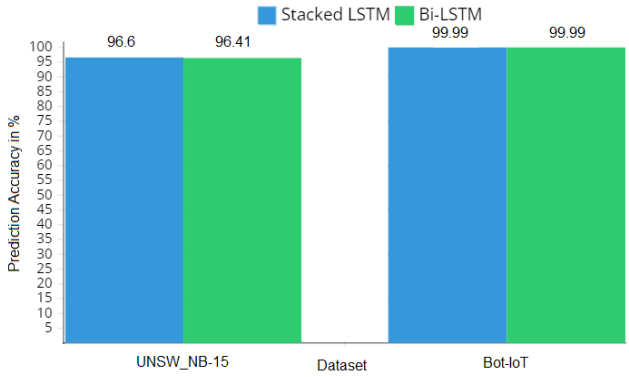


Figure 4.10: NIDS accuracy comparison of Bi-LSTM and Stacked LSTM on both datasets

4.4.11 Comparison with other techniques:

Table 4.17: Comparison of Results on BoT-IoT and UNSW_NB-15 Datasets

BoT-IoT		UNSW_NB-15	
Method	**Accuracy** (%)	**Method**	**Accuracy** (%)
ARM [5]	85.6	FSVM [4]	92
Decision Tree [5]	93.2	GAA [4]	93
DNN [5]	99.9	DNN [5]	99.2
Naive Bayes [5]	72.7	ANN [6]	63.97
Perceptron [5]	63.9	ARM [6]	86.45
-	-	RNN [8]	95.7
-	-	Hybrid [8]	98.7
LBDMIDS	**99.9**	**LBDMIDS**	**96.6**

The results in paper [97, 149, 180, 39] are for binary classification (Attack and Benign) while the results in paper [149] and proposed methodology (LB-DMIDS) is for Multi-Class classification.

4.5 NFDLM: Computationally Efficient DDoS Attack Detection

In this study [134], four deep learning models are utilized for detecting DDoS attacks. We ensured the NFDLM remains computationally efficient by employing feature reduction and selection, along with the use of basic ANN and LSTM. The FS2 model was selected due to its superior performance in both classification accuracy and training time.

4.5.1 Proposed Methodology and Experimental Setup

In this study, four two-staged ML algorithms[151] have been proposed for intrusion detection in IIoT networks. Existing methods include DL systems; however, they either process all data, even if only a portion of it was involved in the attack. However, in this paper, the correlation and mutual information-based feature selection models are used as in [67]. These feature selection methods can select the relevant features. Accuracy is not affected if only the selected features are used to predict the result. Filter-based, wrapper-based and embedded-based feature selections were the three types of feature selection. The filter based has been chosen in this research because it is computationally light, which fulfills our need for making overall model computationally efficient [54]. Workflow for the proposed methodology is given in Fig. 2. The steps taken for building models are explained below.

Dataset Description

The BoT-IoT dataset [76] was developed at the Canberra Cyber Range Lab of the University of New South Wales. The data is separated into four botnet-based assaults and one 'benign' type. The majority of samples were contributed by DDoS and DoS, and for that reason, this dataset has been chosen

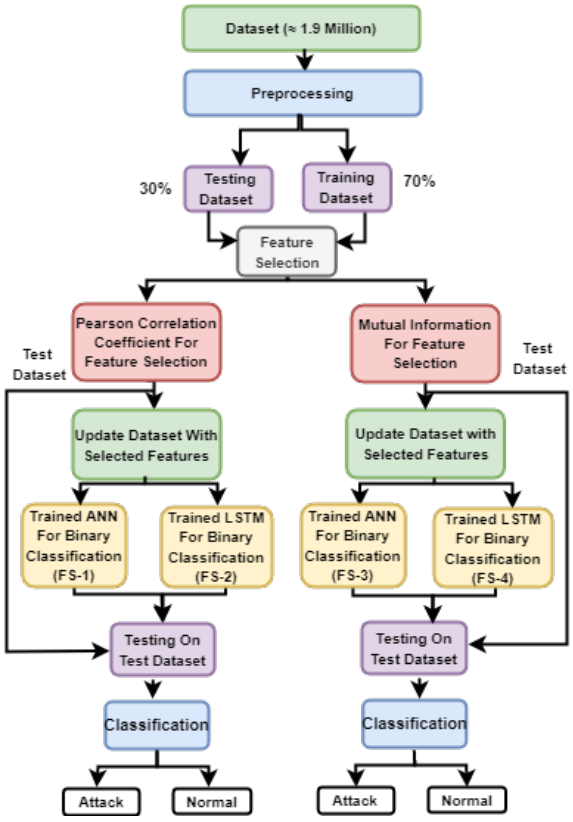


Figure 4.11: Proposed NFDLM workflow diagram

to detect DDoS attacks. Attack distribution for BoT-IoT dataset is given in Fig. 3. In this research, only two categories, i.e. DDoS and benign, are considered due to the negligible presence of other attack categories. Therefore, there is an imbalance in the two categories. The dataset contains 42 features, like bytes, which represent the total number of bytes in transactions, destination IP address and destination port numbers. The "spkts" implies which packet is from the source to the destination count to name a few. The dataset is produced using five IoT devices. These devices are as follows:

- A weather station for measuring air pressure, humidity and temperature.
- Smart fridge used to adjust temperature below certain limit.
- Motion activated lights which switches on or off on the basis of randomly produced signal.

– Smart thermostat used to control building's temperature using air conditioner.

– Remotely activated garage door: using probabilistic input opens or closes the door.

Dataset has 3.6 million samples which includes all the network traffic produced by these devices. In this research, the selected DDoS attack type and benign type are selected from the sample which contains 1,726,624 samples for DDoS attack type and as dataset looks imbalanced. The number of benign samples were only 477.

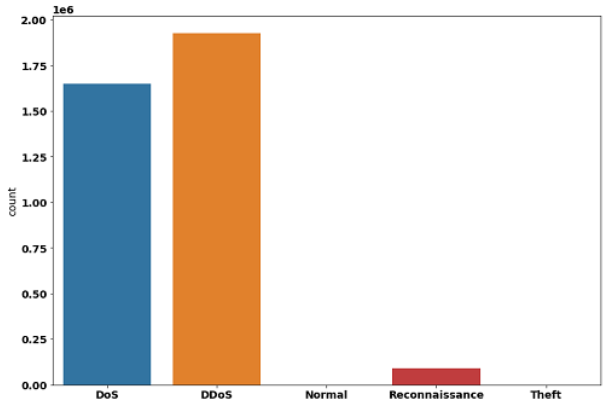


Figure 4.12: Bar graph showing distribution of attack category

4.5.2 Data Preprocessing

In this stage, the dataset is getting preprocessed so that it can be given to the classifier. Firstly all the features of type string object were dropped as their numerical representation in features are also present. There were features like pkSeqID which tells that row numbers so it was dropped as it was not relevant."stime" and "ltime" tells record start time and end time so these were also dropped. After dropping 30 features were remaining. A copy of this updated dataset was saved. The Bot IoT dataset is unbalanced i.e. there

were only 477 samples of benign type in comparison of 19,26,624 samples of DDoS attack type. To deal with data imbalance, “smote” was used which is minority over-sampling technique. After resampling we had 19,26,608 samples each for benign and attack categories.

4.5.3 Data Cleaning

In this research,the Python programming language is used to clean up the data. The categorical variables must be written as numeric values using API Scikit-learn and TensorFlow Keras API. One-hot encoding is used to transform the category variables in terms of numerical values if any.

4.5.4 Scaling and normalizing data

In neural network software, scaling and normalising numeric input characteristics is widely employed to facilitate faster convergence and better outcomes. The Scikit-Learn API has four often used methods: min-max scaler, robust scaler, standard scaler and normalizer. Standard scaler method is used here. To discuss these methods the terms “x” and “xi” are used to represent numerical feature vectors and specific values inside the vectors respectively. The standard scaler is used to turn a feature vector into an approximation of a normal distribution with mean zero and unit variance. The standardization has been used to all the models.

$$StandardScaler(xi, x) = \frac{xi - mean(x)}{stdev(x)} \tag{4.10}$$

4.5.5 Handling Imbalances in Data Labels

The response variables in the Bot-IoT dataset were unbalanced causes the data-imbalances which are a typical occurrence that can be troublesome. There are few, if any malicious flows in network traffic prior to botnet attack then after infection, majority of flow becomes infected. There are a variety of ways that may be utilised to vanish the negative effects of ML classifiers built on data sets which are imbalanced. Some of these tactics include up-sampling minority response variables or down-sampling majority response variables, the use of class weights and synthetic data creation. Synthetic Minority Over-sampling (SMOTE) [20] is employed in this model. It craetes more minority class observations using the KNN. This method has been proved beneficial in various machine learning case studies. SMOTE algorithms are included in the Scikit-Learn API. After train/test split is performed, then resampling using SMOTE is applied only to training data, test data remain untouched.

4.5.6 Feature Selection and Model Building

For each form of assaults, it chooses the most significant features. By lowering the dimensional of the feature, this module is necessary for the creation of a computationally efficient detection system with outstanding detection performance.

Feature Selection is applied to reduce the input dimension [106] which in turn helps our model to converge faster as sometimes some features can have negative impact which leads to increases time to reduce loss function.

4.5.7 Correlation

A variable's In statistical analysis, correlation quantifies the strength and direction of the linear relationship between an independent variable and the target variable. A correlation coefficient of +1 indicates a perfect positive linear relationship, where both variables increase or decrease in tandem. Conversely, a coefficient of –1 signifies a perfect negative linear relationship, implying that as one variable increases, the other decreases proportionally. In the present study, the Pearson correlation coefficient is employed () to evaluate the extent of linear association between feature variables and the target class. This metric is widely recognized for its effectiveness in detecting linear dependencies in supervised learning tasks.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \tag{4.11}$$

$r =$ Pearson's correlation coefficient indicating the strength and direction of a linear relationship

$x_i =$ Individual data points from the x-variable within the sample

$\bar{x} =$ Arithmetic mean of all x-variable observations

$y_i =$ Individual data points from the y-variable within the sample

$\bar{y} =$ Arithmetic mean of all y-variable observations

4.5.8 Mutual information

Mutual information is a metric that assesses how much information can be extracted from one random variable given another. It assesses how much

knowledge of one variable decreases uncertainty about another [107]. It's a two-step process: Firstly, it ranks features according to a set of criteria/metrics, and then the next step is to choose the attributes that have the highest rating.

Table 4.18 shows the naming convention opted for the proposed models. using "Feature selection 1 (FS1) model, which used the correlation coefficient as feature selection and an ANN model as the classifier. The Feature Selection 2 (FS2) model, which used mutual information as feature selection and an ANN model as a classifier. "Feature Selection 3 (FS3) model, which used the correlation coefficient as feature selection and an LSTM model as the classifier. "Feature selection 4 (FS4 model) [150], which used mutual information as feature selection and LSTM model as a class predictor. These two feature-selection approaches will be utilised to select the appropriate feature set for each sort of attack, such as FS-1, FS-2, and so on. After obtaining the most relevant feature sets, the deep learning (DL) algorithms in the "Model Trainer" module will generate the trained model for each type of attack. ANN and LSTM will be employed to predict the class label.

4.5.9 Model Selection & Training

This module develops and assesses a variety of ML methods before selecting the optimal one based on detection accuracy and other evaluation parameters for each of the sub-engines. Parameter tuning is done for all models, and the best four are selected for attack detection and performance evaluation. Firstly, a base ANN model is built without any feature selection for comparative purposes. This model has an input layer with 6 neurons and ReLu as an activation function, one dense layer with 6 neurons and ReLu as the activation function, and an output layer with Sigmoid as the activation function.

The Adam optimiser was used with binary cross-entropy as a loss function. The model was trained with a batch size of 20 and a batch size of 10 epochs.

For the FS1 model, all the parameters were the same as the base ANN model; it just had the addition of a correlation coefficient feature selection to reduce the number of features, which reduced to 9 features only. With a batch size of 20 and 20 epochs, this model was trained.

For the FS2 model, all the parameters were the same as the base ANN model; it just had the addition of Mutual Information feature selection to reduce the number of features to 11 features. This model was trained using 20 epochs and a batch size of 20.

For FS3 model, all preprocessing was the same as the FS1 and FS2 models; feature selection was able to reduce the features to 13 features. Classic LSTM was modeled to maintain an overall computationally efficient architecture. The model had three layers. The input layer had 64 neurons with ReLU as the activation function, while the second layer had 128 neurons with ReLU as the activation function. The output layer was a dense layer with one neuron and a Sigmoid activation function. The model was trained with 50 epochs and a batch size of 32, and the Loss function chosen was binary cross-entropy with the Adam optimiser.

For the FS4 model, all preprocessing was the same as for the FS3 model. Here, mutual information was used as a feature selector, which reduced the number of features to 11. Classic LSTM was implemented to maintain the overall architecture's computationally efficient. The model had three layers. The input layer had 64 neurons with ReLU as the activation function, while the second layer had 128 neurons with ReLU as the activation function. The output layer was a dense layer with one neuron and a sigmoid activation function. The model was trained with 50 epochs and a batch size of 32. The loss function chosen was binary cross-entropy with the Adam optimiser.

In the NFDLM evaluation, feature selection is performed within the cross-validation loop to avoid information leakage. For each fold, Correlation-based or Mutual Information-based feature selection is applied only to the training subset, and the selected features are then used to transform the corresponding validation subset. Thus, the validation data remain unseen during both feature selection and model training.

4.5.10 Results & Discussion

In this experiment, feature selection algorithms were able to reduce the number of features drastically. The feature selector for the FS1 selected 20 features which were highly correlated with other features, so all of them were dropped. The threshold of 65% was selected so that the feature correlating with other features with more than 65% was dropped. The per-protocol average rate per destination IP feature was substantially correlated with the "Rate" feature, which indicates a transaction's total number of packets per second. Because the feature "maximum length" of aggregated data is significantly associated with the mean duration of aggregated records, the "mean" feature was omitted. Similarly, it was found that another 18 features were closely correlated with each other, as can be seen by the heat diagram shown in figure 4.13; all of the highly correlated features were removed. The feature selection in both FS1 and FS3 was the same, as both were applied on the same data set; therefore, the features dropped in both were the same. A heat diagram is given in figure 4.13, which shows the correlation coefficient with other features; the darker the shade, the more the features are highly correlated.

The energies of all features for a feature selector are based on mutual information, and then the top 11 features are selected, which contribute the most to classifying the attack type. First place has gone to the **number of**

Table 4.18: Models comparison

Model	Accuracy	Training Time	Features	Classifier	Feature Selector
FS-1	98.53	1029	9	ANN	Correlation Coefficient
FS-2	99.99	944	11	ANN	Mutual Information
FS-3	99.89	2395	13	LSTM	Correlation Coefficient
FS-4	99.84	2601	11	LSTM	Mutual Information

Table 4.19: Effectiveness comparison

System	Deep Learning Techniques	Task	Dataset Used	Accuracy
Jia et. al. [58]	LSTM	Binary Classification	CIC-DDoS2019	98.9
Li et. al. [81]	LSTM	Binary Classification	CIC-DDoS2019 dataset	98.1
B. Roy et. al. [119]	Bi- LSTM	Binary Classification	UNSW-NB15	95.75
M. A. Ferrag et. al. [32]	RNN	Binary Classification	Bot-IoT dataset	98.20
NFDLM	ANN + Mutual Information	Binary Classification	Bot-IoT dataset	99.99

inbound connections per destination IP, indicating that it had the most important role in attack identification. **Total number of packets per protocol** feature was ranked second, while **total number of bytes per destination IP** was ranked third. There is a graph provided in figure 4.14, which shows the ranking of features from the highest to the lowest. Both FS2 and FS4 were using mutual information as a feature selector on the same dataset; hence, the same features were selected. These features were used in a classifier to detect attacks.

For the FS1 model (ANN + correlation coefficient), the accuracy obtained was 98.53 % (Table 4.18) and the overall training time was 1029 seconds.

For the FS2 model (ANN + Mutual Information), the accuracy obtained in this model was 99.99% (Table 4.18), which was the best of all four models.

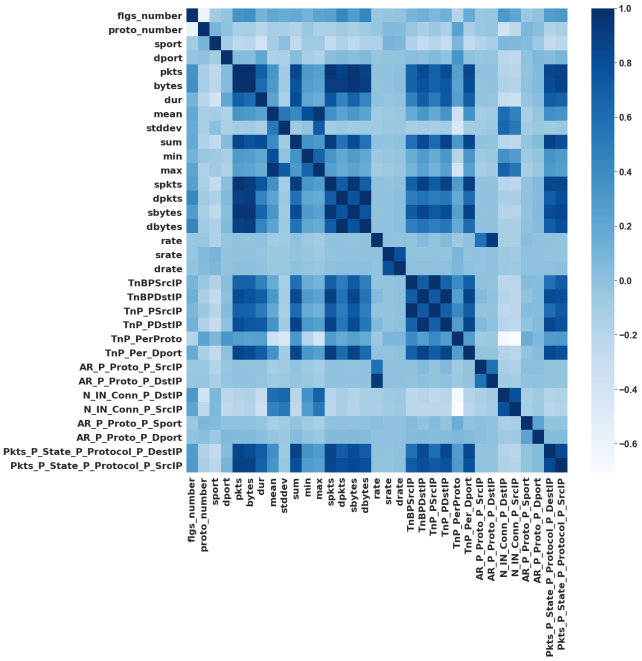


Figure 4.13: Heat diagram for Correlation Coefficient feature selector

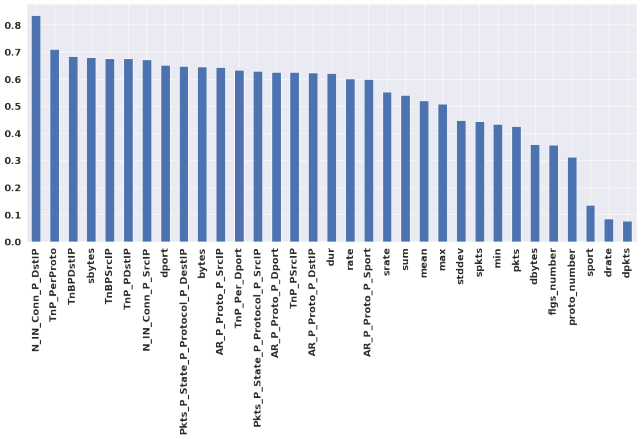


Figure 4.14: Graph showing ranking of features based on Mutual Information

Overall training time took 944 seconds, which was the best among the four models,, and the average epoch time achieved was 47.2 seconds.

For the FS3 model (LSTM + Correlation Coefficient), the total training time was 2395 seconds. On average, the epochs took 47.9 seconds, which is better than FS-1 but less than FS2. A decent accuracy of 99.89 % was achieved, as shown in Table 4.18.

For the FS4 model (LSTM + Mutual Information) the accuracy was 99.84%, but this model took the highest time to train. The total time to train was 2601 seconds, with on average, each epoch taking 52.02 seconds as shown in Table 4.18.

Overall, it was discovered that ANN with "Mutual Information" performed significantly better than the other models based on accuracy and training time. The FS-1 model performed well in terms of feature reduction. Features were reduced to 9, but the accuracy achieved was the poorest of all the models. FS-3 was the second-best model in terms of accuracy. FS-4 performed decently in terms of accuracy, but it took the highest time to train. The Correlation Coefficients feature selector was fast in finding the optimal features in comparison with Mutual Information as a feature selector. Accuracy obtained by the model with the mutual information feature selector was comparatively higher, signifying its ability to select features mainly contributing to identifying DDoS attacks. The proposed model's FS-2, FS-3, and FS-4 gave superior results to recent studies in both classification accuracy and time to train the model for DDoS attack detection (Table 4.18).

4.6 Chapter Summary

This chapter reviewed two deep learning-based intrusion detection systems focused on securing IIoT environments. The first approach proposed a hybrid CNN-LSTM model incorporating zero-shot learning to effectively detect zero-day attacks using the CICIoT2023 dataset. By simulating scenarios where one attack type is excluded during training, the model demonstrated superior performance over standalone CNN and ensemble models. The second work introduced LBDMIDS, an LSTM-based model evaluated on BoT-IoT and UNSW-NB15 datasets, showing high accuracy and F1 scores for

time-series intrusion detection. Both Stacked and Bidirectional LSTM architectures achieved up to 99.99% accuracy on BoT-IoT. While results are promising, further improvements can be achieved through better hardware, advanced optimization, and hybrid modeling strategies to enhance detection in real-world IIoT systems.

Chapter 5

Semantic Attribute based Open-Set IDS for IIoT

An anomaly-based intrusion detection system, despite having a high false alarm rate, is capable of detecting both known and unknown attacks. Much research has been conducted to detect zero-day attacks or unseen attacks using machine learning models. Despite significant advancements, ML models have demonstrated considerable limitations in effectively detecting zero-day attacks. Traditional ML models rely heavily on historical data to learn patterns and classify new instances. However, zero-day attacks are, by definition, novel and previously unseen, meaning their patterns and signatures are absent from the training data. This reliance on historical data leads to a high incidence of false negatives, where new attacks go undetected because the model cannot recognise unfamiliar patterns [24].

To tackle this problem, IDSs have taken advantage of artificial intelligence's capability to learn from the dataset without manually extracting features as it is done in machine learning systems. A deep learning system utilising neural networks is capable of extracting features from a dataset and subsequently carrying out classification and detection tasks. By leveraging these capabilities, deep learning can enhance the model's detection accuracy beyond

what traditional machine learning methods can achieve [153]. This research bases on [136] an open-set based learning methodology and a hybrid neural network (CNN-LSTM) system to reduce the false positive rate and improve zero-day attack price control. "Open-set IDS" refers to Intrusion Detection Systems that can detect previously unseen attacks instead of assuming every test sample belongs to one of the known training classes[166]. The proposed research work uses CICIoT2023 to evaluate the proposed model. The outcome of the open-set-learning based hybrid CNN- LSTM system is compared with stand-alone CNN and ML-based Stacked Ensemble Learning. The result shows that the hybrid model outperforms other models with high detection rate accuracy.

5.1 Methodology and Implementation Details

In Network Intrusion Detection Systems (NIDS)/Network Intrusion Prevention Systems (NIPS) [133] using machine learning, the model is typically trained and evaluated based on established attack categories. Therefore, the evaluation primarily aims to gauge the model's efficacy in identifying instances from known attack classes as malicious.

The methodology proposed in this thesis involves evaluating the model during testing using samples from an attack class that was not part of its training stage, employing a zero-shot learning approach. The training set for the used dataset is designed to include benign traffic as well as $n - 1$ attack classes, intentionally leaving out one specific attack class designated as a "zero-day" attack. In contrast, the testing set contains all attack classes. This setup simulates real-world scenarios where the model encounters previously unseen attack types during testing, as it has not been trained on data from the excluded attack class [179]. Fig. 5.1 below illustrates the scenario of zero-shot

learning.



Figure 5.1: Illustration of the Training and Testing Dataset.

This semantic attribute-assisted open-set is then integrated with a hybrid DL model. Fig. 5.2 below showcases the methodology's framework:

5.1.1 Dataset & Preprocessing

The first crucial task in developing a robust intrusion detection system is to choose a suitable dataset. The dataset must encompass normal and malicious records that accurately reflect the real-world scenarios the model will encounter. The proposed methodology uses the CICIoT 2023 dataset, which was developed by [103]. The dataset contains 33 attacks, which are divided into 7 classes: web-based, denial of service (DoS), distributed denial of service (DDoS), spoofing, Mirai, Reconnaissance & brute force. The number of rows for each class is shown in Fig. 5.3.

1. **Data Encoding:** This process involves transforming the labels within the datasets. Because deep neural networks operate on numerical inputs and the labels are not naturally numerical, a one-hot encoder was employed. This encoder translates labels, like benign or malicious, into numerical forms appropriate for the neural network.

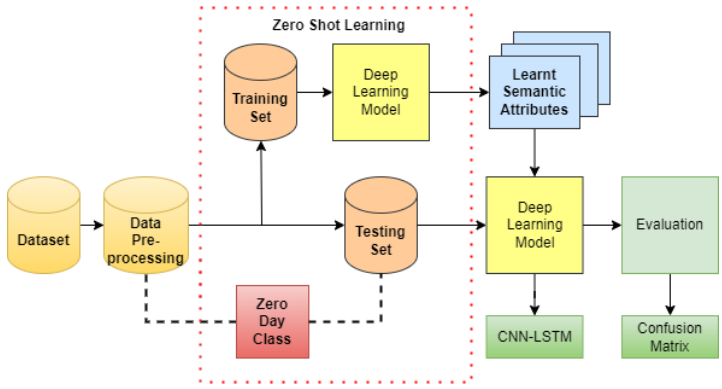


Figure 5.2: Methodology Flowchart.

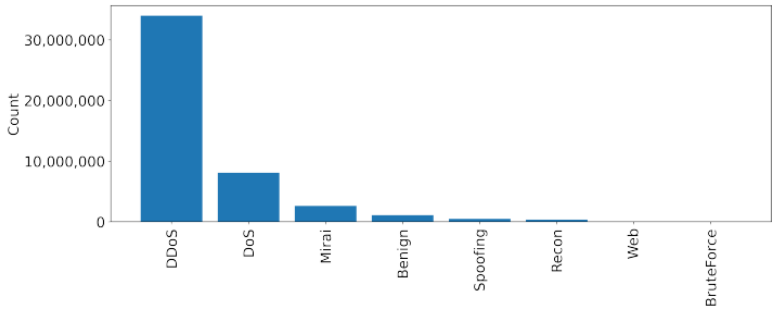


Figure 5.3: Number of rows for each class.

2. **Data Normalisation :** This preprocessing technique is used to optimise data characteristics within a specific range. The differing means and standard deviations of the data extracted from the CSV file can influence the model's learning efficiency. To address this, the Standard Scaler was utilised to adjust the input data, with a mean equal to zero and a standard deviation equal to one.

3. **Data Splitting:** The dataset is split into two sets for training (80%)and testing purposes (20%). Further, the training set is divided into training and validation processes to fine-tune the hyperparameters [147] and improve the model's performance.

5.1.2 Hybrid CNN-LSTM Model

CNNs are proficient at extracting spatial features, while LSTMs excel in capturing temporal characteristics. Leveraging CNN's ability to extract high-level big data, the proposed model starts with a CNN layer. First, the data goes through a convolutional process where filters identify important features to create a unique map. This specification is max-pooled to retain the most important features and then batch-normalised. The resulting results are then fed as input to the LSTM layer to capture physical properties. A layer is added to prevent over-fitting. The first layer uses 32 filters with the ReLU activation function, and then the Max-pooling 1D layer is used to reduce the remaining amount. The second layer uses 64 filters, and the third layer uses 128 filters, each with max pooling. This sequence is then passed to a dense layer of 128 units, preparing the data for compatibility with the LSTM layers. The reshape layer modifies the output to a 2D array that is appropriate for the LSTM layer, specifically with the dimensions (128, 1).

The LSTM layers capture temporal dependencies in data, with the first layer processing 64 units and returning the full sequence of outputs, followed by the second layer processing 32 units and outputting the final temporal features.

Lastly, there are two dense layers in the model for categorisation. With 64 units, the first layer uses ReLU to generate a function. The last Dense layer is designed for multiple classes and uses a soft-max activation function to output the results of each class. The Adam optimiser uses the loss function (categorical cross-entropy) and a learning rate of 0.001 to create the model. Accuracy is the evaluation metric. Based on validation loss, callbacks, more especially, ReduceLROnPlateau, are used to modify the learning rate. Combining the advantages of CNNs and LSTMs, the model is trained and vali-

dated after every epoch to make it resilient to invisible attacks.

5.2 Experimental Setup, Results and Analysis

The subsequent subsections explain the experiment environment and the outcomes in terms of detection rate, recall, precision and F1 scores. A discussion on evaluation against training and validation loss, confusion matrix and ROC (Receiver Operating Characteristic) curve has also been added.

5.2.1 Experimental Setup

The research involves training a hybrid deep learning model through a zero-shot learning technique. The experiments were performed on a machine with 16 GB of DDR6 RAM and an NVIDIA GPU containing 2560 CUDA cores over the course of 50 epochs.

5.2.2 Results and Analysis

In the proposed research, the performance of ML-based Stacked Ensemble learning was compared, which utilises MLP and Random Forest as the baseline classifier, with CNN and hybrid CNN-LSTM models on the CICIoT2023 dataset. The outcomes are presented in Table 5.1 and 5.2. Due to their significantly lower occurrence in the dataset, the Brute-Force attack class and the web class, as a zero-day, were disregarded.

In table 5.1, the Zero-day Attack Class is abbreviated as ZDA Class, and Stacked Ensemble Learning is abbreviated as Stacked EL. It can be seen that the CNN-LSTM hybrid model consistently outperforms both the Stacked Ensemble Learning (Stacked EL) model and the stand-alone CNN model

Table 5.1: Detection Rate Comparison of Different Models

ZDA Class	Stacked EL	CNN	CNN-LSTM
DDoS	67.02%	94.91%	94.95%
DoS	48.95%	81.34%	85.23%
Recon	98.90%	99.60%	99.76%
Spoofing	94.00%	95.87%	95.88%
Mirai	76.43%	92.73%	94.98%

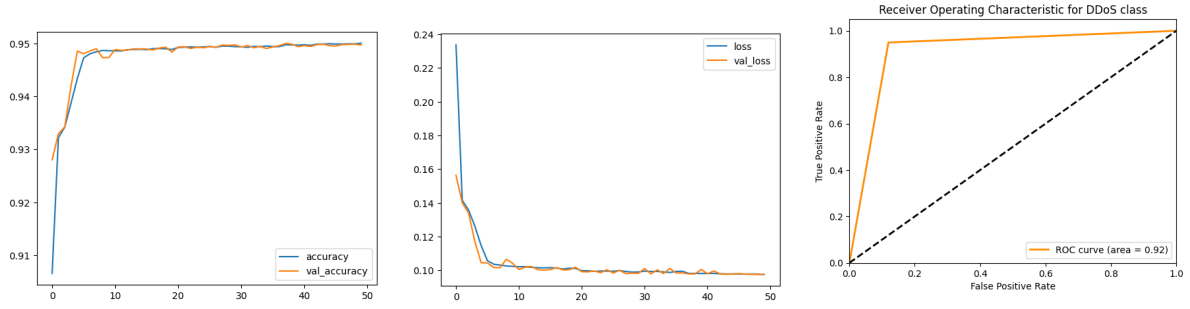
across most attack classes. For instance, in detecting DDoS attacks, the CNN-LSTM model achieves a detection of 94.95%, slightly higher than the CNN model at 94.91%, and substantially better than the Stacked EL model at 67.02%. Similarly, for DoS attacks, the CNN- LSTM model shows a detection rate of 85.23%, outperforming the CNN model's 81.34% and the Stacked EL model's 48.95%. The trend continues for Recon, Spoofing, and Mirai as ZDA classes.

Table 5.2: Metrics for Different Attack Classes based on CNN-LSTM model

ZDA Class	Recall	Precision	F1 Score
DDoS	0.9496	0.9526	0.9484
DoS	0.8929	0.8023	0.8428
Recon	1.000	0.9959	0.9979
Spoofing	0.9600	0.9579	0.9589
Mirai	0.9592	0.9400	0.9495

From Table 5.2, it could be observed that the CNN-LSTM model demonstrates exceptional performance in detecting Recon attacks, achieving perfect recall (1.000), high precision (0.9959), and an outstanding F1 score (0.9979). This indicates the model's remarkable capability in identifying all instances of Reconnaissance attacks with minimal false positives. Similarly, for DDoS, spoofing, and Mirai attacks, the model achieves impressive results, showcasing its robust performance in effectively capturing and accurately classifying

these attacks. However, the performance for DoS attacks, while still notable, is slightly lower, with a recall of 0.8929, a precision of 0.8023, and an F1 score of 0.8428. This suggests that although the model is quite effective in detecting DoS attacks, there is room for improvement, particularly in reducing false positives and increasing overall classification accuracy.



(a) Accuracy and validation accuracy of CNN-LSTM model

(b) Training and validation loss of CNN-LSTM model

(c) ROC curve of CNN-LSTM for Zero-Day DDoS attack detection

Figure 5.4: Performance of CNN-LSTM model in detecting DDoS attacks as Zero-Day Attacks

Fig. 5.4 illustrates the training and validation accuracy of the CNN-LSTM model for detecting DDoS attacks over 50 epochs. As observed, both training and validation accuracies increase rapidly during the initial epochs. The training accuracy begins at approximately 90% and swiftly rises, stabilising around 95% after about 10 epochs. Similarly, the validation accuracy, which starts slightly higher than the training accuracy, follows a parallel trajectory and converges to a similar value of around 95%. The model is effectively learning without experiencing considerable overfitting, as evidenced by the near alignment of the training and validation accuracy curves.

Similarly, Fig. 5.4 illustrates the training and validation loss of the CNN-LSTM model for detecting DDoS attacks over 50 epochs. The graph demonstrates a rapid decrease in both training and validation loss during the initial epochs, indicating the model's quick adaptation to the training data. Initially,

the training loss starts at approximately 0.24 and quickly declines, stabilising around 0.10 after about 10 epochs. The validation loss follows a similar trajectory, starting slightly lower than the training loss and converging to a value of around 0.10. This behaviour indicates that the model has generalised well to the unseen data, maintaining a low error rate during validation. The confusion matrix in Fig. 5.3 demonstrates that the model is particularly effective at identifying DDoS attacks. The high true positive rate for DDoS, coupled with relatively low misclassification rates, suggests that the model can effectively detect an unseen attack. However, there is still room for improvement in reducing the misclassification rates, particularly in distinguishing between DDoS and other types of denial-of-service attacks.

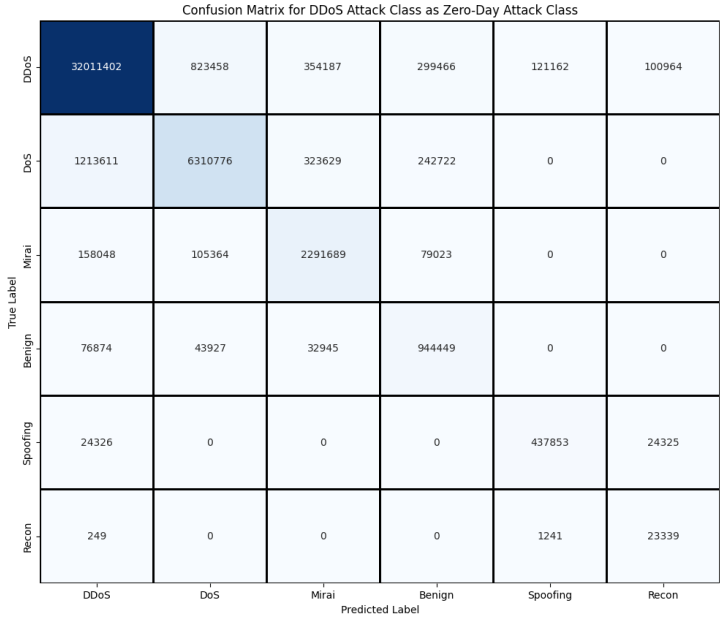


Figure 5.5: Confusion Matrix for DDoS attack as Zero-Day Attack class.

The model's performance was assessed by generating a ROC curve and calculating the AUC for the DDoS attack class, as depicted in Fig. 5.4, which plots the true positive rate against false positive rate. The AUC for the DDoS class is 0.92, demonstrating a high probability that the model will correctly

classify DDoS attacks as zero-day attacks. This high AUC value, along with the high detection rate observed in the confusion matrix, confirms the model's effectiveness in distinguishing DDoS attacks even under zero-day conditions.

Chapter 6

A Hybrid Approach for Zero-Day & Advanced Persistent Threats Detection in IIoT

Cybersecurity developments have witnessed ML methodologies integration, enabling the acquisition of knowledge from data through statistical algorithms. ML models exhibit proficiency in anticipating, categorising, and forecasting events and scenarios by discerning intricate data patterns that may elude domain experts [59] and provide the self-healing in case of cyberattacks. This has led to the adoption of ML to fortify organisational security across various industries. Traditional "signature-based IDS" employs a scanning mechanism to detect recognisable attacks by identifying specific Indicators of Compromise (IOCs). However, these systems face challenges in detecting zero-day attacks, which exploit unknown vulnerabilities before they are addressed, posing a substantial security risk [176]. In contrast, ML-based IDSs [132, 134, 130, 133] leverage behavioural patterns learnt from incoming network packets to identify anomalies or malicious content, going beyond IOCs. These models possess greater potential for detecting zero-day attacks as they learn intricate traffic patterns and do not rely solely on estab-

lished IOCs.

In the IIoT environment, the proposed work focuses on integrating threat intelligence for zero-day attacks and advanced persistent threats [44]. An IIoT ecosystem incorporates linked devices along with critical infrastructure, thus posing special security issues. The study suggests an intelligent IDS specially made to recognise known and zero-day cyberattacks in IIoT systems. The suggested IDS improves threat detection efficiency in IIoT environments by incorporating machine learning methods and adopting a tiered ensemble structure. By identifying complex patterns and indicators of malicious behaviour, the system overcomes the drawbacks of standard signature-based techniques [64]. Data preparation and feature engineering approaches are also employed to optimise the identification of threats. The integration of K-Means clustering offers a flexible solution to the shortcomings of conventional systems in identifying zero-day threats. To determine if a packet belongs to a known attack, an unknown assault, or a benign activity, the study offers a two-step verification process for packet detection in the IIoT context. ML algorithms make the ability to learn from data possible, which empowers the IDS to recognise and understand complicated data patterns that frequently escape domain specialists. By leveraging ML algorithms and analysing network behaviours, these systems enhance detection capabilities, enabling organisations to provide adequate safeguards from the evolving cyberattacks [87]. The research contributes to the field of threat intelligence integration for IIoT by proposing an advanced IDS architecture that strengthens the resilience and effectiveness of security measures in safeguarding critical IIoT systems.

6.1 Proposed Methodology

The proposed work (Fig. 6.1) elucidates a hybrid multi-stage intrusion detection system (HMS-IDS) meticulously designed to cater exclusively for the IIoT networks. The structural framework of the system encompasses four discrete stages, namely data pre-processing, feature engineering, a signature-based IDS, and an anomaly-based IDS. In the data pre-processing stage, raw data undergoes cleaning and formatting to ensure its suitability for analysis. Feature engineering involves extracting relevant features from the pre-processed data, allowing the detection system to capture essential characteristics of network traffic. The signature-based IDS utilises a database of known attack signatures to compare incoming traffic and identify any matches, enabling the detection of familiar or previously documented attacks. Whereas the second stage, unsupervised IDS, focuses on identifying abnormal patterns and behaviours in network traffic.

By employing machine learning techniques, it learns the general properties of the network and detects any deviations that may indicate a potential attack, including previously unseen or unknown attack types. Through the integration of these stages, the proposed architecture aims to deliver a comprehensive and robust IDS for IIoT networks. It effectively identifies and mitigates both known and unknown cyber threats, thereby enhancing the overall security of the network environment.

6.1.1 Data Pre-Processing

The proposed system encompasses an initial data pre-processing stage, aimed at meticulously cleansing and standardising the raw network data. Subsequently, in the feature engineering stage, a comprehensive set of 33 distinct features is extracted, encompassing a wide range of statistical attributes of

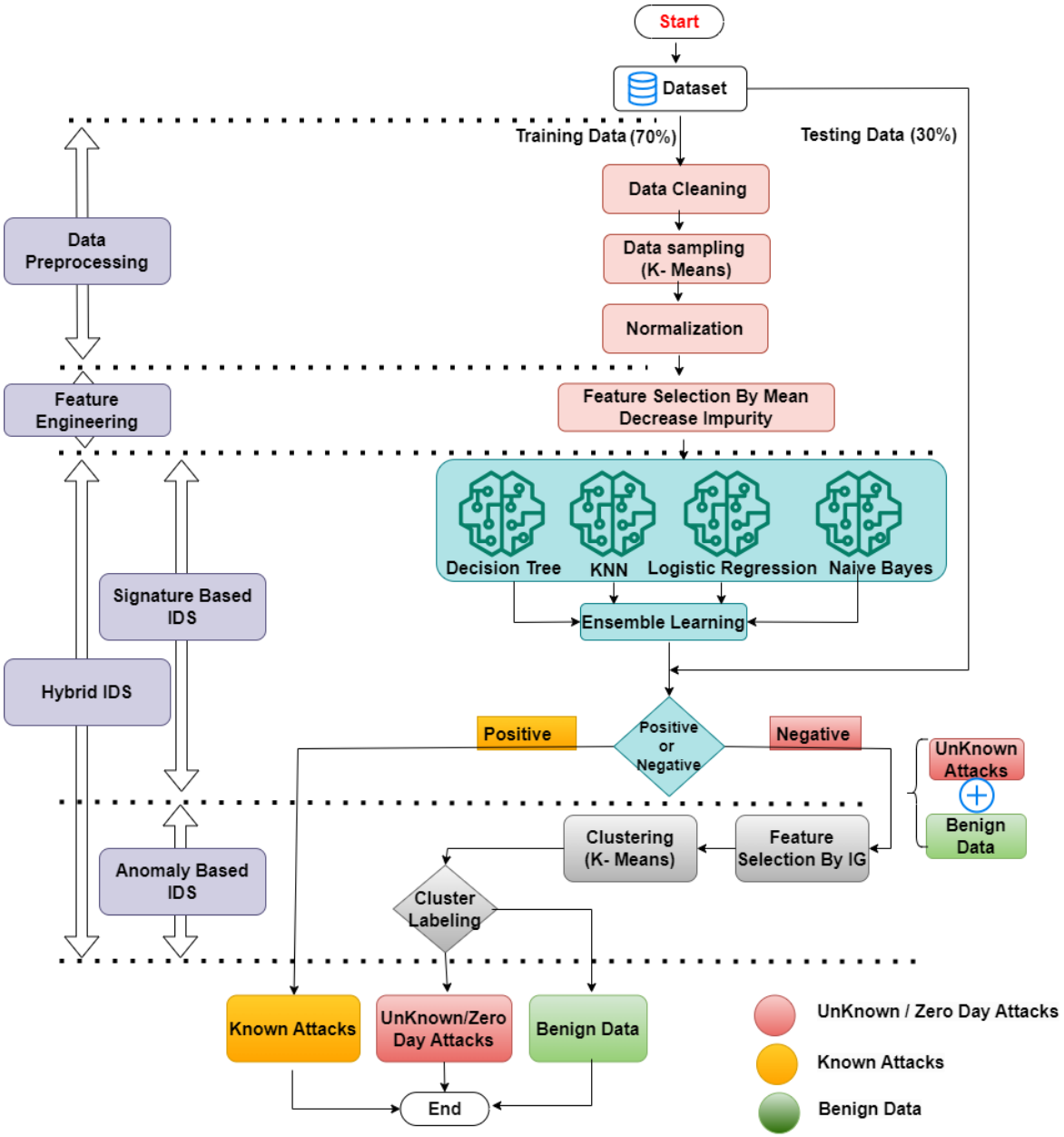


Figure 6.1: Proposed Methodology Diagram of The HMS-IDS

flows, protocols, and packet lengths. These features facilitate efficient identification and examination of prospective vulnerabilities in IIoT networks. The preprocessing of raw data and development of custom features through the feature engineering process improve the accuracy of intrusion detection and network security by finding both known and unknown threats.

6.1.2 Data Sampling

The significance of data sampling in enhancing the efficacy of machine learning (ML) model training on vast volumes of network traffic data is paramount. By creating a reduced subset from the original dataset, sampling methodologies alleviate the computational burden entailed in handling extensive data, thereby facilitating expeditious and efficient model training. Within the proposed system, a cluster sampling approach, predicated on the principles of k-means clustering, is implemented to procure a representative subset of data. This technique ensures that the selected subset adeptly encapsulates the diversity and intrinsic attributes exhibited by the original dataset. During the cluster sampling process, data points are methodically clustered into distinctive groups, from which an appropriately proportionate sample is extracted, thereby culminating in the formation of the representative subset. This approach permits the selection of varied data samples that encompass a myriad of characteristics observed throughout the entirety of the dataset. By harnessing the amalgamation of cluster sampling and k-means clustering, the proposed system endeavours to streamline the training process while optimising the utilisation of network traffic data for the development of ML models.

In contrast to random sampling, where each data sample is selected with equal probability, cluster sampling offers the advantage of generating a highly representative subset by effectively discarding redundant data. This tech-

nique leverages the clustering of data points to identify groups or clusters that share similar characteristics. K-means is widely regarded as the most commonly used clustering algorithm for data sampling.

K-means clustering algorithms utilise distance metrics such as Euclidean, Manhattan, or Mahalanobis distances to partition data points into k clusters. By measuring the dissimilarity between data points, the algorithm identifies the closest centroid for each point, assigning it to the corresponding cluster. This process is repeated iteratively, refining the cluster assignments until convergence is reached. This approach acknowledges that samples within a group share common characteristics or patterns, enabling the selection of representative samples that encapsulate the essential information present in the complete dataset.

The K-means algorithm endeavours to minimise the aggregate of squared distances linking individual data points to their respective assigned cluster centroids. Through a series of iterative steps, it dynamically adjusts the centroids to attain the optimal configuration that aptly captures the inherent structure exhibited within the data. The formula for calculating this sum of squared distances is:

$$\sum_{i=0}^{n_k} \min_{u_j \in C_k} \left(\mathbf{x}_i - u_j\right)^2,$$

6.1.3 Data Normalisation

Upon the completion of the representative dataset acquisition through the utilisation of the k-means method, a subsequent series of data pre-processing procedures is initiated to prepare for the ensuing stages. These ensuing steps encompass a range of data manipulation and transformation techniques aimed at enhancing the quality and relevance of the dataset. As an initial

step, the network traffic datasets are subjected to encoding utilising a label encoder, thereby facilitating the conversion of categorical attributes into numerical representations, enabling the subsequent employment of machine learning algorithms. This process is deemed imperative due to the inherent requirement of numerous machine learning algorithms for numeric inputs, rendering them incapable of directly processing string or categorical attributes. Subsequently, subsequent to the label encoding procedure, the network datasets undergo normalisation utilising the Z-score algorithm, a method indispensable for achieving standardised and comparable data by adjusting each attribute's values relative to the dataset's mean and standard deviation.

This normalisation technique is employed due to the significant variation in range among the collected features in network traffic data. Normalising the data to a common scale is beneficial for ML models, as it can enhance their performance. When working with an unnormalised dataset that exhibits significant differences in feature scales, there is a risk of introducing bias into the machine learning model. This bias occurs because the model tends to prioritise features with larger scales over those with smaller scales, potentially overshadowing the impact of the latter. Consequently, the model's decision-making process may become skewed, relying heavily on the dominant features while undervaluing the contributions of others. By applying the Z-score method, the features within the dataset are transformed to possess a mean value of 0 and a standard deviation of 1. This normalisation process standardises the feature values, aligning them with a common scale.

By incorporating the Z-score technique, the normalisation of each feature value, represented as x_n, is expressed as follows:

$$x_n = \frac{x - \mu}{\sigma}$$

In the feature normalisation process, each original feature value, denoted as x, is transformed using the mean (μ) and standard deviation (σ) of the feature values. This transformation ensures that the features are on a similar scale and have a comparable range, which aids in the accurate analysis and modeling of the data.

6.1.4 Feature Engineering

After undergoing a data pre-processing phase, a dataset of exceptional quality and strong representativeness is generated. Furthermore, by employing effective feature engineering techniques [147], it is possible to attain an optimal list of features that greatly enhances the quality of the datasets. Feature engineering involves selecting, transforming, or creating new features based on domain knowledge and data analysis. Before training the machine learning model, a comprehensive feature engineering method, including Information Gain (IG), is executed to streamline the dataset. This method serves the purpose of facilitating the process of feature selection by effectively discerning and eliminating inconsequential, duplicative, and disruptive attributes, while simultaneously preserving the pivotal ones that exert substantial influence on the desired outcomes [152].

The K-means algorithm operates on a data matrix $(x_1, \ldots, x_n)$, where each x_i represents a data point. In this algorithm, u_j, also called the cluster's (C_k)"centroid", The IG value of the target variable T over a feature X, denoted as such $IG(T|X)$, is defined as the difference between the entropy of the target variable T ($H(T)$) and the conditional entropy of T given X ($H(T|X)$) [175]. It quantifies the importance of feature X to the target variable T, indicating the reduction in uncertainty about T achieved by incorporating feature X in the analysis. In particular, if $IG(T|X) > IG(T|Y)$,

where Y represents another feature, it signifies that the feature X is given higher importance than the target variable T than feature Y [79].

6.1.5 Proposed Hybrid Intrusion Detection System Design

Intrusion detection systems have two prominent classifications, namely signature/rule-based and behaviour/anomaly/heuristics-based. Rule-based IDS is meticulously crafted to discern and acknowledge well-established attack patterns through the employment of machine learning models trained on datasets annotated with labels that correspond to preconceived attack patterns. Although these systems exhibit commendable effectiveness in detecting familiar attacks, they often confront impediments when it comes to discerning uncharted and burgeoning attack patterns that have yet to be catalogued or archived within their databases [145].

In contrast, anomaly heuristics-based IDSs leverage unsupervised learning algorithms to differentiate unknown attack data from normal data. These systems function under the assumption that recently encountered attack data will demonstrate statistical resemblances with known attack data, as opposed to normal data. Nevertheless, a prevalent obstacle encountered by anomaly-based IDSs is their inclination to produce a substantial quantity of false alarms [178]. In order to address the constraints of standalone IDSs, this research presents a groundbreaking hybrid IDS that combines both heuristic and rule-based methodologies. Through the amalgamation of the inherent capabilities of these two methodologies, the hybrid IDS endeavours to furnish an all-encompassing and efficacious detection mechanism, catering to both pre-existing attacks and novel zero-day attacks.

6.1.6 Devising The the Signature-based IDS (To Identify Known Attacks)

Once the steps for feature engineering and data preprocessing are finished, the labeled datasets are utilised to train an ensemble learning model, which serves as the fundamental component of the signature-based IDS. By utilising ensemble learning, which amalgamates multiple individual classifiers, the IDS is capable of augmenting its detection capabilities and enhancing overall accuracy. The signature-based IDS under consideration integrates a heterogeneous ensemble of machine learning algorithms as its foundational learners. Four machine learning (ML) algorithms, particularly Naive Bayes (NB), Decision Tree (DT), Logistic Regression (LR), and k-Nearest Neighbors (KNN), have been chosen based on their unique characteristics and advantages [92]. Every individual-based learner undergoes training using labeled datasets, enabling them to effectively capture distinct characteristics of the underlying data and make unique contributions to it as a whole.

Factors behind the selection of algorithms are as below:

1. DT, NB, KNN, and logistic regression algorithms frequently exhibit superior performance compared to tree-based algorithms when dealing with uncomplicated datasets.
2. They have relatively lower time complexity than tree-based algorithms, especially for NB and logistic regression.
3. They have fewer hyperparameters to tune, making it easier to find a good model.
4. Using different types of base learners can lead to a more diverse ensemble model, which often results in better performance.

- **Ensemble Model:** To enhance the overall model performance, the four base

Algorithm 2 Signature-Based IDS

```
function SignatureBasedIDS(train_data)
    // Feature engineering and data pre-processing
    processed_data ← PreprocessData(train_data)
    // Ensemble learning with base learners
    base_model_DT ← TrainDecisionTreeClassifier(processed_data)
    base_model_NB ← TrainNaiveBayesClassifier(processed_data)
    base_model_KNN ← TrainKNearestNeighborsClassifier(processed_data)
    base_model_LR ← TrainLogisticRegressionClassifier(processed_data)
    // Stacking ensemble to combine base models
    stacked_model ← TrainStackingClassifier(base_model_DT, base_model_NB,
base_model_KNN, base_model_LR, processed_data)
    return stacked_model
```

models are integrated using stacking, an ensemble learning technique. Stacking combines the predictions of multiple base learners, leveraging their strengths and generalisability [91]. To leverage the predictions of the four base learners (DT, NB, KNN, and logistic regression), the stacking ensemble learning technique is employed. Stacking involves utilising the estimated output labels from these base learners as input features for training a meta-learner [91]. By employing the stacking technique, the proposed system aims to leverage the collective knowledge from all four base learners to minimise individual errors and enhance the overall reliability and robustness of the meta-classifier. In this approach, the best-performing base model among the four is selected as the algorithm for constructing the meta-learner.

6.1.7 An Anomaly-based IDS (To Cluster Unknown Attacks & Benign Data)

The signature-based IDS introduced in this study demonstrates effective detection capabilities for various types of known attacks. Despite the effectiveness of the proposed signature-based IDS in detecting known attacks, there remains a vulnerability to zero-day attacks, which are not captured by the existing known attack patterns and may be mistakenly classified as normal system behaviour.

As a result, instances labeled as "normal" by the signature-based IDS are regarded as potentially suspicious, as they may include unknown attack samples that were not captured by the known attack patterns. Subsequently, a novel architecture for an anomaly-based IDS is devised, focusing on the analysis and processing of the potentially suspicious instances to detect zero-day attacks.

Once the feature engineering stage is completed, the optimised dataset resulting from the feature selection process using the IG method is utilised for training the anomaly-based IDS. The initial tier of the proposed anomaly-based IDS involves the utilisation of an enhanced version of the k-means algorithm known as cluster labeling (CL) k-means. Given the extensive range of instances present in network traffic datasets, it is crucial to employ an adequate number of clusters to effectively differentiate between benign and attack data.

The following are the major steps in the suggested CL-k-means method:

1. **K-Means Clustering:** Dividing the dataset into an ample number of clusters is achieved through the utilisation of the k-means algorithm.

$$\text{argmin}_{C_1,C_2,\ldots,C_k} \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

   where μ_i is the centroid of cluster C_i.

2. **labeling Clusters:** The clusters are labeled based on the majority class label of the data samples they contain. Each cluster is assigned the class label ("normal" or "attack") that represents the majority of instances within that cluster.

$$M_i = \text{argmax}_{\text{class}} \sum_{x \in C_i} I(\text{label}(x) = \text{class})$$

   Where $I(\cdot)$ is the indicator function?

3. **Assigning Labels to Test Samples:** Each sample in the test set is assigned a label ("normal" or "attack") based on the label of the cluster to which it belongs, determined by the k-means algorithm

$$\text{label}(x_i) = M_j \quad \text{where} \quad x_i \in C_j$$

4. **Computing Confidence or Clustering Probability:** To assess the confidence or clustering probability of each test sample i within its assigned cluster, the percentage of majority class samples in that cluster is computed.

$$\text{Confidence}(x_i) = \frac{\text{Number of majority class samples in } C_j}{\text{Total number of samples in } C_j} \times 100$$

The mentioned approach leverages the k-means algorithm for real-time demands in distinguishing between attack and normal data in IIoT-level systems. The efficiency of k-means, with a linear time complexity$O(nkt)$, is a key factor in this choice, where n is the size of the data, k is the number of clusters, and t is the number of iterations[99].

Algorithm 3 Anomaly-Based IDS

```
function AnomalyBasedIDS(train_data)
    // Feature engineering and data pre-processing with Information Gain
    processed_data ← PreprocessDataWithInformationGain(train_data)
    // CL-k-means clustering
    clusters ← KMeansClustering(processed_data)
    //labeling clusters based on the majority class
    labeled_clusters ← LabelClusters(clusters)
    // Assigning labels to test samples based on cluster membership
    test_labels ← AssignTestLabels(processed_data, labeled_clusters)
    // Calculate clustering probability for each test sample
    clustering_probabilities ← ComputeClusteringProbabilities(processed_data, labeled_clusters)
    return test_labels, clustering_probabilities
```

To further decrease the training time of the model, minibatch k-means is employed, which employs randomly selected subsets, known as minibatches, during

Algorithm 4 K-Means-based Data Classification for IIoT-level Systems

procedure KMEANSBASEDCLASSIFICATION(D, k, t)
    // Step 1: Apply k-means algorithm
    $centroids \leftarrow$ INITIALIZERANDOMCENTROIDS(D, k)
    for $iteration = 1$ to t **do**
        $clusters \leftarrow$ ASSIGNSAMPLESTOCLUSTERS(D, $centroids$)
        $centroids \leftarrow$ UPDATECENTROIDS($clusters$)
    // Step 2: Label the clusters
    $cluster_labels \leftarrow []$
    for each $cluster$ **in** $clusters$ **do**
        $cluster_label \leftarrow$ MAJORITYCLASSLABEL($cluster$)
        $cluster_labels$.append($cluster_label$)
    // Step 3: Assign labels to test samples
    $labeled_samples \leftarrow []$
    for each $sample$ **in** $test_set$ **do**
        $nearest_cluster \leftarrow$ FINDNEARESTCLUSTER($sample$, $centroids$)
        $label \leftarrow cluster_labels[nearest_cluster]$
        $labeled_samples$.append($label$)
    // Step 4: Compute clustering probability (optional)
    $clustering_probabilities \leftarrow$ COMPUTECLUSTERINGPROBABILITIES($test_set$, $centroids$, $clusters$)
    return $labeled_samples$

each iteration of training [30]. Furthermore, k-means provides the advantage of convergence assurance and seamless adaptation to new samples.

The proposed CL-k-means technique with biased classifiers has numerous benefits in terms of obtaining high accuracy and efficiency compared to other unsupervised anomaly detection algorithms like one-class SVM (OCSVM) [164] and isolation forest (iForest).

1. The CL-k-means model, as proposed, demonstrates its ability to utilise a large number of clusters, allowing for effective capturing of various attack and normal patterns present in the data samples. This unique feature enhances the model's generalisability and improves its capability to accurately model data patterns compared to binary models employed in other outlier detection techniques like iForest and OC-SVM.

2. The CL-k-means model can integrate the BO-GP optimisation technique to automatically estimate the ideal number of clusters (k) required for effective data analysis. This adaptive strategy empowers the proposed model to adapt to diverse datasets and tasks by dynamically adjusting the complexity of the learnt data patterns. Leveraging this adaptive mechanism enhances the model's performance and flexibility in accurately capturing the inherent structures within the data.

3. By employing the mini-batch k-means algorithm, the IDS effectively accomplishes noteworthy decreases in execution time, thereby fulfilling the demanding real-time necessities of IIoT environments. This optimisation technique enables the model to perform data processing in smaller subsets, referred to as mini-batches, within each iteration. As a result, it reduces the computational workload and improves the efficiency of the system. The IDS is capable of expeditiously analysing and reacting to network traffic, thereby

guaranteeing the timely and efficient identification of potential intrusions in IIoT environments.

The presence of attack patterns that closely resemble normal patterns poses a challenge in accurately distinguishing between them. Furthermore, during specific legitimate network events such as crowd events, the collected samples may exhibit significantly different patterns compared to existing normal patterns, leading to potential misclassification as attack samples. Moreover, while k-means is often preferred for developing IDSs in the context of IIoT due to its computational efficiency, it is important to acknowledge the inherent limitation of k-means in accurately modeling complex data distributions. This limitation should be carefully addressed for improved performance and reliability [164].

Given sufficient time and resources, it is possible to explore alternative clustering algorithms that are better suited for specific data shapes and distributions. By replacing k-means with such algorithms while retaining the cluster labeling technique, the system's functionality as a whole has room for improvement. It was planned to investigate the potential of online learning techniques to continually update the learning models based on emerging attack patterns. By incorporating this capability into the IDS, it is expected to enhance its generalisability and accuracy over time, as it will be able to adapt and respond to new and evolving threats more effectively.

6.2 Results and Observations

To assess the efficacy of the proposed IDS for accurately identifying known attacks, the ML models incorporated within the signature-based IDS undergo an extensive training and evaluation procedure utilising a 10-fold cross-validation

approach on the CIC-ToN-IoT dataset. This rigorous approach ensures a robust and comprehensive assessment of the models' intrusion detection capabilities by partitioning the dataset into ten equally sized subsets. Subsequently, the models are trained on nine subsets while their performances are meticulously evaluated on the remaining subsets. Through the repetition of this process for all possible combinations of subsets, the evaluation yields reliable metrics regarding the IDS's proficiency in detecting known intrusions across diverse sets of data samples.

6.2.1 Performance Evaluation of all individual ML models on CIC-ToN-IoT dataset

The confusion matrix depicted in Fig. 6.2 for signature-based IDS is an essential tool for assessing a classification model's strengths and weaknesses since it enables you to fine-tune it, determine the right threshold, or decide between other models depending on your unique objectives and needs and gives a clear image of how well the model performs when separating different classes or categories. The training and testing performance evaluation of the individual ML models like K-Nearest Neighbors, Decision Tree, Random Forest, and Naive Bayes is also shown in Fig. 6.3 and Fig. 6.4 with the respective training and testing time metrics (Fig. 6.5).

6.2.2 Performance Evaluation of Ensemble Learning models on CIC-ToN-IoT dataset

Based on the tabulated results in Table 6.1, it is visible that the ensemble approach has not only increased the detection accuracy significantly (99.49%) but also the other performance metrics like F1-score, recall, precision, training time, and testing time. In conjunction with assessing the IDS's effectiveness in detecting diverse

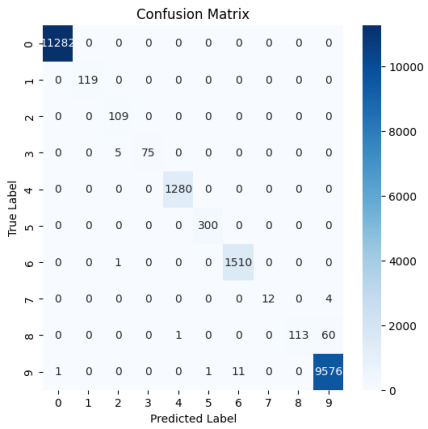


Figure 6.2: Confusion Matrix for Signature-Based IDS

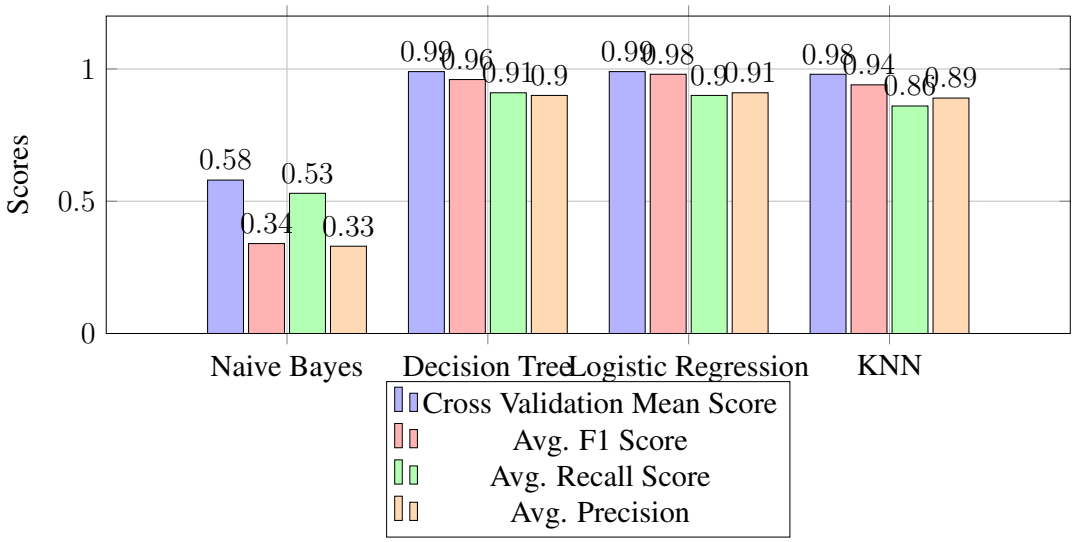


Figure 6.3: Signature-Based IDS's Training Accuracy

attack types (Table 6.2 & 6.3), this model is specifically tailored to discriminate between normal and abnormal network traffic data, assigning a categorical label of either "normal" or "abnormal" to each instance. Notably, the proposed IDS attains a remarkable accuracy rate of 99% during the training phase of the binary classification model (Fig.6.3) and performs the same during the testing phase (Fig. 6.4). This aspect affords users the flexibility to opt for either binary classifiers or multi-classifiers, based on their unique prerequisites and preferences.

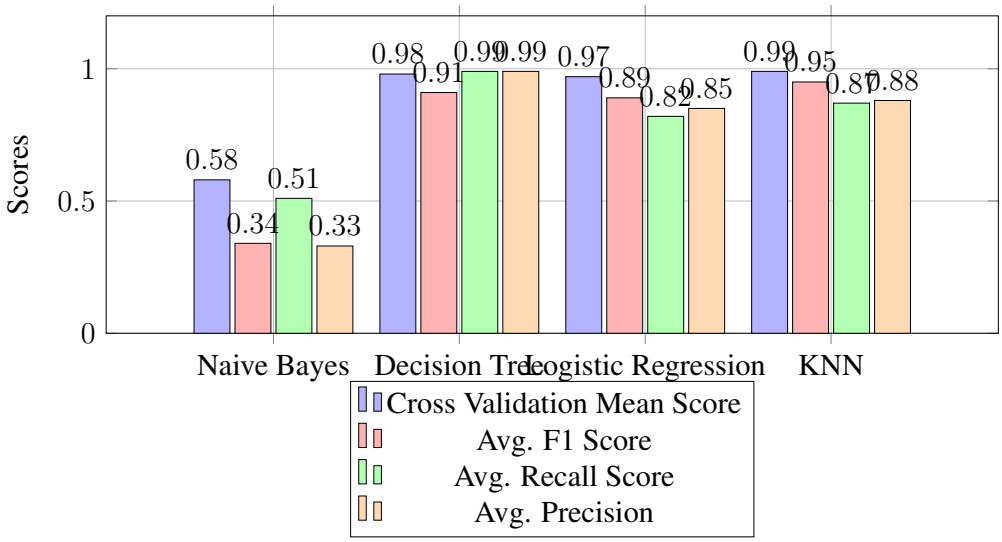
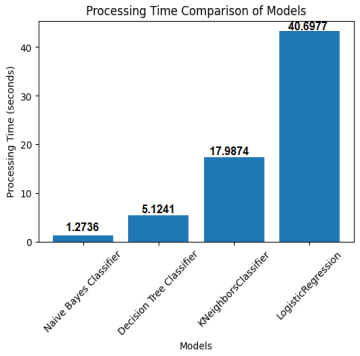
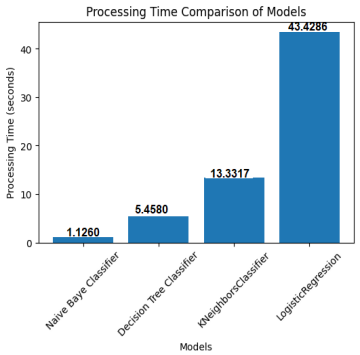


Figure 6.4: Signature-Based IDS's Testing Accuracies



(a) Training Time (Model-Wise)



(b) Testing Time (Model-Wise)

Figure 6.5: Signature-Based IDS's Training and Testing Time-Model-Wise

Test-Model	Cross Validation Mean Score	Avg. F1 Score	Avg. Recall Score	Avg. Precision	Avg. Accuracy (%)	Training Time (seconds)	Testing Time (seconds)
Ensemble	0.99	0.98	0.92	0.94	99.49	16.786	8.885

Table 6.1: Ensemble-Based Model Accuracy

Attack	Precision	Recall	F1-Score	Support
Benign	1.00	1.00	1.00	11241
Xss	1.00	1.00	1.00	120
Password	0.96	1.00	0.98	111
Injection	0.95	0.97	0.96	87
Scanning	1.00	1.00	1.00	1255
Backdoor	1.00	0.99	0.99	285
Ransomware	0.99	1.00	0.99	1566
MITM	1.00	0.86	0.93	22
DDoS	0.97	0.54	0.69	173
DoS	0.99	1.00	0.99	9600
Accuracy			1.00	24460
macro avg	0.99	0.93	0.95	24460
weighted avg	1.00	1.00	0.99	24460

Table 6.2: Ensemble Model's Training Multi-Classification Report

6.2.3 Performance Evaluation of K-means Clustering on CIC-ToN-IoT dataset

The confusion matrix depicted in Fig. 6.6 for anomaly-based IDS helps in assessing the clustering model's strengths and weaknesses since it enables you to fine-tune it, determine the right threshold, or decide between other models depending on your unique objectives and needs and gives a clear image of how well the the model performs when separating different classes or categories Based on the tabulated results in Table 6.4, the implementation of the proposed IG feature

Class	Precision	Recall	F1-Score	Support
Benign	1.00	1.00	1.00	11241
Xss	1.00	1.00	1.00	120
Password	0.70	1.00	0.83	111
Injection	0.89	0.50	0.64	87
Scanning	1.00	1.00	1.00	1255
Backdoor	1.00	0.97	0.99	285
Ransomware	0.99	1.00	0.99	1566
MITM	1.00	0.88	0.93	22
DDoS	1.00	0.56	0.72	173
DoS	0.99	1.00	0.99	9600
Accuracy			0.9949	24460
Macro Avg	0.96	0.89	0.91	24460
Weighted Avg	0.99	0.99	0.99	24460

Table 6.3: Ensemble Model's Testing Multi-Classification Report

selection method and mean decrease impurity method, utilising the optimised correlation threshold (α = 0.49), led to the identification and selection of 15 features from the original set of 85 features. Consequently, the F1 score of the IDS, as applied to the CIC-ToN-IoT dataset, exhibited a noteworthy improvement, reaching a value to 0.99 and the detection accuracy for Zero-day attacks to 99.49%. These findings underscore the crucial role played by the selected features in augmenting the accuracy and efficacy of the IDS, as evidenced by the F1-score metric, thereby enabling the system to accurately detect and classify intrusions.

It is important to clarify the construction of the unknown-attack evaluation reported in Table 6.4. The 98.93% detection accuracy corresponds to the anomaly-based (CL-k-means) stage of HMS-IDS and was obtained using an **intra-dataset held-out class protocol** on the CIC-ToN-IoT dataset. Specifically, selected attack categories were deliberately excluded from the supervised training phase of the signature-based IDS stage, so that the signature stage had no prior knowledge of

those classes. These held-out attack samples were then presented exclusively to the unsupervised anomaly stage during testing, thereby simulating zero-day or previously unseen attack behaviour. The held-out classes were drawn from the same CIC-ToN-IoT dataset, and not from an independent or cross-domain source. The anomaly stage employs CL-k-means clustering with Information Gain feature selection (15 features selected at correlation threshold $\alpha = 0.49$) and assigns cluster labels via majority-class voting. Under this protocol, the stage achieves 98.93% detection accuracy, with precision 0.98, recall 0.99, and F1-score 0.99 for both benign and attack classes (Table 6.4). It is acknowledged that a cross-dataset evaluation, wherein the anomaly stage is trained on CIC-ToN-IoT and tested on attack classes from a different dataset such as CICIoT2023, would provide a stronger zero-day generalisation claim. This is identified as a direction for future work (Section 8)

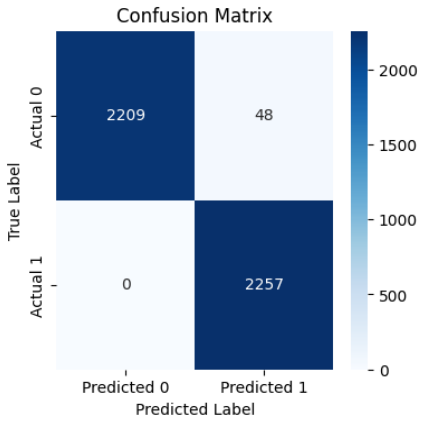


Figure 6.6: Confusion Matrix for Anomaly-Based IDS

Class	Accuracy	Precision	Recall	F1-score	Support
Benign	98.93	1.00	0.98	0.99	2257
Attack	98.93	0.98	0.99	0.99	2257

Table 6.4: Anomaly-Based Model's Detection Accuracy

6.2.4 Performance evaluation of signature based on different datasets

Table[6.5] provides a comprehensive performance evaluation of a signature-based IDS across various datasets. The evaluation encompasses different machine learning models, namely Naive Bayes, Decision Tree, KNeighbors, Logistic Regression, and an Ensemble model. The assessment is conducted based on key metrics such as Cross-Validation (CV) Mean Score, Accuracy, Precision, Recall, and F1-Score.

Dataset	Model	CV Mean Score	Accuracy	Precision	Recall	F1-Score
CIC TON IOT	Naive Bayes	0.6085	0.6126	0.82	0.61	0.65
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9742	0.9768	0.93	0.83	0.86
	LogisticRegression	0.9909	0.9937	0.99	0.99	0.99
	Ensemble	0.9926	0.9931	0.99	0.99	0.99
CIC IDS 2017	Naive Bayes	0.8850	0.9039	0.88	0.70	0.75
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9981	0.9986	0.99	0.99	0.99
	LogisticRegression	0.9969	0.9968	0.99	0.99	0.99
	Ensemble	0.9988	0.9991	1.0	1.0	1.0
NSL KDD	Naive Bayes	0.9395	0.9428	0.94	0.90	0.91
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9979	0.9977	1.0	1.0	1.0
	LogisticRegression	0.9977	0.9980	1.0	1.0	1.0
	Ensemble	0.9989	0.9992	1.0	1.0	1.0
KDD CUP 99	Naive Bayes	0.9395	0.9428	0.82	0.61	0.65
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9979	0.9977	0.98	0.98	0.98
	LogisticRegression	0.9977	0.9980	0.99	0.99	0.99
	Ensemble	0.9989	0.9992	0.99	0.99	0.99

Table 6.5: Results of Signature-Based IDS on different datasets

6.2.5 Performance evaluation of Anomaly-Based IDS on different datasets

The table[6.6] below outlines the performance metrics of Anomaly-Based IDS across various datasets. The metrics include Accuracy, Precision, Recall, F1-Score, True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN).

Anomaly-Based IDS Metrics								
Dataset	Accuracy	Precision	Recall	F1-Score	TP	FP	TN	FN
CIC TON IOT	0.9722	0.9722	0.9722	0.9722	13398	20	15082	2
CIC IDS 2017	0.9722	0.9722	0.9722	0.9722	17215	557	19525	495
NSL KDD	0.9992	0.9992	0.9992	0.9992	13398	20	15082	2
KDD Cup 99	0.9722	0.9772	0.9792	0.9992	34	2	36	0

Table 6.6: Anomaly-Based IDS Metrics for Different Datasets

6.2.6 Testing Time comparison for Signature and Anomaly-based IDS on different datasets

Table [6.7] presents a comparative analysis of the performance evaluation for testing time in signature-based and anomaly-based intrusion detection systems (IDS) across various datasets. The metrics considered include the time taken for support cluster creation, classification time, and the overall testing time for both types of IDS. This comprehensive evaluation provides insights into the efficiency of both signature-based and anomaly-based IDS across diverse datasets. The comparison includes the time taken for critical processes such as Support Cluster Creation and

Classification, contributing to a holistic understanding of the testing time performance in different scenarios.

Dataset	Signature Based IDS (in sec)	Anomaly Based IDS (in sec)		Support
		Cluster Creation Time	Classification Time	
CIC-ToN-IoT	14.42	0.16	0.34	28502
CIC IDS 2017	25.16	75.58	427.07	37792
NSL KDD	59.31	3.76	193.22	28502
KDD CUP 99	31.45	0.98	6.41	72

Table 6.7: Time comparison for different datasets

6.2.7 Runtime Complexity

During the training phase of the hybrid IDS, a high-performance server machine is utilised due to its computational capabilities and ample resources. This facilitates the optimization of training the IDS models by leveraging massive data sets and sophisticated algorithms. In the context of the testing phase, it is necessary to deploy and execute the IDS within the operational IIoT systems. This deployment facilitates the capability to monitor and detect network traffic in real-time. This methodology guarantees the seamless integration of the IDS within the IIoT ecosystem, facilitating prompt detection and response to potential security breaches. In order to guarantee smooth functionality within IIoT systems, the suggested IDS prioritizes the creation of models that possess minimal runtime complexity. By optimizing the computational time and reducing the latency of the IDS, the objective of the system is to attain efficient and prompt identification of attacks, while minimizing the impact on the performance of IIoT devices and networks.

In the implementation phase of the Hybrid Intrusion Detection System (HIDS), a stacked model is employed to process every test sample. This model combines four supervised machine learning (ML) algorithms, namely decision trees (DT),

k-nearest neighbors (KNN), logistic regression (LR), and Naive Bayes (NB). The utilization of this stacked model, in conjunction with the CL-k-means algorithm, facilitates efficient examination and categorization of network traffic. The runtime complexity of the signature-based IDS, the CL-k-means algorithm and the overall runtime complexity of the proposed IDS are given below:

1. **Runtime Complexity of Signature-based IDS** The upper bound on the runtime complexity of the signature-based IDS, which incorporates four supervised machine learning algorithms (decision trees, k-nearest neighbors, logistic regression, and Naive Bayes), can be expressed as O(df t).

Here:

- d: represents the maximum depth of the decision trees.

- f: denotes the number of features in the dataset.

- t: signifies the number of decision trees used in the ensemble.

2. **Runtime Complexity of CL-k-means Algorithm (O(fk)):** The CL-k-means algorithm, incorporated in the proposed IDS, has a runtime complexity of O(fk).

Here:

- f: represents the number of features in the dataset.

- k: represents the number of clusters used in the algorithm.

3. **Overall Runtime Complexity of the Proposed IDS:** The overall runtime complexity of the proposed IDS is relatively low, specifically with a maximum

of O(2df t + f k). This takes into account both the signature-based IDS and the CL-k-means algorithm.

The proposed IDS prioritises creating models with minimal runtime complexity to optimize computational time and reduce latency. The overall efficiency and suitability of the IDS for real-time applications are evaluated, considering computation and categorization time for data samples.

6.3 Chapter Summary

The principle idea behind the Hybrid IDS proposed for its application in IIoT networks involves multiple tiers, each serving a distinct purpose to enhance detection capabilities for both known and unknown attacks. The first tier is a signature-based IDS that uses rule-based detection mechanisms to identify predefined attack patterns documented in databases. The second tier is an anomaly-based IDS that employs unsupervised learning techniques to identify deviations from normal behavior, enabling the detection of unknown / Zero Day attacks.

The proposed hybrid IDS has been comprehensively evaluated on a real-world dataset and has demonstrated efficacy and robustness in detecting various types of attacks targeting IIoT networks. The third tier of the hybrid IDS, which includes the CL-k- means unsupervised model, is specifically designed to address the detection of unknown or zero-day attacks. It uses the k-means clustering algorithm to group similar network traffic patterns and identify deviations from established norms. The proposed IDS uses a multi-tiered approach that includes the CL-k-means model. It has a high accuracy rate and F1-score, which shows that it can find both known and unknown attacks in IIoT networks. The system's performance evaluations on real-world datasets validate its pragmatic utility and potential for enhancing the security of IIoT environments.

Chapter 7

An ML-based Second Line of Defense for Intrusion Prevention

IDS and IPS both share and even update a shared knowledge base while having slightly diverse roles. The IDS detects intrusions and may generate a passive alert, which can be provided as input to the IPS for further action. IPSs are the extensions of IDSs [11] since they monitor systems or network activities and stop/block intrusions by taking a set of actions from the prevention engine. IPSs, unlike IDSs, are installed in-line and can block intrusions before they are noticed by an user. For example, IPS can discard malicious packets, transmit alerts, reset the connection, fix transmission faults, clear out undesirable network and transport layer choices and so on [111]. The two stages of defensive security prevention and detection are as depicted in Fig. 7.1. To maximise network security, they frequently cooperate or might be included in a single defensive system.

In this research work, an Intrusion Prevention System (IPS) has been proposed which uses two-level classifiers for dropping data packets in two phases. Phase 1 consists of model formation for the level 1 classifier using Decision Tree (DT) and the level 2 classifier using Random Forest (RF) and calculation of the threshold

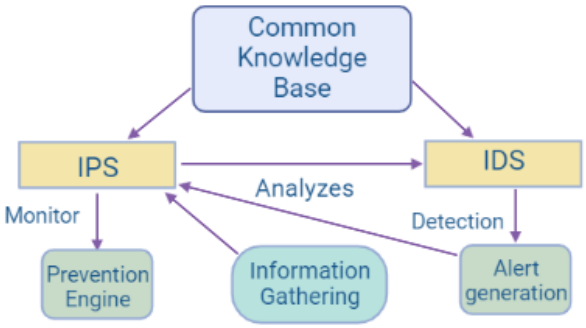


Figure 7.1: IDS To IPS

value (θ). Phase 2 deals with splitting the testing dataset based on the threshold value (θ).

7.1 Methodology

In this proposed Intrusion Prevention System (IPS), two level classifiers for dropping of data packets are implemented. This two level classification is done in two phases. Phase one consists of models formation for level 1 classifier and level 2 classifier. Level 1 classifier comprises of Decision Tree (DT) whereas Level 2 classifier uses Random Forest (RF). Intrusion Prevention System (IPS) which automatically adapts to system changes and further reduces training time and increases the accuracy of the system. In this system, calculation of (θ) is dealt by first finding the execution time of each data point in the dataset then finding the average time. Phase two deals with splitting of testing dataset on the basis of threshold value (θ).

7.1.1 Datapoint Duration and Threshold Calculation

In the proposed MLAPS methodology, the duration of a datapoint refers to the processing or inference time required by the IPS pipeline to classify a given traffic instance after feature extraction. It does not refer to the total duration of an attack or the lifetime of a network flow. During offline evaluation, the processing duration of each datapoint is calculated by recording the start and end timestamps of the classification process. If t_s denotes the timestamp before classification and t_e denotes the timestamp after classification, then the processing duration D_i for the ith datapoint is calculated as:

$$D_i = t_e - t_s$$

The threshold θ is estimated from the execution-time distribution observed during training or validation. This threshold is then used to divide traffic instances into different processing paths. Instances requiring lower processing time can be handled by the computationally efficient classifier, while instances requiring higher processing time or more detailed inspection can be forwarded to the more robust classifier. In a live IPS environment, the traffic instance is unseen before arrival; however, its processing duration can still be measured at runtime. When an incoming packet or flow enters the IPS pipeline, a start timestamp is recorded before preprocessing and classification. After the IPS generates a decision, an end timestamp is recorded. The difference between these timestamps gives the processing duration of that unseen instance. Therefore, θ is not computed from future unseen traffic in advance; rather, it is learned from historical training or validation data and applied during deployment. The measured runtime duration of each incoming instance is then compared with the predefined threshold θ for routing and prevention decision-making.

7.1.2 Dataset & Data Preprocessing

The UNSW-NB 15 data set was generated in the Australian Centre for Cyber Security's (ACCS) Cyber Range Lab, which is a mix of genuine modern regular activities and synthetic contemporary threat behaviours. Analysis, backdoors, fuzzers, DoS, exploits, reconnaissance, generics, shellcode, and worms are among the nine types of attacks. This dataset comprises 49 features, and the total records in the dataset is 175,341, containing various attack and benign data.

Table 7.1: Dataset Features

Feature type	**Feature Name**
Nominal	'proto', 'service', 'state', 'attack_cat'
Integer	'sbytes', 'sttl', 'sloss', 'swin', 'stcpb','dtcpb', 'dwin', 'trans_depth', 'ct_srv_src', 'dloss', 'ct_state_ttl', 'ct_dst_ltm', 'ct_src_dport_ltm', 'dttl', 'ct_dst_src_ltm', 'ct_dst_sport_ltm','ct_ftp_cmd', 'ct_flw_http_mthd', 'dbytes','ct_srv_dst'
Float	'dur', 'tcprtt', 'synack', 'ackdat'
Binary	'is_sm_ips_ports', 'is_ftp_login'

For Binary Classification, feature label has two labels as 19,488 records as normal data and 61685 records as attack data.

For Multiclass Classification, feature attack_cat has 9 labels Analysis(564), Backdoor(99), Dos(1,791), Exploits(16,187), Fuzzers Multi labels(1,731), Generic(39,496),

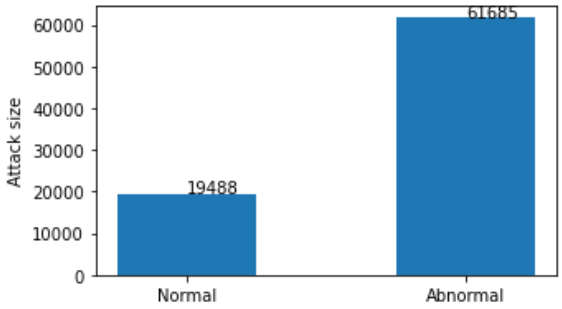


Figure 7.2: Binary classification (UNSW-NB15)

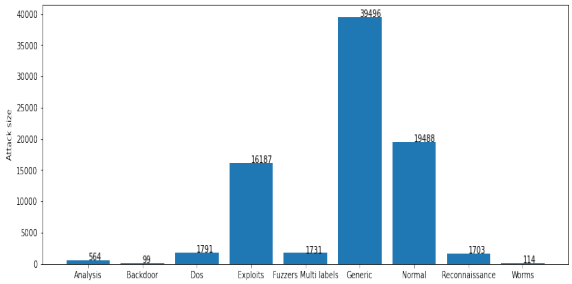


Figure 7.3: Multiclass classification (UNSW-NB15)

Normal(19,488), Reconnaissance(1,703), Worms(114).

7.1.3 Data Preprocessing and Normalization

Data must be cleaned up and prepared for a machine learning model in order for the model to be more accurate and efficient. It's essential to perform data preparation to clean the data and to make it ready for a machine learning model, which accuracy and efficiency of the model. As dataset might contain null or void values. Data Prepossessing moves from data collection, data formatting to data transformation. Normalization is responsible for scaling of data, as it makes data more of use for classification. Feature Selection is one of the major aspect to improve model accuracy and as well as makes the computational cost low. In this research, for Feature Selection, Correlation matrix is used. figures 7.4, 7.5, 7.6, 7.7 are made to depict the correlation between the features.

Dataset's feature 'select' consists of values '-' so entire rows are deleted from dataset. Variant numeric data types are converted into single numeric data type. Nominal/categorical data is dealt using **one-hot encoding** i.e., features which lie in this category are **proto**, **service**, **state**. Categorical parameters will prepare individual columns for every distinct labels in the above mentioned category.

Total features after encoding are 61. All numeric datatype features are normalized using **MinMaxScalar()** with range(0,1). The largest feature is scaled to one and the smallest feature to zero in another method of scaling data. The MinMax Scaler reduces the inputted data range, which is typically 0 to 1. Data is changed by scaling properties to a preset range. Without altering the underlying distribution's form, it modifies the values to fall inside a specified value range.

Binary labels are formed using **LabelEncoder()**, whereas multiclass labels are formed using **one-hot-encoding** and **LabelEncoding**.

7.1.4 Feature Selection

Correlation matrix is formed and features with correlation value less than 0.3 are removed from dataset. This strategy is applied for binary data and multiclass data selection.

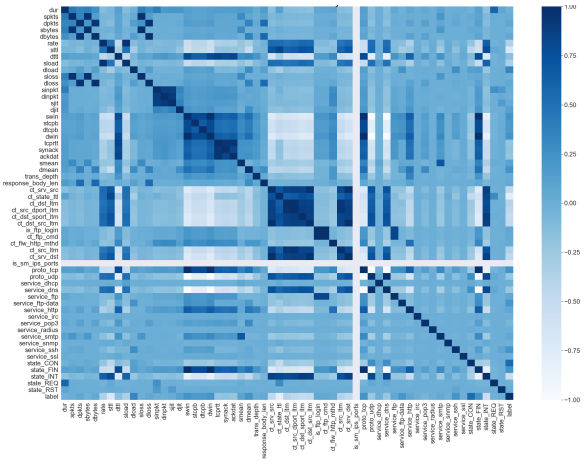


Figure 7.4: Correlation matrix Binary Data

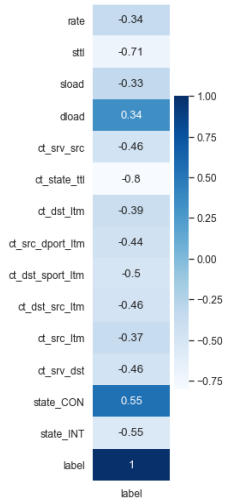


Figure 7.5: Feature Selection Binary Data

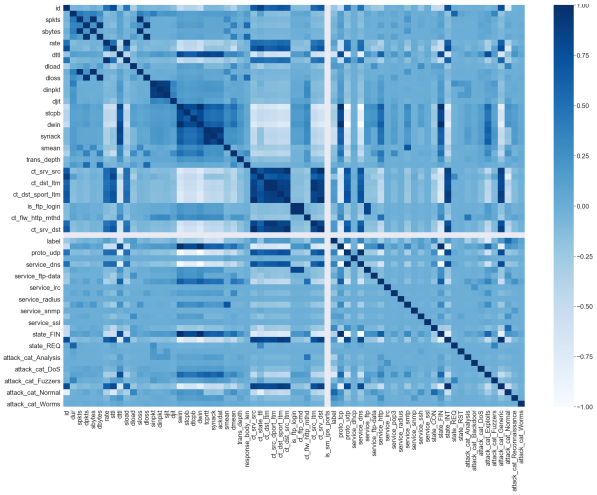


Figure 7.6: Correlation matrix Multiclass data

7.1.5 Observation

Seven ML models have been implemented for both binary and multi-class classification. For each model, its accuracy, precision, recall, f1-score and time to execute the model are noted. Since the concern is over the trade-off between the accuracy and time to execute the model so models with high accuracy and less execution time are noted for both binary classification as well as multi-class classification. It is observed that Decision Tree takes less time compared to other models and also

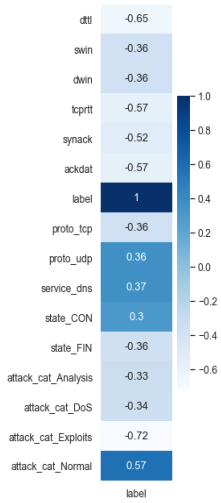


Figure 7.7: Feature Selection Multiclass Data

has high accuracy. Whereas the Random Forest model has the highest accuracy.

7.1.6 Selection of Models

Binary classification and Multi-class classification on UNSW-NB15 are depicted by Tables III & IV.

Table 7.2: Binary classification

Model	Accuracy	Precision	Recall	F1	Time(s)
LR	0.977	0.98	0.98	0.98	0.00267
NB	0.74	0.87	0.75	0.76	0.01101
KNN	0.982	0.98	0.98	0.98	4.27056
DT	0.980	0.98	0.98	0.98	0.00444
RF	0.984	0.98	0.98	0.98	0.04627
AdaB	0.981	0.98	0.98	0.98	0.07368
SVM	0.966	0.96	0.96	0.96	0.75300

So for Intrusion Prevention System Models to be used are Decision Tree & Random Forest.

Table 7.3: Multiclass classification

Model	Accuracy	Precision	Recall	F1	Time(s)
LR	0.974	0.97	0.97	0.97	0.00763
NB	0.951	0.95	0.95	0.95	0.04636
KNN	0.972	0.97	0.97	0.97	17.2074
DT	0.970	0.97	0.97	0.97	0.00661
RF	0.973	0.97	0.97	0.97	0.07719
AdaB	0.751	0.63	0.75	0.67	0.22904
SVM	0.973	0.97	0.97	0.97	1.26248

7.2 Performance Analysis of the Proposed Intrusion Prevention System

In contrast, IPSs are said to be extensions of IDSs [11] since they monitor system or network activities to stop/block intrusions. IPS, unlike IDS, are installed in-line and can block intrusions before they are noticed. For example, IPS can discard malicious packets, transmit alerts, reset the connection, fix transmission faults, clear out undesirable network and transport layer choices and so on [161].
The proposed Intrusion Prevention System consists of two phases. Phase one consists of model formation for the level one classifier, Decision Tree, and the level two classifier, Random Forest. Calculation of (θ) is dealt with by first finding the execution time of each data point in the dataset, then finding the average time. Phase two deals with the splitting of the testing dataset based on the threshold value (θ).

7.2.1 Phase 1: Formation of models and calculation of threshold value

Firstly, both models are trained using Decision Tree & Random Forest. The threshold value (θ) is calculated by finding the average time taken for classifying the

training dataset.

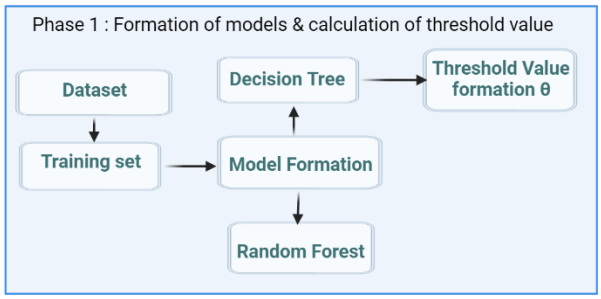


Figure 7.8: Phase 1: IPS

7.2.2 Phase 2: Prevention - Dropping of packets

Now, the Decision Tree Model is used for the Level 1 classifier, whereas Random Forest is used for the Level 2 classifier. If the value is less than or equal to the threshold value θ, then the L1 classifier is used; otherwise stream is forwarded to the L2 classifier. Both L1 and L2 classifiers are used for the prevention of attacks.

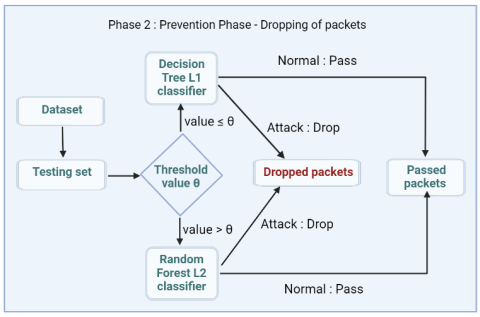


Figure 7.9: Phase 2 : IPS

It is noted that the accuracy for both the binary and multiclass classifier results is slightly higher. Performance metrics for level classifiers (Decision Tree model and Random Forest model) for binary and multiclass classification are depicted below.

Table 7.4: Binary classifiers

ML Model	Accuracy	Precision	Recall	F1
DT(L1)	0.981	0.98	0.98	0.98
RF(L2)	0.982	0.98	0.98	0.98

Table 7.5: Multi-class classifiers

ML Model	Accuracy	Precision	Recall	F1
DT(L1)	0.970	0.97	0.97	0.97
RF(L2)	0.972	0.97	0.97	0.97

7.3 Chapter Summary

This chapter introduces a computationally efficient and adaptive Intrusion Prevention System (IPS) for IoT & IIoT networks, designed to improve accuracy and reduce training time with minimal data loss. The system calculates thresholds using the average execution time from the training set. Unlike traditional models that trade off between accuracy and computational cost, the proposed hybrid approach offers a balanced solution. While effective, the model was tested on a limited scale, with future work suggested in expanding attack coverage and exploring deep learning and semi-supervised techniques for enhanced performance and scalability.

Chapter 8

Discussion and Future Direction

This chapter presents a reflective overview of the key outcomes and implications of the research. It discusses how the findings align with the original research objectives, highlights the unique contributions made to the field of **Industrial IoT Cybersecurity**, and acknowledges the study's limitations. Additionally, it outlines future research directions that can further enhance the proposed models and address the unresolved challenges observed during experimentation. The insights from this chapter aim to guide future work and practical implementations in both academic and industrial contexts.

8.1 Insights and Observations

This research presents a unified approach and delivers a strategic trifecta from TMAP, to diverse detection models, and lastly culminating in a computationally efficient protection framework, underscoring the critical and escalating need for specific cybersecurity solutions within Industrial IoT (IIoT) environments. Through the implementation of various ML and DL models, such as HMS-IDS or hybrid IDS for known and unknown attack detection, NFDLM for DDoS At-

tack Detection, LBDMIDS for LSTM-based Intrusion Detection Systems, GANIBOT for SGAN-based Botnet attack detection, Open-Set based neural networks for Zero-day attack detection, and MLAPS for attack prevention, the study demonstrates that advanced detection methods can significantly improve accuracy, adaptability, and real-time performance. The TMAP framework introduced in this research provides a structured, industry-agnostic, six-step process for threat modelling and attack path analysis using STRIDE and CVSS, thereby filling a vital gap in IIoT cybersecurity. Experimental results from case studies in industrial and power systems underline its real-world applicability.

Additionally, these models like MLAPS, NFDLM, GANIBOT, LBDMIDS, ELA-NIDS and HMS-IDS highlighted significant improvements in detection accuracy (up to 99.99% in certain cases), particularly with hybrid or multi-tiered architectures. However, models such as SGAN showed limitations in early stages (45–59% accuracy), emphasizing the need for further tuning and dataset enhancement. Real-time deployment, federated learning integration, and zero-day attack detection were identified as critical directions, demonstrating a clear progression in both academic and practical cybersecurity development for IIoT systems.

8.1.1 Unified Case Study for Attack Path Analysis and Detection Response

To strengthen the practical interpretation of the proposed IDS framework, this section presents a unified case study demonstrating the complete workflow from threat identification to attack path construction and detection response in an Industrial Internet of Things (IIoT) environment. The purpose of this case study is not to introduce a new model, but to show how the proposed IDS models operate together in a realistic attack scenario.

Case Study Scenario

Consider a smart manufacturing plant consisting of programmable logic controllers (PLCs), industrial sensors, actuators, supervisory control and data acquisition (SCADA) servers, edge gateways, and a cloud-based monitoring dashboard. These devices communicate through industrial Ethernet and IoT protocols. The plant uses the proposed IDS framework at the IIoT gateway and monitoring server to observe network traffic and detect malicious activity.

In this scenario (Fig. 8.1), an external attacker attempts to compromise the IIoT network by first identifying exposed services, then exploiting a vulnerable gateway, moving laterally toward internal devices, and finally launching a denial-of-service attack against the monitoring server. This scenario is representative of multi-stage IIoT attacks where the attacker does not rely on a single attack vector, but gradually progresses through multiple steps.

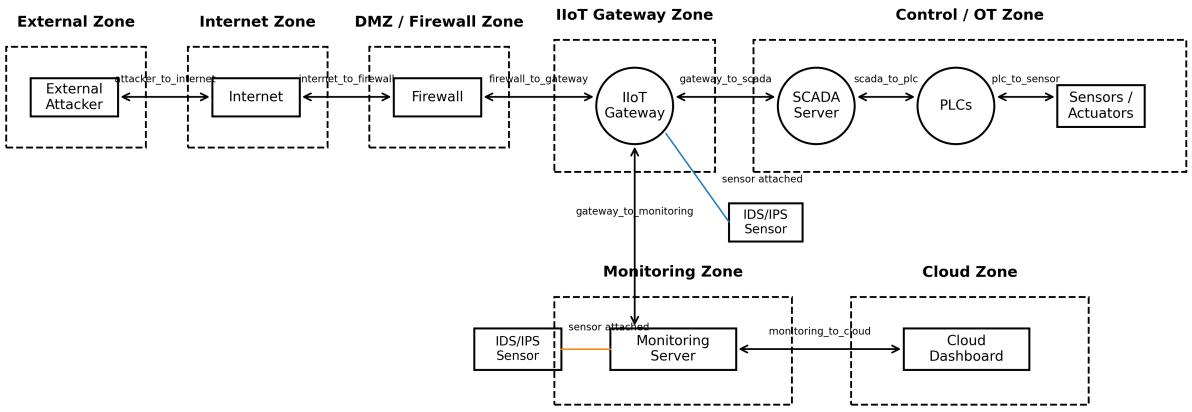


Figure 8.1: Data Flow Diagram for Unified IIoT Case Study

Threat Identification

The first stage of the workflow is the identification of potential threats and vulnerable assets. In the considered IIoT environment, the major assets and possible

threats are summarized in Table 8.1.

Table 8.1: Threat identification in the IIoT case study

Asset / Component	Possible Threat	Attack Objective	Observable Network Behaviour
IIoT gateway	Scanning, password attack, exploitation	Identify open ports and gain unauthorized access	High connection attempts, repeated login failures, abnormal request frequency
SCADA server	Injection, command manipulation, MITM	Manipulate monitoring or control data	Unusual command sequence, abnormal payload patterns, unexpected protocol use
PLCs and sensors	Lateral movement, unauthorized access	Reach internal industrial devices	New internal connections, abnormal source-destination communication
Monitoring server	DDoS / DoS	Disrupt availability of monitoring services	Sudden traffic surge, repeated requests, high packet rate
Unknown attack vector	Zero-day attack	Evade known signatures	Behaviour deviates from normal traffic but does not match known attack patterns

The threat identification stage provides the input for attack path construction. Each identified threat is associated with possible attacker actions and observable network indicators that can be monitored by the proposed IDS.

Attack Path Construction

Based on the identified threats, the attack path is constructed as a sequence of attacker actions. The attack path shows 8.2 how the attacker may move from an external entry point to internal IIoT assets:

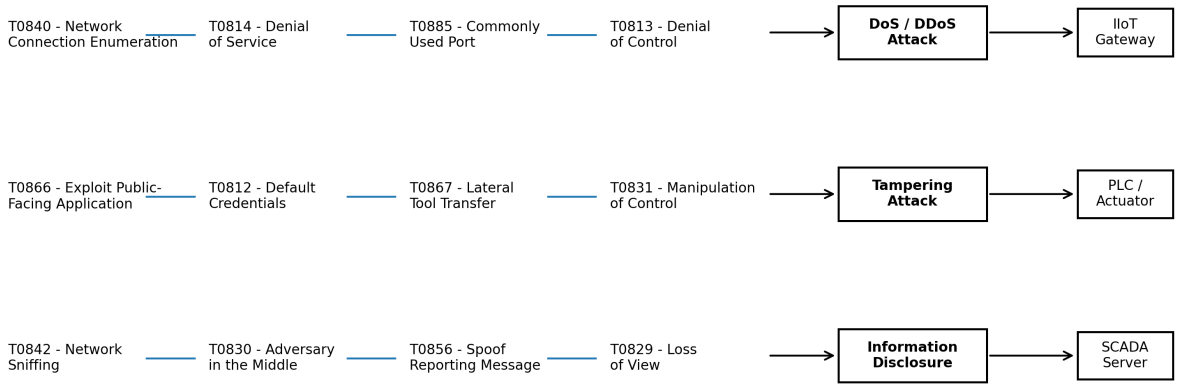


Figure 8.2: Attack Paths for Unified IIoT Case Study

The complete attack path is described as follows. First, in the reconnaissance stage, the attacker scans the IIoT gateway to identify open ports, active services, and possible vulnerabilities. This stage may generate abnormal scanning patterns and repeated connection attempts. Second, in the initial compromise stage, the attacker attempts password-based access or injection-based exploitation. This stage may appear as repeated failed login attempts, unusual payloads, or abnormal protocol requests. Third, during lateral movement, the attacker attempts to communicate with internal PLCs, sensors, or SCADA systems, creating unusual internal communication paths that differ from normal device-to-device behaviour. Fourth, the attacker may attempt command manipulation or man-in-the-middle activity, resulting in abnormal command sequences, unexpected packet contents, or changes in communication frequency. Finally, the attacker launches a DoS or DDoS attack against the monitoring server or IIoT gateway to reduce system availability.

Mapping of Attack Path to IDS Detection Modules

The proposed IDS framework detects different stages of the attack path using different detection components. Table 8.2 presents the mapping between attack path stages, detection indicators, and the corresponding IDS response.

This mapping shows that the proposed IDS models are complementary. Signature-based detection is effective in the early identification of known attacks, while anomaly-based and open-set detection improve the ability to identify unknown or zero-day behaviours. The hybrid IDS combines both approaches to provide a more complete detection mechanism for multi-stage IIoT attacks.

Table 8.2: Attack path mapping with IDS detection and response

Attack Path Stage	Threat Type	Detection Indicator	Proposed Detection Module	Detection Response
Reconnaissance	Scanning	Repeated connection attempts, port probing, abnormal flow count	Signature-based IDS / ELA-NIDS	Generate alert and increase monitoring priority for source IP
Initial compromise	Password attack, injection, exploitation	Repeated login failures, abnormal payload, suspicious request pattern	Signature-based IDS / HMS-IDS Stage 1	Classify known attack and trigger access-control response
Lateral movement	Unauthorized internal communication	New source-destination pair, unusual internal traffic flow	Anomaly-based IDS / CL-k-means stage	Flag as anomalous and isolate suspicious flow for inspection
Command manipulation / MITM	MITM, command injection	Unexpected command sequence, altered communication pattern	Hybrid IDS / HMS-IDS	Correlate signature and anomaly outputs; raise high-severity alert
Service disruption	DoS / DDoS	High packet rate, sudden traffic burst, repeated requests	NFDLM / MLAPS / HMS-IDS	Drop malicious packets, throttle traffic, and notify administrator
Unknown attack behaviour	Zero-day attack	Deviation from normal behaviour without known signature match	Open-set IDS and anomaly IDS	Mark as unknown attack and forward for deeper analysis

Detection Response Workflow

The detection response workflow explains how the system reacts after detecting suspicious activity at different stages of the attack path. The workflow consists of the following steps:

1. **Traffic monitoring:** Network traffic from IIoT devices, gateways, and servers is continuously monitored.
2. **Feature extraction and preprocessing:** Relevant flow-level and packet-level features are extracted and normalized before classification.
3. **Signature-based detection:** Known attacks such as scanning, password attacks, injection, DoS, DDoS, ransomware, backdoor, and MITM are identified using supervised learning models.
4. **Anomaly-based detection:** If the traffic does not match known attack patterns but deviates from normal behaviour, the anomaly detection module identifies it as suspicious.

5. **Attack path correlation:** Alerts from different stages are correlated to determine whether the activity represents an isolated event or a multi-stage attack path.

6. **Response action:** Depending on the severity of the detected activity, the system may generate an alert, block malicious traffic, isolate the affected node, throttle traffic, or forward the event for administrator review.

7. **Feedback and model update:** Confirmed attack instances can be added to the knowledge base to improve future detection of similar attack paths.

Proposed Unified Workflow for IIoT Cybersecurity

The complete workflow of the case study can be summarized as follows:

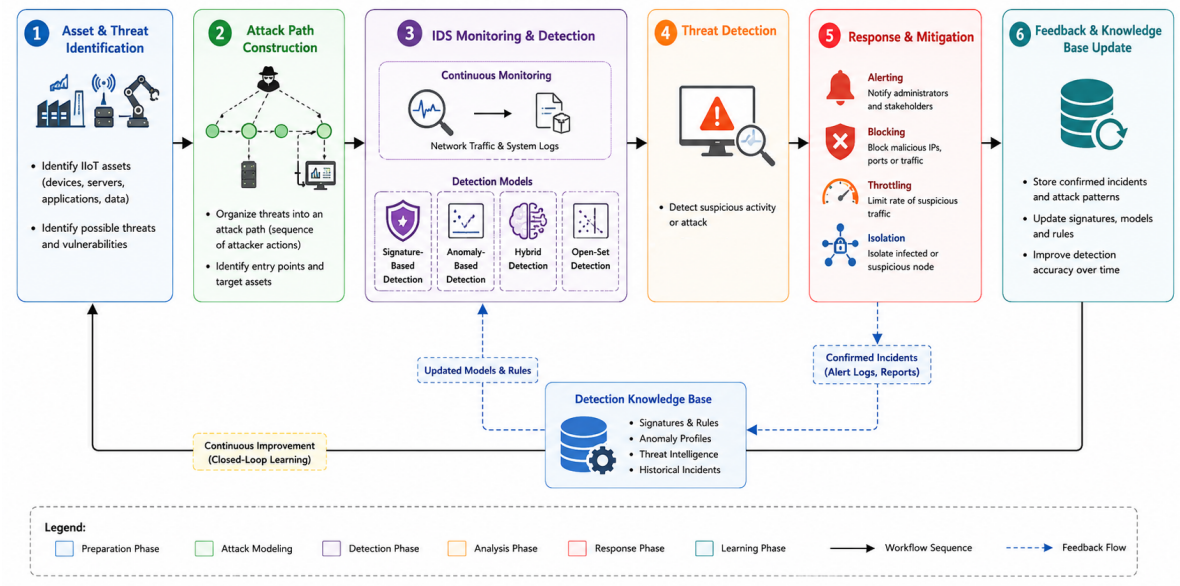


Figure 8.3: Proposed Workflow for Unified IIoT Cybersecurity

In the proposed framework (Fig. 8.3), the workflow begins by identifying the assets and possible threats in the IIoT environment. These threats are then

organized into an attack path that represents the likely sequence of attacker actions. The IDS continuously monitors network traffic and applies signature-based, anomaly-based, hybrid, and open-set detection models depending on the nature of the observed behaviour. Once a threat is detected, the system generates an appropriate response such as alerting, blocking, throttling, or isolating the suspicious node. Finally, feedback from confirmed incidents can be used to update the detection knowledge base.

Discussion

The unified case study demonstrates that attack path analysis improves the interpretability and operational usefulness of the proposed IDS framework. Instead of treating each alert as an isolated event, the attack path view connects multiple suspicious activities into a meaningful sequence. This helps in understanding how an attacker progresses through the IIoT network and how different IDS components contribute to detection at each stage.

The case study also shows that no single detection method is sufficient for all attack stages. Signature-based IDS is effective for known attacks, anomaly-based IDS is useful for detecting deviations from normal behaviour, and open-set or zero-day detection is required for unseen atThe complete attack path is described as follows. First, in the reconnaissance stage, the attacker scans the IIoT gateway to identify open ports, active services, and possible vulnerabilities. This stage may generate abnormal scanning patterns and repeated connection attempts. Second, in the initial compromise stage, the attacker attempts password-based access or injection-based exploitation. This stage may appear as repeated failed login attempts, unusual payloads, or abnor- mal protocol requests. Third, during lateral movement, the attacker attempts to communicate with internal PLCs, sensors, or SCADA systems, creating unusual internal communication paths that differ from

normal device-to-device behaviour. Fourth, the attacker may attempt command manipulation or man-in-the-middle activity, resulting in abnormal command sequences, unexpected packet contents, or changes in communication frequency. Finally, the attacker launches a DoS or DDoS attack against the monitoring server or IIoT gateway to reduce system availability. Therefore, the proposed hybrid and multi-stage IDS framework is suitable for practical IIoT environments where attacks may involve both known and unknown behaviours.

Overall, the attack path analysis confirms that the proposed IDS framework can support the complete security workflow from threat identification to attack path construction and detection response.

8.1.2 Cross-Chapter Comparative Summary of Proposed IDS/IPS Models

The proposed models span multiple IIoT threat domains, each evaluated on a dataset suited to its design objective. Table 8.3 consolidates all six Detect-stage models and MLAPS, covering datasets, attack types, performance metrics, computational complexity, and deployment scenarios. Since each model targets a distinct threat category, direct apples-to-apples comparison is not feasible; results should therefore be interpreted as complementary rather than competing. Together, they form a *portfolio* covering the full IIoT threat landscape from known intrusions and botnet traffic to zero-day threats. Specifically, ELA-NIDS and LBDMIDS handle known attack classification; GANIBOT addresses botnet detection with minimal labels; NFDLM provides computationally efficient DDoS detection for edge nodes; ZDA enables zero-day open-set detection; HMS-IDS unifies signature and anomaly-based detection, and MLAPS completes the Detect, and Respond pipeline per NIST CSF 2.0. Cross-dataset evaluation on a single benchmark (e.g., CIC-ToN-IoT) remains a priority for future work.

Table 8.3: Cross-chapter comparative summary of proposed IDS/IPS models

Model	Paradigm	Dataset	Attack Scope	Architecture	Performance	Complexity	Deployment Scenario
ELA-NIDS	Signature IDS	NF-ToN-IoT-v2	DDoS, DoS, Injection, MITM, XSS, Ransomware	Ensemble ML (XGBoost, RF, ET, CatBoost)	Acc 96.01%, F1 96.32%	Low–Med	General IIoT monitoring and large-scale industrial networks
GANIBOT	Semi-supervised IDS	N-BaIoT	Mirai, Bashlite botnet attacks	SGAN + LASSO feature selection	Binary 99.88%, Multi 59%	High training cost	Botnet detection with limited labelled data
LBDMIDS	Deep IDS	BoT-IoT, UNSW-NB15	DDoS, DoS, Recon, Theft	Stacked LSTM / Bi-LSTM	99.99% (BoT-IoT), 96.6% (UNSW)	Medium–High	Time-series IoT/I-IoT intrusion detection
NFDLM	Computationally efficient IDS	BoT-IoT	Binary DDoS detection	ANN + LSTM + MI/Correlation FS	Acc 99.99%, F1 0.9999	Low	Resource-constrained IIoT edge environments
Open-Set IDS	Novel-class IDS	CICIoT2023	Seen and unseen attacks	CNN-LSTM with semantic attributes	85–99% class-wise detection	Medium–High	Novel / zero-day attack detection
HMS-IDS	Hybrid IDS	CIC-ToN-IoT + benchmark datasets	Known + unknown attacks	Signature + anomaly multi-stage pipeline	Known 99.49%, Unknown 98.93%	Medium	Comprehensive IIoT protection for known and zero-day threats
MLAPS	Intrusion Prevention	UNSW-NB15	9 attack categories	Two-stage DT + RF IPS	97–98% accuracy	Low	Real-time inline IPS and packet-drop prevention

Note 1: Acc. = accuracy; Prec. = precision; F1 = F1-score; DT = Decision Tree; RF = Random Forest; CV = cross-validation; n = samples; k = clusters/trees; t = iterations; n_f = features.

Note 2: The proposed models are evaluated using different datasets, attack categories, preprocessing pipelines, and metrics. Therefore, the reported results should not be interpreted as direct head-to-head comparisons. The models collectively represent complementary IDS/IPS strategies for different IIoT cybersecurity requirements.

8.2 Contributions of this Thesis

This thesis makes the following key contributions in IIoT cybersecurity:

1. Development of a Structured Threat Modeling Framework: The TMAP framework addresses the lack of a standardized approach for threat modeling and attack path analysis in IIoT, using STRIDE and CVSS to assess and prioritize vulnerabilities across diverse industrial environments.

2. Development of Advanced, Hybrid, and Zero-Day-Aware IDS/IPS Architectures for Secure IIoT Environments: This research proposes and evaluates multiple advanced, computationally efficient, and hybrid IDS/IPS models

for securing Industrial Internet of Things (IIoT) environments against both known and emerging cyber threats. The developed architectures, including LBDMIDS, MLAPS, NFDLM, the Open-Set Learning-based Hybrid CNN-LSTM model, and the Hybrid Multi-Stage IDS, integrate machine learning, deep learning, feature selection, and hybrid learning strategies to enhance intrusion detection accuracy, scalability, and training efficiency.

The proposed models collectively address a broad spectrum of attack scenarios, including DDoS, botnet activity, reconnaissance, injection attacks, and zero-day threats, while supporting both signature-based and anomaly-based detection paradigms. In particular, the research introduces novel zero-day attack detection mechanisms through open-set learning and multi-stage decision frameworks capable of identifying previously unseen attacks that evade traditional signature-based systems. Experimental evaluations conducted on benchmark IIoT cybersecurity datasets demonstrate strong detection performance, with the Hybrid Multi-Stage IDS achieving 99.49% accuracy for known attacks and 98.93% accuracy for zero-day threats. Furthermore, the proposed frameworks emphasize practical deployment considerations such as reduced computational complexity, computationally efficient feature engineering, real-time applicability, and adaptability to evolving IIoT infrastructures.

Collectively, the proposed IDS/IPS frameworks contribute toward the development of IIoT cybersecurity solutions aligned with selected NIST Cybersecurity Framework (CSF) functions, primarily **Identify**, **Detect**, and **Respond**. Table 8.3 consolidates the comparative analysis of all seven proposed models, summarizing their datasets, detection capabilities, computational requirements, and deployment scenarios across the overall research framework.

3. Scalable and Industry-Agnostic Solutions: The methods proposed are not constrained to any single domain, enhancing their adaptability and potential

deployment in varied IIoT sectors such as manufacturing and power.

4. Semi-Supervised and Federated Learning Integration Potential: The study opens avenues for future privacy-aware and collaborative IDS development, aligning with real-world deployment needs and data labeling constraints.

8.3 Limitations of this Study

Despite the promising findings and contributions, several limitations were encountered during the course of this study:

- **Lack of Industrial-Scale Evaluation:** Many models, including MLAPS, NFDLM, and LBDMIDS, were not tested in live industrial settings or under large-scale, diverse attack conditions, which may affect generalizability.
- **Dataset Limitations:** The use of datasets such as CICIDS2017 and BoT-IoT, while popular, may not fully reflect the complexity and evolving threat landscape of real-world IIoT networks.
- **Limited Multi-Class Attack Detection:** While some models showed high binary classification accuracy, their performance on multi-class or complex attack vectors remains an area for improvement, as seen in SGAN's initial results.
- **Computational Constraints:** Due to limited access to high-performance computing resources, experiments on training time and scalability were constrained, affecting deep model optimization and epoch duration.
- **Model Interpretability:** Certain deep learning models used in this study, while accurate, operate as black boxes, making it difficult to interpret or justify predictions in sensitive industrial environments.

- **Limitations of NIST CSF Alignment:** The proposed IDS/IPS frameworks are primarily aligned with selected NIST CSF functions, namely **Identify**, **Detect**, and **Respond**. Other functions such as **Govern**, **Protect** and **Recover**, including automated recovery and enterprise-wide resilience mechanisms, are beyond the current scope of this research.

- **Adversarial Robustness:** The proposed ML/DL-based IDS models were evaluated under the assumption that network traffic is drawn from unperturbed distributions. Adversarial evasion attacks, wherein an attacker deliberately crafts malicious network flows to bypass a trained classifier, are not addressed. The exposure of models such as ELA-NIDS, LBDMIDS, and HMS-IDS to such perturbations remains an open concern, particularly since feature-engineered representations may be susceptible to input-space manipulation in real deployments.

- **Concept Drift:** The proposed models were trained and evaluated on static benchmark datasets and do not incorporate mechanisms for detecting or adapting to concept drift, the gradual shift in underlying traffic distributions that degrades model performance over time. In live IIoT environments, both benign traffic patterns and attack methodologies evolve continuously, which may reduce the long-term effectiveness of the trained models without periodic retraining or adaptation.

These limitations highlight areas where further research and experimentation are needed, particularly in industrial validation, real-time deployment, and the expansion of datasets and learning techniques.

8.4 Future Research Directions

Future research should focus on expanding the **Govern**, **Protect**, and **Recover** phases of the NIST CSF 2.0 framework, incorporating federated learning for privacy preservation, optimizing models for real-time deployment, and broadening datasets to better capture the evolving IIoT threat landscape. The subsections below outline the key directions identified from this research.

8.4.1 Open-Source IIoT Cybersecurity Assessment Tool

A key future extension is an open-source cybersecurity assessment platform for IIoT environments, targeting resource-constrained stakeholders such as SMEs where commercial tools may be unavailable. The tool will consolidate the thesis contributions, threat modeling, attack path analysis, IDS, zero-day detection, and IPS, into a reusable platform for system administrators, analysts, and researchers.

The development roadmap spans five phases:

1. Asset identification and STRIDE-based threat modeling via DFDs;
2. Attack path generation and visualization;
3. Integration of ML-based IDS modules for known and zero-day detection;
4. Intrusion prevention support via alert generation or packet dropping; and
5. Dashboard visualization, reporting, and containerized deployment with IIoT gateway interfaces.

Core functionalities include risk scoring, IDS evaluation, zero-day detection, and automated security reporting, all aligned with the thesis methodology.

The tool is envisioned as modular and extensible, where each component (threat modeling, IDS, IPS) can be updated independently. Public datasets, documented

APIs, version control, and containerized deployment will ensure reproducibility and long-term maintainability. It is not intended to replace commercial platforms but to provide a research-driven, computationally efficient assessment tool accessible to academia, industry, and the wider cybersecurity community.

8.4.2 Dynamic Threat Modeling for Evolving IIoT Environments

The current TMAP framework provides a static, design-time assessment using the Purdue Model, DFDs, STRIDE, MITRE ATT&CK ICS, and CVSS. However, IIoT environments are inherently dynamic, devices are added or removed, firmware is updated, and communication flows evolve continuously.

Future work will extend TMAP toward **dynamic threat modeling** by periodically refreshing DFD elements, STRIDE categories, ATT&CK mappings, attack paths, and CVSS scores using runtime telemetry, network monitoring logs, and live vulnerability feeds. This shifts the framework from a one-time assessment toward near-real-time cybersecurity posture evaluation, and naturally complements the open-source tool roadmap described above.

8.4.3 Improved Generative Models for Multiclass IDS

Although GANIBOT demonstrates improvement over baseline ANN and CNN models, its multiclass accuracy indicates a need for more diverse synthetic sample generation. Future work will explore class-conditional GANs, Wasserstein GANs, auxiliary-classifier GANs, diversity-aware regularization, and hybrid augmentation strategies combining GAN-based generation with clustering or oversampling. These improvements aim to produce more representative class boundaries for fine-grained attack categories, reducing missed detections and alert fatigue in opera-

tional IDS deployments.

8.4.4 Edge-Hardware Latency Evaluation of HMS-IDS

The current HMS-IDS evaluation provides theoretical runtime complexity analysis. Future work will empirically measure per-stage detection latency, covering preprocessing, signature-based classification, anomaly clustering, and deep-learning inference, on representative edge hardware such as Raspberry Pi 4, NVIDIA Jetson Nano, and industrial IoT gateways. Throughput, memory usage, and energy consumption will also be profiled.

Optimization strategies including early-exit decision logic, model compression, feature reduction, and edge-cloud partitioning will be investigated. Traffic confidently resolved at the signature stage can bypass deeper stages, reducing average latency while preserving robustness against zero-day threats.

8.4.5 Adversarial Robustness and Concept Drift Mitigation

Future work will investigate the resilience of the proposed IDS/IPS models against adversarial evasion, wherein attackers design perturbed network flows to bypass ML-based classifiers. Adversarial training via FGSM and PGD, along with ensemble-based defences, will be explored to harden the models.

Separately, concept drift, the gradual degradation of model performance as IIoT traffic patterns and attack strategies evolve, must be addressed for long-term viability. Promising directions include online learning, Hoeffding-tree-based adaptive classifiers, and drift detection mechanisms such as ADWIN and Page-Hinkley tests, supported by periodic retraining pipelines. Together, these capabilities will improve the practical resilience and longevity of the proposed frameworks in real-

world IIoT deployments.

8.5 Code and Reproducibility

To support transparency, reproducibility, and further research, future extensions of this work may include the public release of the source code, trained model configurations, preprocessing scripts, and experimental settings used in the proposed IDS/IPS frameworks. The availability of these resources would enable other researchers to validate the reported results, compare the proposed models with alternative approaches, and extend the work to additional datasets and deployment environments.

All source codes, dataset split indices, and reproducibility artifacts associated with this thesis are available at `https://github.com/kumarsaurabhiiita/Phd_work` under the MIT License.

Publications

8.6 Journal (Included in Thesis)

1. Saurabh, Kumar, D. Gajjala, K. Kaipa, R. Khondoker, R. Vyas, and O. P. Vyas, “TMAP: A Threat Modeling and Attack Path Analysis Framework for Industrial IoT Systems (A Case Study of IoM and IoP),” Published In Springer’s AJSE (SCIE Indexed, Q1 Journal (SJR)). DOI: https://doi.org/10.1007/s13369-023-08600-3
   Metrics: (SCIE Indexed, Q1 Journal, Impact Factor 2.9 (2024)

2. Saurabh, Kumar, U. Singh, V. Sharma, R. Khondoker, R. Vyas, and O. P. Vyas, “Threat Intelligence Integration for Advanced Persistent Threats and Zero-Day Exploits in IIoT,” Published In Springer’s AJSE (SCIE Indexed, Q1 Journal (SJR)). DOI: https://doi.org/10.1007/s13369-024-08935-5
   Metrics: (SCIE Indexed, Q1 Journal, Impact Factor 2.9 (2024)

8.7 Conferences (Included in Thesis)

1. K. Saurabh, U. Singh, V. Vamshi, R. Vyas and O. P. Vyas, ”ELA-NIDS: An Improved Ensemble Learning Network Intrusion Detection System for IIoT,” 2024 IEEE 21st India Council International Conference (INDICON), Kharagpur, India, 2024, pp. 1-6, doi: 10.1109/INDICON63790.2024.10958487.

2. K. Saurabh, U. Singh, A. Mishra, R. Vyas and O. P. Vyas, ”Zero-Shot Based Hybrid Neural Network System for Enhancing Zero-Day Attack Detection,” 2024 IEEE 21st India Council International Conference (INDICON), Kharagpur, India, 2024, pp. 1-6, doi: 10.1109/INDICON63790.2024.10958486.

3. Saurabh, K., Sood, S., Kumar, P. A., Singh, U., Vyas, R., Vyas, O. P., & Khondoker, R. (2022, June). LBDMIDS: LSTM based deep learning model for intrusion detection systems for IOT networks. In 2022 IEEE World AI IoT Congress (AIIoT) (pp. 753-759). IEEE.

4. Saurabh, K., Singh, A., Singh, U., Vyas, O. P., & Khondoker, R. (2022, August). GANIBOT: A Network Flow Based Semi Supervised Generative Adversarial Networks Model for IoT Botnets Detection. In 2022 IEEE International Conference on Omni-layer Intelligent Systems (COINS) (pp. 1-5). IEEE.

5. Saurabh, K., Kumar, T., Singh, U., Vyas, O. P., & Khondoker, R. (2022, June). NFDLM: A computationally efficient Network Flow based Deep Learning Model for DDoS Attack Detection in IoT Domains. In 2022 IEEE World AI IoT Congress (AIIoT) (pp. 736-742). IEEE.

6. Saurabh, Kumar, S. Singh, R. Vyas, O. P. Vyas, and R. Khondoker, “MLAPS: A Machine Learning based Second Line of Defense for Attack Prevention in IoT Network,” in 2022 IEEE 19th India Council International Conference (INDICON), 2022, pp. 1–6. doi: 10.1109/INDICON56171.2022.10039777.

Bibliography

[1] Fereshteh Abbasi, Marjan Naderan, and Seyed Enayatallah Alavi. "Anomaly detection in Internet of Things using feature selection and classification based on Logistic Regression and Artificial Neural Network on N-BaIoT dataset". In: *2021 5th International Conference on Internet of Things and Applications (IoT)*. 2021, pp. 1–7. DOI: 10.1109/IoT52625.2021.9469605.

[2] N. Abdalgawad et al. "Generative Deep Learning to Detect Cyberattacks for the IoT-23 Dataset". In: *IEEE Access* 10 (2022), pp. 6430–6441. DOI: 10.1109/ACCESS.2021.3140015.

[3] MS Abirami, Umaretiya Yash, and Sonal Singh. "Building an ensemble learning based algorithm for improving intrusion detection system". In: *Artificial Intelligence and Evolutionary Computations in Engineering Systems*. Springer, 2020, pp. 635–649.

[4] R. Ahmad et al. "Zero-day attack detection: a systematic literature review". In: *Artificial Intelligence Review* 56.10 (2023), pp. 10733–10811.

[5] A. Aldaej. "Enhancing Cyber Security in Modern Internet of Things (IoT) Using Intrusion Prevention Algorithm for IoT (IPAI)". In: *IEEE Access* (2019). DOI: 10.1109/ACCESS.2019.2893445.

[6] Khattab M Ali Alheeti and Klaus McDonald-Maier. "Intelligent intrusion detection in external communication systems for autonomous vehicles". In: *Systems Science & Control Engineering* 6.1 (2018), pp. 48–56.

[7] Hasan Alkahtani and Theyazn HH Aldhyani. "Intrusion detection system to advance internet of things infrastructure-based deep learning algorithms". In: *Complexity* 2021 (2021).

[8] Hadeel A. Almurshid et al. "A Holistic Intelligent Cryptojacking Malware Detection System". In: *IEEE Access* 12 (2024), pp. 161417–161439. DOI: 10.1109/ACCESS.2024.3488192.

[9] A. S. Alqahtani, K. A. Abuhasel, and M. Alquraish. "On implementing a powerful intrusion prevention system focused on big data". In: *Journal of Supercomputing* 77 (2021), pp. 14039–14052. DOI: 10.1007/s11227-021-03856-8.

[10] Abdullah Alsaedi et al. "TON_IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems". In: *IEEE Access* 8 (2020), pp. 165130–165150. DOI: 10.1109/ACCESS.2020.3022862.

[11] S. Anwar et al. "From intrusion detection to an intrusion response system: Fundamentals, requirements, and future directions". In: *Algorithms* 10.2 (2017), p. 39.

[12] P. Freitas De Araujo-Filho et al. "An efficient intrusion prevention system for CAN: Hindering cyber-attacks with a low-cost platform". In: *IEEE Access* 9 (2021), pp. 166855–166869. DOI: 10.1109/access.2021.3136147.

[13] Arctic Wolf. *Understanding and Implementing the NIST CSF 2.0 Cybersecurity Framework*. 2024. URL: https://arcticwolf.com/understanding-and-implementing-the-nist-csf-2-0-cybersecurity-framework/ (visited on 06/03/2026).

[14] Junaid Ashraf et al. "Novel Deep Learning-Enabled LSTM Autoencoder Architecture for Discovering Anomalous Events From Intelligent Transportation Systems". In: *IEEE Transactions on Intelligent Transportation Systems* 22.7 (July 2021), pp. 4507–4518. DOI: 10.1109/TITS.2020.3017882.

[15] Rabeea Basir et al. "Fog Computing Enabling Industrial Internet of Things: State-of-the-Art and Research Challenges". In: *Sensors* 19 (Nov. 2019), p. 4807. DOI: 10.3390/s19214807.

[16] Hugh Boyes et al. "The industrial internet of things (IIoT): An analysis framework". In: *Computers in Industry* 101 (2018), pp. 1–12. ISSN: 0166-3615. DOI: https://doi.org/10.1016/j.compind.2018.04.015. URL: https://www.sciencedirect.com/science/article/pii/S0166361517307285.

[17] Mehmet Bozdal, Mohammad Samie, and Ian Jennions. "A Survey on CAN Bus Protocol: Attacks, Challenges, and Potential Solutions". In: *2018 International Conference on Computing, Electronics & Communications Engineering (iCCECE)*. 2018, pp. 201–205. DOI: 10.1109/iCCECOME.2018.8658720.

[18] Philipp Brauner et al. "A Computer Science Perspective on Digital Transformation in Production". In: *ACM Trans. Internet Things* 3.2 (Feb. 2022). ISSN: 2691-1914. DOI: 10.1145/3502265. URL: https://doi.org/10.1145/3502265.

[19] Luca Caviglione et al. “Tight arms race: Overview of current malware threats and trends in their detection”. In: *IEEE Access* 9 (2020), pp. 5371–5396.

[20] Nitesh V Chawla et al. “SMOTE: synthetic minority over-sampling technique”. In: *Journal of artificial intelligence research* 16 (2002), pp. 321–357.

[21] Tianqi Chen and Carlos Guestrin. “Xgboost: A scalable tree boosting system”. In: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*. 2016, pp. 785–794.

[22] Zhenxiang Chen et al. “Machine learning based mobile malware detection using highly imbalanced network traffic”. In: *Information Sciences* 433 (2018), pp. 346–364.

[23] *CVSS Version 2.0 Calculator*. https://nvd.nist.gov/vuln-metrics/cvss/v2-calculator. Accessed: 29 June 2025.

[24] D. I. Edeh. “Network intrusion detection system using deep learning technique”. M.S. thesis. Turku, Finland: University of Turku, 2021.

[25] Cellect Energy. *IT Security for Utility-Scale Battery Energy Storage Systems — Cellect Energy*. 2025. URL: https://cellectenergy.com/blog/it-security-bess.

[26] European Commission. *NIS2 Directive: new rules on cybersecurity of network and information systems*. Accessed: 2025-06-28. European Commission. 2025. URL: https://digital-strategy.ec.europa.eu/en/policies/nis2-directive.

[27] Gregory Falco, Carlos Caldera, and Howard Shrobe. “IIoT Cybersecurity Risk Modeling for SCADA Systems”. In: *IEEE Internet of Things Journal* 5.6 (2018), pp. 4486–4495. DOI: 10.1109/JIOT.2018.2822842.

[28] Yulin Fan et al. “Iotdefender: A federated transfer learning intrusion detection framework for 5g iot”. In: *2020 IEEE 14th International Conference on Big Data Science and Engineering (BigDataSE)*. IEEE. 2020, pp. 88–95.

[29] Federal Office for Information Security (BSI). *General information on Critical Infrastructures (KRITIS)*. Accessed: 2025-06-28. BSI. 2025. URL: https://www.bsi.bund.de/EN/Themen/Regulierte-Wirtschaft/Kritische-Infrastrukturen/Allgemeine-Infos-zu-KRITIS/allgemeine-infos-zu-kritis_node.html.

[30] Ali Feizollah et al. “Comparative study of k-means and mini batch k-means clustering algorithms in android malware detection using network traffic analysis”. In: *2014 International Symposium on Biometrics and Security Technologies (ISBAST)*. 2014, pp. 193–197. DOI: 10.1109/ISBAST.2014.7013120.

[31] Fang Feng et al. “Anomaly detection in ad-hoc networks based on deep learning model: A plug and play device”. In: *Ad Hoc Networks* 84 (2019), pp. 82–89. ISSN: 1570-8705. DOI: https://doi.org/10.1016/j.adhoc.2018.09.014. URL: https://www.sciencedirect.com/science/article/pii/S1570870518306887.

[32] Mohamed Amine Ferrag et al. “Deep learning for cyber security intrusion detection: Approaches, datasets, and comparative study”. In: *Journal of Information Security and Applications* 50 (2020), p. 102419.

[33] Mohamed Amine Ferrag et al. “Deep Learning-Based Intrusion Detection for Distributed Denial of Service Attack in Agriculture 4.0”. In: *Electronics* 10.11 (2021). ISSN: 2079-9292. DOI: 10.3390/electronics10111257. URL: https://www.mdpi.com/2079-9292/10/11/1257.

[34] Mohamed Amine Ferrag et al. “Rdtids: Rules and decision tree-based intrusion detection system for internet-of-things networks”. In: *Future internet* 12.3 (2020), p. 44.

[35] Santiago Figueroa-Lorenzo, Javier Añorga, and Saioa Arrizabalaga. “A Survey of IIoT Protocols: A Measure of Vulnerability Risk Analysis Based on CVSS”. In: *ACM Computing Surveys* 53 (Apr. 2020), pp. 1–53. DOI: 10.1145/3381038.

[36] “FortiGuard Labs Threat Landscape Report Highlights Tenfold Increase in Ransomware”. In: (Available: https://www.fortinet.com/content/dam/fortinet/assets/threat-reports/report-threat-landscape-2021.pdf [Accessed: 9 April 2022]).

[37] Abdallah R Gad, Ahmed A Nashat, and Tamer M Barkat. “Intrusion detection system using machine learning for vehicular ad hoc networks based on ToN-IoT dataset”. In: *IEEE Access* 9 (2021), pp. 142206–142217.

[38] J. F. Canola Garcia and G. E. T. Blandon. “A deep learning-based intrusion detection and prevention system for detecting and preventing denial-of-service attacks”. In: *IEEE Access* 10 (2022), pp. 83043–83060. DOI: 10.1109/access.2022.3196642.

[39] Ming Ge et al. “Deep Learning-Based Intrusion Detection for IoT Networks”. In: *2019 IEEE 24th Pacific Rim International Symposium on Dependable Computing (PRDC)*. 2019, pp. 256–25609. DOI: 10.1109/PRDC47002.2019.00056.

[40] Pierre Geurts, Damien Ernst, and Louis Wehenkel. “Extremely randomized trees”. In: *Machine learning* 63 (2006), pp. 3–42.

[41] Nandita Goyal et al. “IoT based Blockchain System for Security from Identity Theft in Industrial Automation”. In: *2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies*. 2024, pp. 1–4. DOI: 10.1109/TQCEBT59414.2024.10545083.

[42] Alka Gupta and Lalit Sen Sharma. “Detecting attacks in high-speed networks: Issues and solutions”. In: *Information Security Journal: A Global Perspective* 29.2 (2020), pp. 51–61.

[43] Hacking and Countermeasure Research Lab. *Car-Hacking Dataset*. https://sites.google.com/a/hksecurity.net/ocslab/Datasets/CAN-intrusion-dataset. Accessed: Oct. 1, 2021.

[44] Talal Halabi et al. “Protecting the Internet of vehicles against advanced persistent threats: A bayesian Stackelberg game”. In: *IEEE Transactions on Reliability* 70.3 (2021), pp. 970–985.

[45] A. Halbouni et al. “CNN-LSTM: hybrid deep neural network for network intrusion detection system”. In: *IEEE Access* 10 (2022), pp. 99837–99849.

[46] Vikas Hassija et al. “A Survey on IoT Security: Application Areas, Security Threats, and Solution Architectures”. In: *IEEE Access* 7 (2019), pp. 82721–82743. DOI: 10.1109/ACCESS.2019.2924045.

[47] Hanan Hindy et al. “Utilising deep learning techniques for effective zero-day attack detection”. In: *Electronics* 9.10 (2020), p. 1684.

[48] “How does IoT affect big data?” In: (Available: https://www.iotacommunications.com/blog/iot-big-data/ [Online] [Accessed: 11 April 2022]).

[49] Ren-Hung Hwang et al. “An unsupervised deep learning model for early network traffic anomaly detection”. In: *IEEE Access* 8 (2020), pp. 30387–30399.

[50] i-SCOOP. *The CIA Triad of confidentiality, Integrity and Availability*. URL: https://www.i-scoop.eu/cybersecurity/cia-confidentiality-integrity-availability-security/.

[51] IN Groupe. *NIS-2: What Impact for Companies and Organizations? Cyber Security in Europe*. 2024. URL: https://ingroupe.com/insights/nis-2-what-impact-for-companies-organizations-cyber-security-europe/ (visited on 06/03/2026).

[52] International Electrotechnical Commission (IEC). *Functional Safety*. Accessed: 2025-06-28. 2025. URL: https://iec.ch/functional-safety.

[53] International Society of Automation (ISA). *ISA/IEC 62443 Series of Standards*. Accessed: 2025-06-27. ISA. 2024. URL: https://www.isa.org/standards-and-publications/isa-standards/isa-iec-62443-series-of-standards.

[54] Sana Ullah Jan et al. "Toward a Lightweight Intrusion Detection System for the Internet of Things". In: *IEEE Access* 7 (2019), pp. 42450–42471. DOI: 10.1109/ACCESS.2019.2907965.

[55] Pls Jayalaxmi et al. "A Taxonomy of Security Issues in Industrial Internet-of-Things: Scoping Review for Existing Solutions, Future Implications, and Research Challenges". In: *IEEE Access* 9 (2021), pp. 25344–25359. DOI: 10.1109/ACCESS.2021.3057766.

[56] Mohammad Jbair et al. "Threat modelling for industrial cyber physical systems in the era of smart manufacturing". In: *Computers in Industry* 137 (2022), p. 103611. ISSN: 0166-3615. DOI: https://doi.org/10.1016/j.compind.2022.103611.

[57] Jegeib. *Microsoft Threat Modeling Tool Overview - Azure*. URL: https://learn.microsoft.com/en-us/azure/security/develop/threat-modeling-tool.

[58] Yizhen Jia et al. "Flowguard: an intelligent edge defense mechanism against IoT DDoS attacks". In: *IEEE Internet of Things Journal* 7.10 (2020), pp. 9552–9562.

[59] Ming Jin et al. "A Survey on Graph Neural Networks for Time Series: Forecasting, Classification, Imputation, and Anomaly Detection". In: *arXiv preprint arXiv:2307.03759* (2023).

[60] Pankaj Kumar Keserwani, Mahesh Chandra Govil, and Emmanuel S. Pilli. "An Optimal Intrusion Detection System using GWO-CSA-DSAE Model". In: *Cyber-Physical Systems* 7.4 (2021), pp. 197–220.

[61] Pankaj Kumar Keserwani et al. "A smart anomaly-based intrusion detection system for the Internet of Things (IoT) network using GWO–PSO–RF model". In: *Journal of Reliable Intelligent Environments* 7.1 (2021), pp. 3–21.

[62] Adnan Shahid Khan et al. "A spectrogram image-based network anomaly detection system using deep convolutional neural network". In: *IEEE Access* 9 (2021), pp. 87079–87093.

[63] Rahamatullah Khondoker et al. “Addressing Industry 4.0 Security by Software-Defined Networking”. In: *Guide to Security in SDN and NFV: Challenges, Opportunities, and Applications*. Ed. by Shao Ying Zhu et al. Cham: Springer International Publishing, 2017, pp. 229–251. ISBN: 978-3-319-64653-4. DOI: 10.1007/978-3-319-64653-4_9. URL: https://doi.org/10.1007/978-3-319-64653-4_9.

[64] Ansam Khraisat et al. “Survey of intrusion detection systems: techniques, datasets and challenges”. In: *Cybersecurity* 2.1 (2019), pp. 1–22.

[65] Jiyeon Kim et al. “Intelligent Detection of IoT Botnets Using Machine Learning and Deep Learning”. In: *Applied Sciences* 10.19 (2020). ISSN: 2076-3417. DOI: 10.3390/app10197009. URL: https://www.mdpi.com/2076-3417/10/19/7009.

[66] Kyoung Ho Kim, Kyounggon Kim, and Huy Kang Kim. “STRIDE-based threat modeling and DREAD evaluation for the distributed control system in the oil refinery”. In: *ETRI Journal* 44.6 (2022), pp. 991–1003. DOI: https://doi.org/10.4218/etrij.2021-0181.

[67] Maciej Koch-Janusz and Zohar Ringel. “Mutual information, neural networks and the renormalization group”. In: *Nature Physics* 14.6 (2018), pp. 578–582.

[68] Constantinos Kolias et al. “DDoS in the IoT: Mirai and other botnets”. In: *Computer* 50.7 (2017), pp. 80–84.

[69] Anton Konev et al. “A Survey on Threat-Modeling Techniques: Protected Objects and Classification of Threats”. In: *Symmetry* 14.3 (2022). ISSN: 2073-8994. DOI: 10.3390/sym14030549.

[70] Nickolaos Koroniotis. “Designing an effective network forensic framework for the investigation of botnets in the Internet of Things”. Ph.D. Dissertation. The University of New South Wales Australia, 2020.

[71] Nickolaos Koroniotis and Nour Moustafa. “Enhancing network forensics with particle swarm and deep learning: The particle deep framework”. In: *arXiv preprint arXiv:2005.00722* (2020).

[72] Nickolaos Koroniotis, Nour Moustafa, and Elena Sitnikova. “A new network forensic framework based on deep learning for Internet of Things networks: A particle deep framework”. In: *Future Generation Computer Systems* 110 (2020), pp. 91–106.

[73] Nickolaos Koroniotis et al. “A Holistic Review of Cybersecurity and Reliability Perspectives in Smart Airports”. In: *IEEE Access* (2020).

[74] Nickolaos Koroniotis et al. *Towards Developing Network forensic mechanism for Botnet Activities in the IoT based on Machine Learning Techniques*. Unpublished or conference preprint (details not available). 2017.

[75] Nickolaos Koroniotis et al. "Towards developing network forensic mechanism for botnet activities in the IoT based on machine learning techniques". In: *International Conference on Mobile Networks and Management*. Springer, Cham, 2017, pp. 30–44.

[76] Nickolaos Koroniotis et al. "Towards the development of realistic botnet dataset in the internet of things for network forensic analytics: Bot-iot dataset". In: *Future Generation Computer Systems* 100 (2019), pp. 779–796.

[77] Bagesh Kumar et al. "A fast learning algorithm for One-Class Slab Support Vector Machines". In: *Knowledge-Based Systems* 228 (2021), p. 107267. ISSN: 0950-7051. DOI: https://doi.org/10.1016/j.knosys.2021.107267. URL: https://www.sciencedirect.com/science/article/pii/S0950705121005293.

[78] Vikash Kumar and Ditipriya Sinha. "A robust intelligent zero-day cyber-attack detection technique". In: *Complex & Intelligent Systems* 7.5 (2021), pp. 2211–2234.

[79] Arash Habibi Lashkari et al. "Characterization of Tor Traffic using Time based Features". In: *Proceedings of the 3rd International Conference on Information Systems Security and Privacy - Volume 1: ICISSP,* INSTICC. SciTePress, 2017, pp. 253–262. ISBN: 978-989-758-209-7. DOI: 10.5220/0006105602530262.

[80] Björn Leander, Aida Čaušević, and Hans Hansson. "Cybersecurity Challenges in Large Industrial IoT Systems". In: *2019 24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*. 2019, pp. 1035–1042. DOI: 10.1109/ETFA.2019.8869162.

[81] Jiabin Li et al. "RTVD: A real-time volumetric detection scheme for DDoS in the Internet of Things". In: *IEEE Access* 8 (2020), pp. 36191–36201.

[82] Yanmiao Li et al. "Robust detection for network intrusion of industrial IoT based on multi-CNN fusion". In: *Measurement* 154 (2020), p. 107450.

[83] Zhipeng Li et al. "Zero-shot learning for intrusion detection via attribute representation". In: *Neural Information Processing: 26th International Conference, ICONIP 2019, Sydney, NSW, Australia, December 12–15, 2019, Proceedings, Part I 26*. Springer. 2019, pp. 352–364.

[84] Chao Liang et al. “Intrusion detection system for Internet of Things based on a machine learning approach”. In: *2019 International Conference on Vision Towards Emerging Trends in Communication and Networking (ViTECoN)*. IEEE, 2019.

[85] Dr. Surita Maini and Dr.. Ashwani Kumar Aggarwal. “Camera Position Estimation using 2D Image Dataset”. In: 2018. URL: https://api.semanticscholar.org/CorpusID:225091809.

[86] Aaisha Makkar et al. “SecureIIoT Environment: Federated Learning Empowered Approach for Securing IIoT From Data Breach”. In: *IEEE Transactions on Industrial Informatics* 18.9 (2022), pp. 6406–6414. DOI: 10.1109/TII.2022.3149902.

[87] Michal Markevych and Maurice Dawson. “A Review of Enhancing Intrusion Detection Systems for Cybersecurity Using Artificial Intelligence (AI)”. In: *International conference KNOWLEDGE-BASED ORGANIZATION*. Vol. 29. 3. 2023, pp. 30–37.

[88] Mangesh Matke, Kumar Saurabh, and Uphar Singh. “An Empirical Evaluation of Machine Learning Algorithms for Intrusion Detection in IIoT Networks”. In: *2023 IEEE 20th India Council International Conference (INDICON)*. 2023, pp. 1353–1358. DOI: 10.1109/INDICON59947.2023.10440779.

[89] Yair Meidan et al. “N-BaIoT—Network-Based Detection of IoT Botnet Attacks Using Deep Autoencoders”. In: *IEEE Pervasive Computing* 17.3 (2018), pp. 12–22. DOI: 10.1109/MPRV.2018.03367731.

[90] Sri Harsha Mekala et al. “Cybersecurity for Industrial IoT (IIoT): Threats, countermeasures, challenges and future directions”. In: *Computer Communications* 208 (2023), pp. 294–320. ISSN: 0140-3664. DOI: https://doi.org/10.1016/j.comcom.2023.06.020. URL: https://www.sciencedirect.com/science/article/pii/S0140366423002189.

[91] Mohanad Mohammed et al. “Using stacking ensemble for microarray-based cancer classification”. In: *2018 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE)*. IEEE. 2018, pp. 1–8.

[92] Sohrab Mokhtari et al. “A machine learning approach for anomaly detection in industrial control systems based on measurement data”. In: *Electronics* 10.4 (2021), p. 407.

[93] Nour Moustafa. “New generations of internet of things datasets for cybersecurity applications based machine learning: TON_IoT datasets”. In: *Proceedings of the eResearch Australasia Conference, Brisbane, Australia*. 2019, pp. 21–25.

[94] Nour Moustafa et al. “Novel Geometric Area Analysis Technique for Anomaly Detection Using Trapezoidal Area Estimation on Large-Scale Networks”. In: *IEEE Transactions on Big Data* (2017).

[95] Nour Moustafa, Jill Slay, et al. “Big Data Analytics for Intrusion Detection System: Statistical Decision-Making Using Finite Dirichlet Mixture Models”. In: *Data Analytics and Decision Support for Cybersecurity*. Springer, Cham, 2017, pp. 127–156.

[96] Nour Moustafa and Jill Slay. “The Evaluation of Network Anomaly Detection Systems: Statistical Analysis of the UNSW-NB15 Dataset and the Comparison with the KDD99 Dataset”. In: *Information Security Journal: A Global Perspective* (2016), pp. 1–14.

[97] Nour Moustafa and Jill Slay. “UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)”. In: *2015 Military Communications and Information Systems Conference (MilCIS)*. 2015, pp. 1–6. DOI: 10.1109/MilCIS.2015.7348942.

[98] R Muthukrishnan and R Rohini. “LASSO: A feature selection technique in predictive modeling for machine learning”. In: *2016 IEEE International Conference on Advances in Computer Applications (ICACA)*. 2016, pp. 18–20. DOI: 10.1109/ICACA.2016.7887916.

[99] Shi Na, Liu Xumin, and Guan Yong. “Research on k-means clustering algorithm: An improved k-means clustering algorithm”. In: *2010 Third International Symposium on intelligent information technology and security informatics*. Ieee. 2010, pp. 63–67.

[100] S. Naseer et al. “Enhanced network anomaly detection based on deep neural networks”. In: *IEEE Access* 6 (2018), pp. 48231–48246.

[101] National Institute of Standards and Technology (NIST). *Cybersecurity Framework (CSF)*. Accessed: 2025-06-28. NIST. 2025. URL: https://www.nist.gov/cyberframework.

[102] *National Vulnerability Database (NVD) - Vulnerability Search*. https://nvd.nist.gov/vuln/search. Accessed: 29 June 2025. 2025.

[103] E. C. P. Neto et al. “CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment”. In: *Sensors* 23.13 (2023), p. 5941.

[104] “Number of connected devices worldwide 2030”. In: (Available: https://www.statista.com/statistics/802690/w connected-devices-by-access-technology/ [Accessed: 28 Feb 2022]).

[105] Augustus Odena. “Semi-supervised learning with generative adversarial networks”. In: *arXiv preprint arXiv:1606.01583* (2016).

[106] Hamed Haddad Pajouh et al. “A Two-Layer Dimension Reduction and Two-Tier Classification Model for Anomaly-Based Intrusion Detection in IoT Backbone Networks”. In: *IEEE Transactions on Emerging Topics in Computing* 7.2 (2019), pp. 314–323. DOI: 10.1109/TETC.2016.2633228.

[107] Hanchuan Peng, Fuhui Long, and Chris Ding. “Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy”. In: *IEEE Transactions on pattern analysis and machine intelligence* 27.8 (2005), pp. 1226–1238.

[108] Jan Pennekamp et al. “Towards an Infrastructure Enabling the Internet of Production”. In: *2019 IEEE International Conference on Industrial Cyber Physical Systems (ICPS)*. 2019, pp. 31–37. DOI: 10.1109/ICPHYS.2019.8780276.

[109] Phoenix Contact GmbH & Co. KG. “IEC 62443 – the industrial cybersecurity standard”. In: (2025). Accessed: 2025-06-28. URL: https://www.phoenixcontact.com/en-pc/iec-62443-the-industrial-cybersecurity-standard#ex-vwf3n45.

[110] Gabriel Arquelau Pimenta Rodrigues et al. “Cybersecurity and network forensics: Analysis of malicious traffic towards a honeynet with deep packet inspection”. In: *Applied Sciences* 7.10 (2017), p. 1082.

[111] M. Poongodi et al. “Intrusion Prevention System for DDoS Attack on VANET With reCAPTCHA Controller Using Information Based Metrics”. In: *IEEE Access* 7 (2019), pp. 158481–158491. DOI: 10.1109/ACCESS.2019.2945682.

[112] Bradley Potteiger, Goncalo Martins, and Xenofon Koutsoukos. “Software and Attack Centric Integrated Threat Modeling for Quantitative Risk Assessment”. In: HotSos ’16 (2016), pp. 99–108. URL: https://doi.org/10.1145/2898375.2898390.

[113] Liudmila Prokhorenkova et al. “CatBoost: unbiased boosting with categorical features”. In: *Advances in neural information processing systems* 31 (2018).

[114] Manoj Kumar Putchala. “Deep learning approach for intrusion detection system (ids) in the internet of things (iot) network using gated recurrent neural networks (gru)”. In: (2017).

[115] Md Mamunur Rashid et al. “Cyberattacks Detection in IoT-Based Smart City Applications Using Machine Learning Techniques”. In: *International Journal of Environmental Research and Public Health* 17.24 (2020). ISSN: 1660-4601. DOI: 10.3390/ijerph17249347. URL: https://www.mdpi.com/1660-4601/17/24/9347.

[116] Shailendra Rathore and Jong Hyuk Park. “Semi-supervised learning based distributed attack detection framework for IoT”. In: *Applied Soft Computing* 72 (2018), pp. 79–89. ISSN: 1568-4946. DOI: https://doi.org/10.1016/j.asoc.2018.05.049. URL: https://www.sciencedirect.com/science/article/pii/S1568494618303508.

[117] Nagarathna Ravi and S Mercy Shalinie. “Learning-driven detection and mitigation of DDoS attack in IoT via SDN-cloud architecture”. In: *IEEE Internet of Things Journal* 7.4 (2020), pp. 3559–3570.

[118] Sireesha Rodda. “Network intrusion detection systems using neural networks”. In: *Information systems design and intelligent applications*. Springer, 2018, pp. 903–908.

[119] Bipraneel Roy and Hon Cheung. “A deep learning approach for intrusion detection in internet of things using bi-directional long short-term memory recurrent neural network”. In: *2018 28th international telecommunication networks and applications conference (ITNAC)*. IEEE. 2018, pp. 1–6.

[120] Ulya Sabeel et al. “Building an intrusion detection system to detect atypical cyberattack flows”. In: *IEEE Access* 9 (2021), pp. 94352–94370.

[121] Mohamed Salb et al. “Enhancing Internet of Things Network Security Using Hybrid CNN and XGBoost Model Tuned via Modified Reptile Search Algorithm”. In: *Applied Sciences* 13.23 (2023). ISSN: 2076-3417. DOI: 10.3390/app132312687.

[122] Tim Salimans et al. “Improved Techniques for Training GANs”. In: *Advances in Neural Information Processing Systems*. Ed. by D. Lee et al. Vol. 29. Curran Associates, Inc., 2016. URL: https://proceedings.neurips.cc/paper/2016/file/8a3363abe792db2d8761d6403605aeb7-Paper.pdf.

[123] Ahmed Samy, Haining Yu, and Hongli Zhang. “Fog-Based Attack Detection Framework for Internet of Things Using Deep Learning”. In: *IEEE Access* 8 (2020), pp. 74571–74585. DOI: 10.1109/ACCESS.2020.2988854.

[124] T. Saranya et al. “Performance analysis of machine learning algorithms in intrusion detection system: A review”. In: *Procedia Computer Science* 171 (2020), pp. 1251–1260.

[125] M. Sarhan et al. “From zero-shot machine learning to zero-day attack detection”. In: *International Journal of Information Security* 22.4 (2023), pp. 947–959.

[126] Mohanad Sarhan, Siamak Layeghy, and Marius Portmann. "Evaluating Standard Feature Sets Towards Increased Generalisability and Explainability of ML-Based Network Intrusion Detection". In: *Big Data Research* 30 (2022), p. 100359. ISSN: 2214-5796. URL: https://www.sciencedirect.com/science/article/pii/S2214579622000533.

[127] Mohanad Sarhan et al. "NetFlow Datasets for Machine Learning-Based Network Intrusion Detection Systems". In: *Big Data Technologies and Applications: 10th EAI International Conference (BDTA 2020) and 13th EAI International Conference on Wireless Internet (WiCON 2020)*. Springer Nature, 2020, p. 117.

[128] K. Saurabh, V. Sharma, U. Singh, et al. "HMS-IDS: Threat Intelligence Integration for Zero-Day Exploits and Advanced Persistent Threats in IIoT". In: *Arabian Journal for Science and Engineering* (2024). DOI: 10.1007/s13369024089355.

[129] Kumar Saurabh et al. "ELA-NIDS: An Improved Ensemble Learning Network Intrusion Detection System for IIoT". In: *2024 IEEE 21st India Council International Conference (INDICON)*. 2024, pp. 1–6. DOI: 10.1109/INDICON63790.2024.10958487.

[130] Kumar Saurabh et al. "GANIBOT: A Network Flow Based Semi Supervised Generative Adversarial Networks Model for IoT Botnets Detection". In: *2022 IEEE International Conference on Omni-layer Intelligent Systems (COINS)*. IEEE. 2022, pp. 1–5.

[131] Kumar Saurabh et al. "LBDMIDS: LSTM Based Deep Learning Model for Intrusion Detection Systems for IoT Networks". In: (2022). DOI: 10.48550/ARXIV.2207.00424. URL: https://arxiv.org/abs/2207.00424.

[132] Kumar Saurabh et al. "Lbdmids: LSTM based deep learning model for intrusion detection systems for IOT networks". In: *2022 IEEE World AI IoT Congress (AIIoT)*. IEEE. 2022, pp. 753–759.

[133] Kumar Saurabh et al. "MLAPS: A Machine Learning based Second Line of Defense for Attack Prevention in IoT Network". In: *2022 IEEE 19th India Council International Conference (INDICON)*. IEEE. 2022, pp. 1–6.

[134] Kumar Saurabh et al. "NFDLM: A Lightweight Network Flow based Deep Learning Model for DDoS Attack Detection in IoT Domains". In: IEEE. 2022, pp. 736–742.

[135] Kumar Saurabh et al. *TMAP: A Threat Modeling and Attack Path Analysis Framework for Industrial IoT Systems (A Case Study of IoM and IoP)*. 2023. arXiv: 2312.15319 [cs.CR].

[136] Kumar Saurabh et al. “Zero-Shot Based Hybrid Neural Network System for Enhancing Zero-Day Attack Detection”. In: *2024 IEEE 21st India Council International Conference (INDICON)*. 2024, pp. 1–6. DOI: 10.1109/INDICON63790.2024.10958486.

[137] K. Sayed and H.A. Gabbar. “Chapter 18 - SCADA and smart energy grid control automation”. In: *Smart Energy Grid Engineering*. Ed. by Hossam A. Gabbar. Academic Press, 2017, pp. 481–514. ISBN: 978-0-12-805343-0. URL: https://www.sciencedirect.com/science/article/pii/B9780128053430000188.

[138] *Secure Architecture for Industrial Control Systems. https://www.sans.org/white-papers/36327/.* May 2023. URL: https://www.sans.org/white-papers/36327/.

[139] W. Seo and W. Pak. “Real-Time Network Intrusion Prevention System Based on Hybrid Machine Learning”. In: *IEEE Access* 9 (2021), pp. 46386–46397. DOI: 10.1109/ACCESS.2021.3066620.

[140] Kamalakanta Sethi et al. “Attention based multi-agent intrusion detection systems using reinforcement learning”. In: *Journal of Information Security and Applications* 61 (2021), p. 102923.

[141] Shahab Shamshirband et al. “Computational intelligence intrusion detection techniques in mobile cloud computing environments: Review, taxonomy, and open research issues”. In: *Journal of Information Security and Applications* 55 (2020), p. 102582.

[142] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A Ghorbani. “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization”. In: *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*. SciTePress. 2018, pp. 108–116. URL: https://www.unb.ca/cic/datasets/ids-2017.html.

[143] Iman Sharafaldin et al. “Developing Realistic Distributed Denial of Service (DDoS) Attack Dataset and Taxonomy”. In: *2019 International Carnahan Conference on Security Technology (ICCST)*. 2019, pp. 1–8. DOI: 10.1109/CCST.2019.8888419.

[144] Sagar Sharma. *Activation Functions in Neural Networks*. Accessed: June 9, 2026. Sept. 6, 2017. URL: https://towardsdatascience.com/activation-functions-neural-networks-1cbd9f8d91d6.

[145] Gautham Shetty and Krishnan Kocheril Raman. “Performance Analysis of a Snort-based Intrusion Detection System for Wireless Sensor Networks”. In: *Journal homepage: www. ijrpr. com ISSN* 2582 (), p. 7421.

[146] Nataliya Shevchenko et al. “Threat Modeling: A Summary of Available Methods”. In: (2018).

[147] Uphar Singh et al. “Hyperparameter Tuning for LSTM and ARIMA Time Series Model: A Comparative Study”. In: *2023 IEEE 4th Annual Flagship India Council International Subsections Conference (INDISCON)*. 2023, pp. 1–6. DOI: 10.1109/INDISCON58499.2023.10270325.

[148] Uphar Singh et al. “Optimizing Hyperparameters of Deep Learning Models for Stock Price Prediction”. In: *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*. 2024, pp. 1–7. DOI: 10.1109/ICCCNT61001.2024.10724813.

[149] S. Smys, Abul Basar, and Haoxiang Wang. “Hybrid intrusion detection system for Internet of Things (IoT)”. In: *Journal of ISMAC* 2.4 (2020), pp. 190–199.

[150] Yan Naung Soe et al. “Machine learning-based IoT-botnet attack detection with sequential architecture”. In: *Sensors* 20.16 (2020), p. 4372.

[151] S Shinly Swarna Sugi and S Raja Ratna. “Investigation of machine learning techniques in intrusion detection system for IoT network”. In: *2020 3rd International Conference on Intelligent Sustainable Systems (ICISS)*. IEEE. 2020, pp. 1164–1167.

[152] Lin Sun et al. “Feature selection using symmetric uncertainty and hybrid optimization for high-dimensional data”. In: *International Journal of Machine Learning and Cybernetics* (2023), pp. 1–22.

[153] P. Sun et al. “DL-IDS: Extracting features using CNN-LSTM hybrid network for intrusion detection system”. In: *Security and Communication Networks* 2020 (2020).

[154] Yarom Talmi. *Achieving NIS2 Compliance: 10 Steps to Follow*. Apr. 2024. URL: https://cybeready.com/achieving-nis2-compliance-10-steps-to-follow/.

[155] Bayu Adhi Tama and Kyung-Hyune Rhee. “Classifier ensemble design with rotation forest to enhance attack detection of IDS in wireless network”. In: *2016 11th Asia Joint Conference on Information Security (AsiaJCIS)*. IEEE. 2016, pp. 87–91.

[156] Mahbod Tavallaee et al. “A detailed analysis of the KDD CUP 99 data set”. In: *2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications*. 2009, pp. 1–6. DOI: 10.1109/CISDA.2009.5356528.

[157] Abebe Tesfahun and D Lalitha Bhaskari. “Intrusion detection using random forests classifier with SMOTE and feature reduction”. In: *2013 International Conference on Cloud & Ubiquitous Computing & Emerging Technologies*. IEEE. 2013, pp. 127–132.

[158] Robert Tibshirani. “Regression shrinkage and selection via the lasso”. In: *Journal of the Royal Statistical Society: Series B (Methodological)* 58.1 (1996), pp. 267–288.

[159] Firdevs Sevde Toker, Kevser Ovaz Akpinar, and Ibrahim OZCELIK. “MITRE ICS Attack Simulation and Detection on EtherCAT Based Drinking Water System”. In: *2021 9th International Symposium on Digital Forensics and Security (ISDFS)*. 2021, pp. 1–6. DOI: 10.1109/ISDFS52919.2021.9486331.

[160] Pablo Torres et al. “An analysis of recurrent neural networks for botnet detection behavior”. In: *2016 IEEE biennial congress of Argentina (ARGENCON)*. IEEE. 2016, pp. 1–6.

[161] D. Tripathi et al. “Towards analyzing the impact of intrusion prevention and response on cyber-physical system availability: A case study of NPP”. In: *Annals of Nuclear Energy* 168 (2022), p. 108863. DOI: 10.1016/j.anucene.2021.108863.

[162] Imtiaz Ullah and Qusay H Mahmoud. “Design and development of a deep learning-based model for anomaly detection in IoT networks”. In: *IEEE Access* 9 (2021), pp. 103906–103926.

[163] R. K. Vigneswaran et al. “Evaluating shallow and deep neural networks for network intrusion detection systems in cyber security”. In: *2018 9th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*. IEEE, 2018, pp. 1–6.

[164] Aditya Vikram and Mohana. “Anomaly detection in Network Traffic Using Unsupervised Machine learning Approach”. In: *2020 5th International Conference on Communication and Electronics Systems (ICCES)*. 2020, pp. 476–479. DOI: 10.1109/ICCES48766.2020.9137987.

[165] Amakiri Welekwe. “Threat Modeling Guide”. In: *Comparitech* (Sept. 2021). URL: https://www.comparitech.com/net-admin/threat-modeling-guide/.

[166] Y. Xian, B. Schiele, and Z. Akata. “Zero-shot learning—the good, the bad and the ugly”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2017, pp. 4582–4591.

[167] Liang Xiao et al. “IoT Security Techniques Based on Machine Learning: How Do IoT Devices Use AI to Enhance Security?” In: *IEEE Signal Processing Magazine* 35.5 (2018), pp. 41–49. DOI: 10.1109/MSP.2018.2825478.

[168] P Viswanath Y C A Padmanabha Reddy and B Eswara Reddy. “Semi-supervised learning: a brief review”. In: *International Journal of Engineering & Technology* 7.1.8 (2018), pp. 81–85. ISSN: 2227-524X. DOI: 10.14419/ijet.v7i1.8.9977.

[169] Li Yang, Abdallah Moubayed, and Abdallah Shami. “MTH-IDS: A multitiered hybrid intrusion detection system for internet of vehicles”. In: *IEEE Internet of Things Journal* 9.1 (2021), pp. 616–632.

[170] Li Yang and Abdallah Shami. “On hyperparameter optimization of machine learning algorithms: Theory and practice”. In: *Neurocomputing* 415 (2020), pp. 295–316. ISSN: 0925-2312. DOI: https://doi.org/10.1016/j.neucom.2020.07.061.

[171] Li Yang et al. “LCCDE: A decision-based ensemble framework for intrusion detection in the internet of vehicles”. In: *GLOBECOM 2022-2022 IEEE Global Communications Conference*. IEEE. 2022, pp. 3545–3550.

[172] Li Yang et al. “Tree-based intelligent intrusion detection system in internet of vehicles”. In: *2019 IEEE global communications conference (GLOBECOM)*. IEEE. 2019, pp. 1–6.

[173] Yun Yang et al. “GAN-Based Semi-Supervised Learning Approach for Clinical Decision Support in Health-IoT Platform”. In: *IEEE Access* 7 (2019), pp. 8048–8057. DOI: 10.1109/ACCESS.2018.2888816.

[174] Shuochao Yao et al. “Sensegan: Enabling deep learning for internet of things with a semi-supervised framework”. In: vol. 2. 3. ACM New York, NY, USA, 2018, pp. 1–21.

[175] Lei Yu and Huan Liu. “Efficiently Handling Feature Redundancy in High-Dimensional Data”. In: *Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. KDD ’03. Washington, D.C.: Association for Computing Machinery, 2003, pp. 685–690. ISBN: 1581137370. DOI: 10.1145/956750.956840.

[176] Umme Zahoora et al. “Zero-day ransomware attack detection using deep contractive autoencoder and voting based ensemble classifier”. In: *Applied Intelligence* 52.12 (2022), pp. 13941–13960.

[177] Saman Taghavi Zargar, James Joshi, and David Tipper. “A survey of defense mechanisms against distributed denial of service (DDoS) flooding attacks”. In: *IEEE communications surveys & tutorials* 15.4 (2013), pp. 2046–2069.

[178] Chuanlei Zhang et al. “Unsupervised anomaly detection based on deep autoencoding and clustering”. In: *Security and Communication Networks* 2021 (2021), pp. 1–8.

[179] Z. Zhang et al. “Unknown attack detection based on zero-shot learning”. In: *IEEE Access* 8 (2020), pp. 193981–193991.

[180] Ming Zhong, Yajin Zhou, and Gang Chen. “Sequential model based intrusion detection system for IoT servers using deep learning methods”. In: *Sensors* 21.4 (2021), p. 1113.